

**Faculty of Engineering Department of
Informatics Engineering Artificial
Intelligence Engineering and Data Science**



كلية هندسة الذكاء الاصطناعي
FACULTY OF ARTIFICIAL INTELLIGENCE ENGINEERING

Multimodal Deep Learning Framework for Brain Tumor Prediction

Senior1 Project- Completed the requirements for obtaining a bachelor's degree in Informatics
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قسم هندسة الذكاء الاصطناعي وعلوم البيانات

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Supervisor Certification

I certify that the preparation of this graduation project entitled “Multimodal Deep Learning Framework for Brain Tumor Prediction”, prepared by Rafah Sanjakdar, was carried out under my supervision at the Department of Artificial Intelligence Engineering and Data Science, Faculty of Artificial Intelligence Engineering, Syrian Private University, in partial fulfillment of the requirements for the Bachelor’s Degree in Informatics Engineering – Artificial Intelligence Engineering and Data Science.

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Abstract

Brain tumor classification plays a critical role in supporting accurate diagnosis and treatment planning. Magnetic Resonance Imaging (MRI) is one of the most widely used imaging techniques for detecting brain tumors due to its high resolution and detailed representation of soft tissues. However, manual interpretation of MRI scans is time-consuming and highly dependent on the expertise of medical specialists.

This project proposes a deep learning–based framework for the classification of brain tumor MRI images into **low-grade** and **high-grade** tumors. The proposed system employs **three different deep learning models** to analyze MRI images and automatically extract discriminative features for accurate tumor grading. The models are trained and evaluated using a publicly available dataset, with appropriate preprocessing techniques applied to enhance image quality and improve classification performance.

The experimental results demonstrate promising performance in terms of accuracy and reliability in distinguishing between low-grade and high-grade brain tumors. Three different deep learning models were trained and evaluated using standard performance metrics, followed by the application of an **ensemble stacking method** to combine their outputs and enhance the overall classification performance. The proposed framework aims to assist medical professionals by providing an efficient and reliable tool for brain tumor classification, contributing to improved diagnostic accuracy and clinical decision-making.

LIST OF TABLES

Metric	Mean $\pm$ Std	Minimum	Maximum
AUC-ROC	0.9613 $\pm$ 0.0302	0.9095	0.9825
F1-Score	0.8242 $\pm$ 0.0487	0.7689	0.8697
Accuracy	0.8526 $\pm$ 0.0502	0.7895	0.8947

Table (1): Cross-Validation Performance of DualStreamMIL-3D (Mean $\pm$ Std over 5 Folds)

Model	Mean HGG Probability	Standard Deviation	Minimum	Maximum
ResNet50-3D	0.8884	0.1195	0.1652	0.9999
SwinUNETR-3D	0.5799	0.4406	0.0023	1
DualStreamMIL-3D	0.4847	0.153	0.1113	0.9497

Table (2): Out-of-Fold HGG Probability Statistics Across Models

Base Model	Feature Name	Coefficient	Absolute Value
ResNet50-3D	hgg_prob_resnet	0.5369	0.537
SwinUNETR-3D	hgg_prob_swin	4.0634	4.063
DualStreamMIL-3D	hgg_prob_mil	0.89	0.89
—	Intercept	-2.4049	—

Table (3): Learned combination weights of the logistic regression meta-learner used to integrate base model predictions.

Metric	Value
Accuracy	0.8105 (81.05%)
Precision	0.9643 (96.43%)
Recall	0.7714 (77.14%)
F1-Score	0.8571 (85.71%)
AUC-ROC	0.9126 (91.26%)

Table (4): Overall Performance Metrics of the Ensemble Meta-Learner

Class	Precision	Recall	F1-Score	Support
LGG (0)	0.5897	0.92	0.7188	75
HGG (1)	0.9643	0.7714	0.8571	210

Table (5): Per-Class Performance Metrics of the Ensemble Meta-Learner

	Predicted LGG	Predicted HGG
Actual LGG	69	6
Actual HGG	48	162

Table (6): Confusion Matrix of the Ensemble Meta-Learner

List of Equations

(Equation 1)

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN}$$

(Equation 2)

$$precision = \sum_{c=1}^c \frac{1}{c} \frac{TcP}{Tc \cdot p_c + FP}$$

(Equation 3)

$$recall = \sum_{c=1}^c \frac{1}{c} \frac{TcP}{Tc \cdot p_c + FN}$$

(Equation 4)

$$F1 = \sum_{c=1}^c \frac{1}{c} \frac{precision \cdot recall}{precision + recall}$$

(Equation 5)

$$\omega_s = \frac{N}{\mathbf{k} \cdot (\mathbf{n}_c + \epsilon)}$$

LIST OF FIGURES

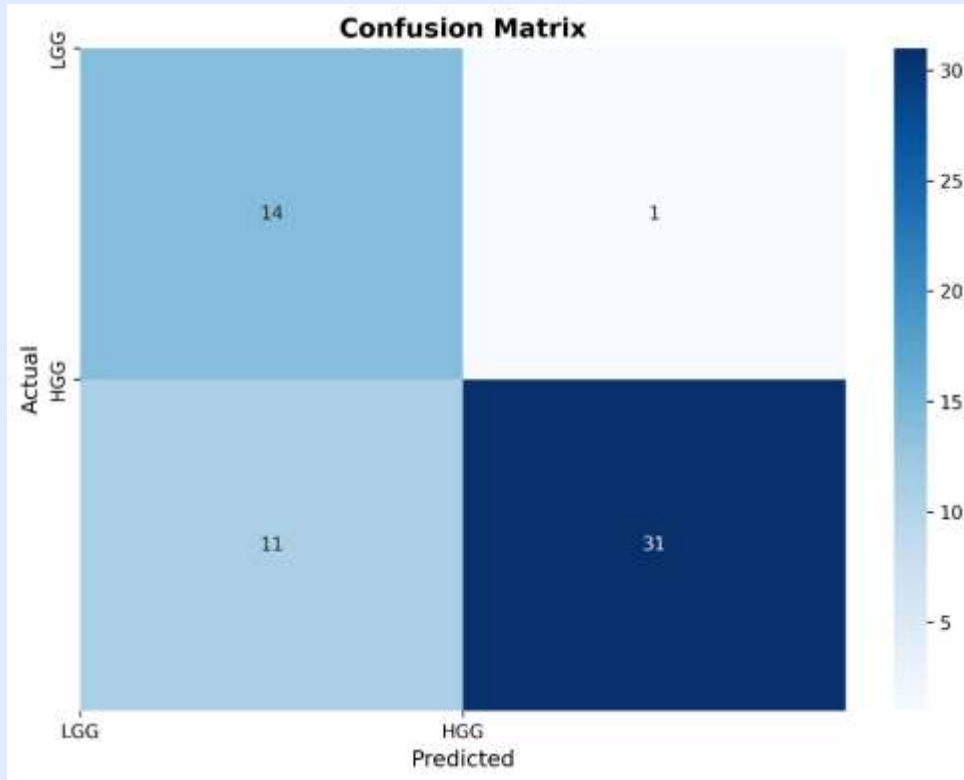


Figure 1: Confusion matrix of DualStreamMIL-3D for Fold 0 at the best validation epoch (epoch 12), showing strong bias toward LGG predictions and poor sensitivity to HGG cases.

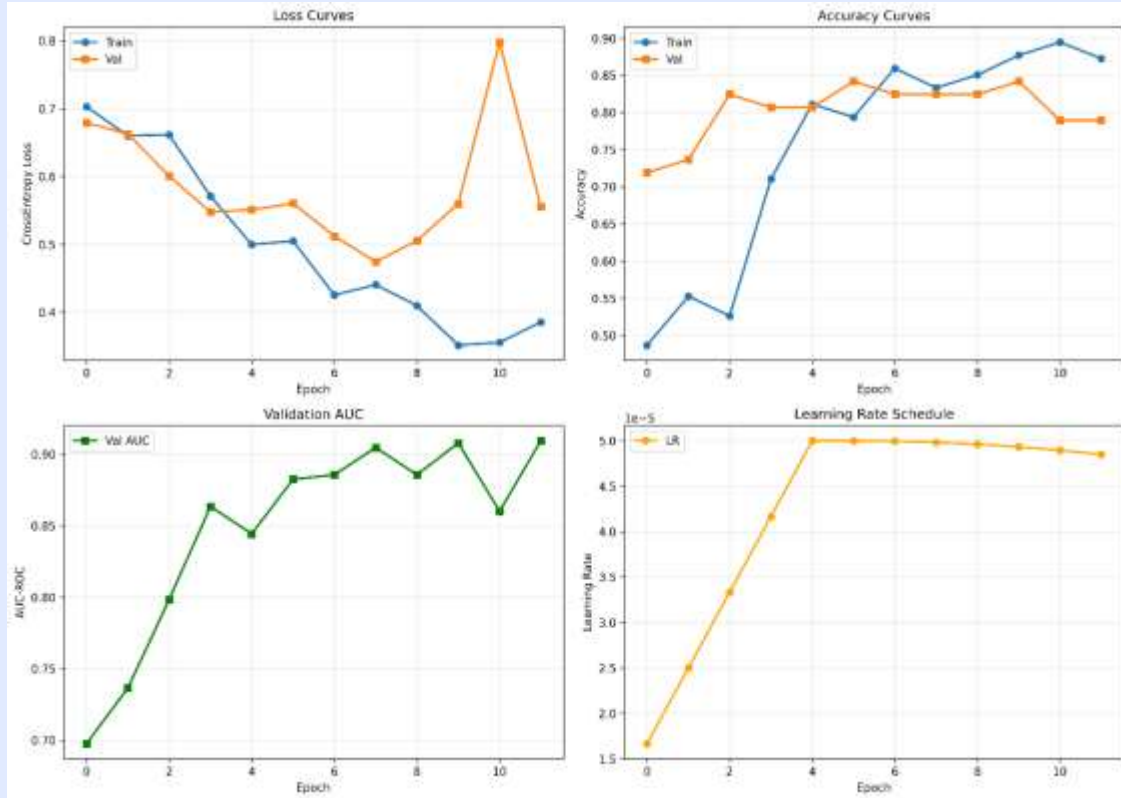


Figure 2: Learning curves of DualStreamMIL-3D for Fold 0, illustrating divergence between training and validation performance indicative of overfitting.

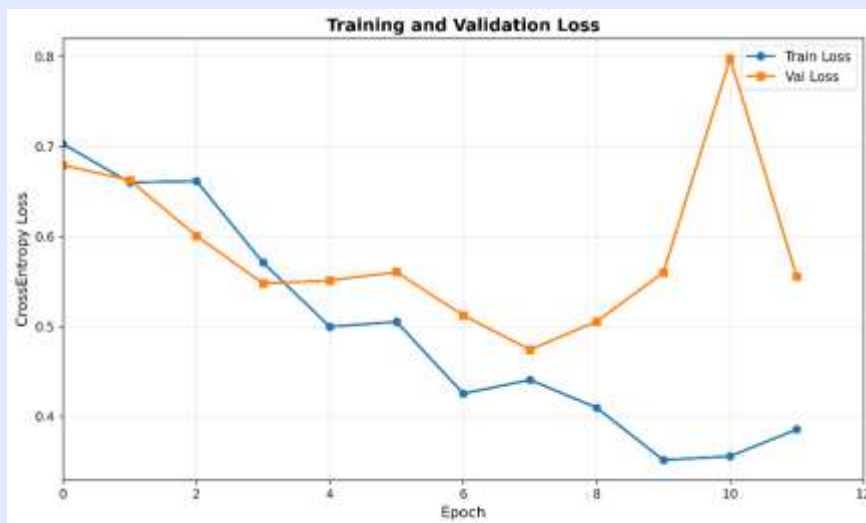


Figure 3: Training and validation loss curves for DualStreamMIL-3D in Fold 0, showing unstable validation loss despite decreasing training loss.

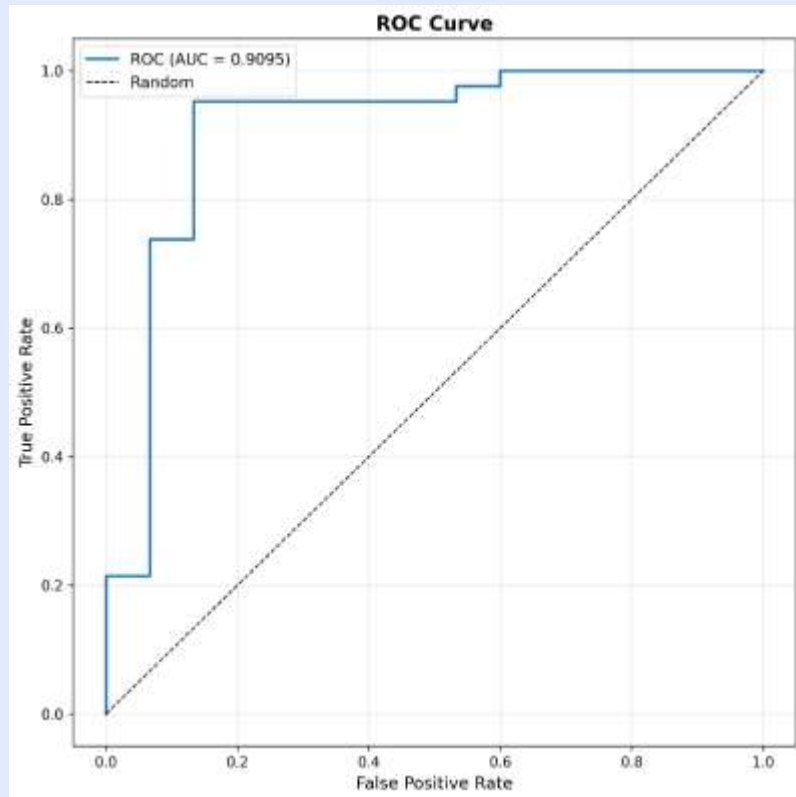


Figure 4: ROC curve of DualStreamMIL-3D for Fold 0 at the best validation epoch (epoch 12), with an AUC of 0.9095.

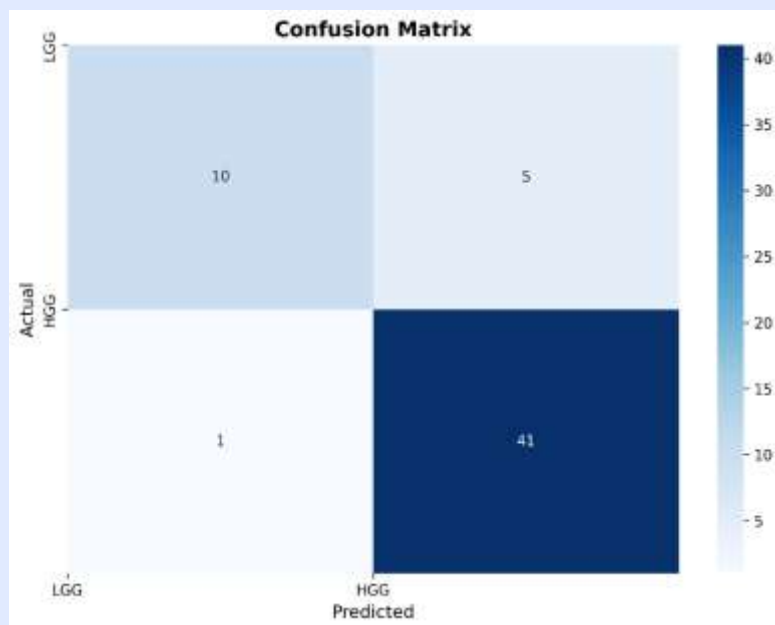


Figure 5: Confusion matrix of DualStreamMIL-3D for Fold 1 at the best validation epoch (epoch 22), showing improved balance between LGG and HGG predictions.

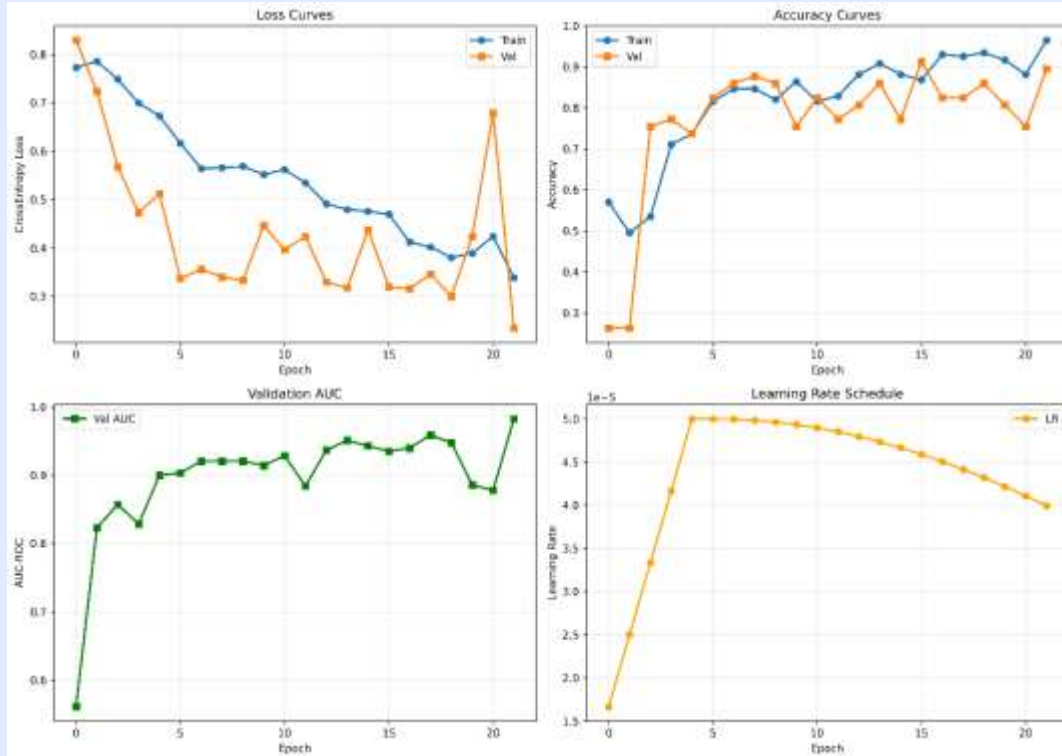


Figure 6: Learning curves of DualStreamMIL-3D for Fold 1, illustrating improved but still imperfect generalization behavior.

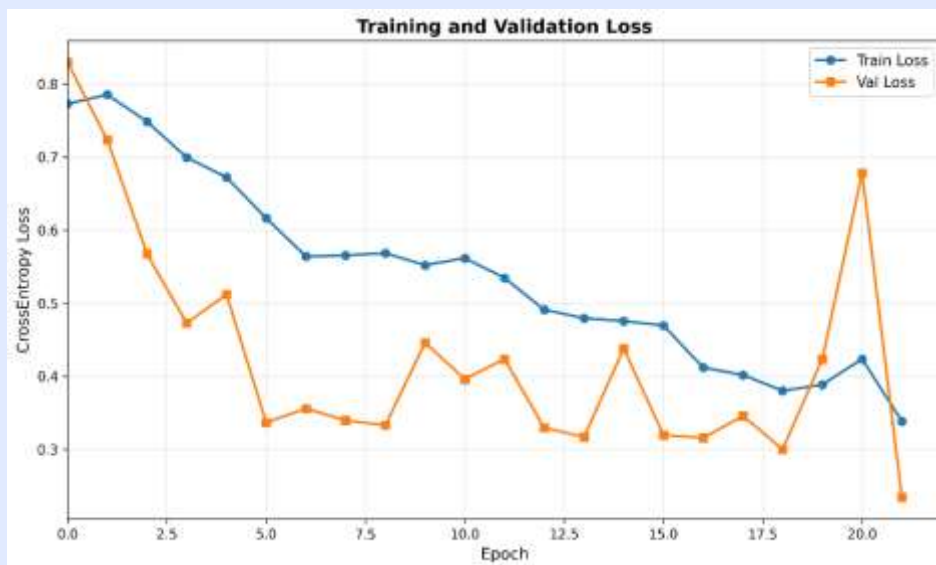


Figure 7: Training and validation loss curves for DualStreamMIL-3D in Fold 1, showing partial stabilization with residual overfitting.

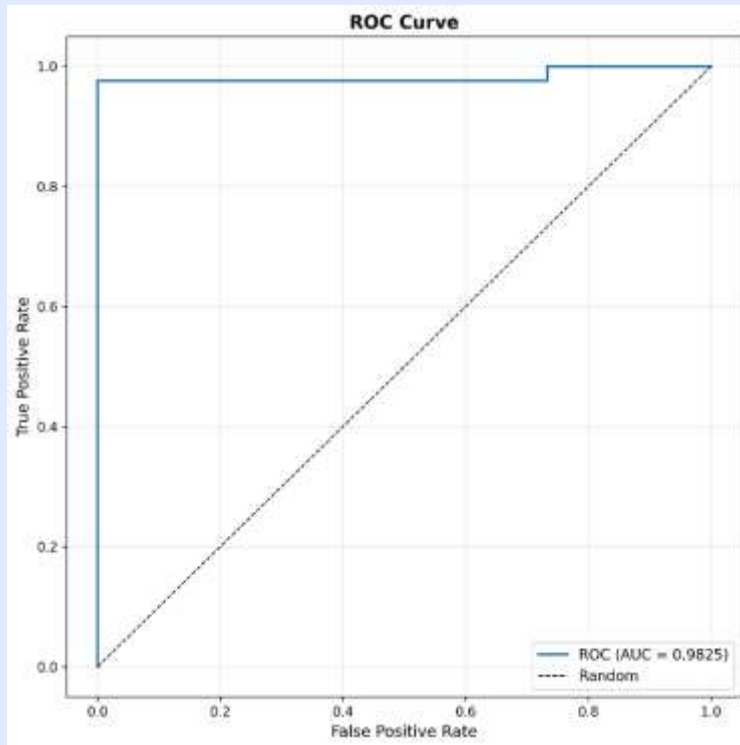


Figure 8: ROC curve of DualStreamMIL-3D for Fold 1 at the best validation epoch (epoch 22), achieving an AUC of 0.9825.

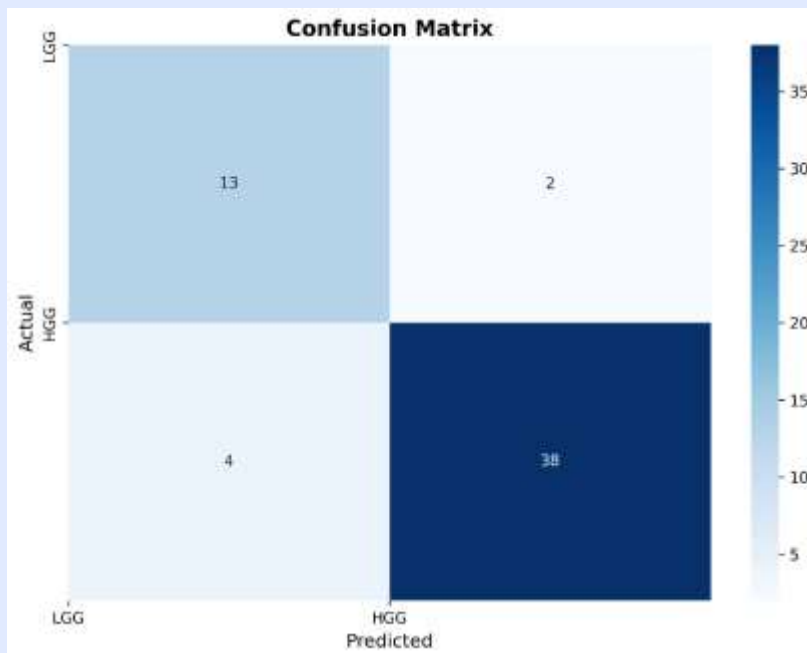


Figure 9: Confusion matrix of DualStreamMIL-3D for Fold 2 at the best validation epoch (epoch 14), showing perfect LGG classification and reduced HGG sensitivity.

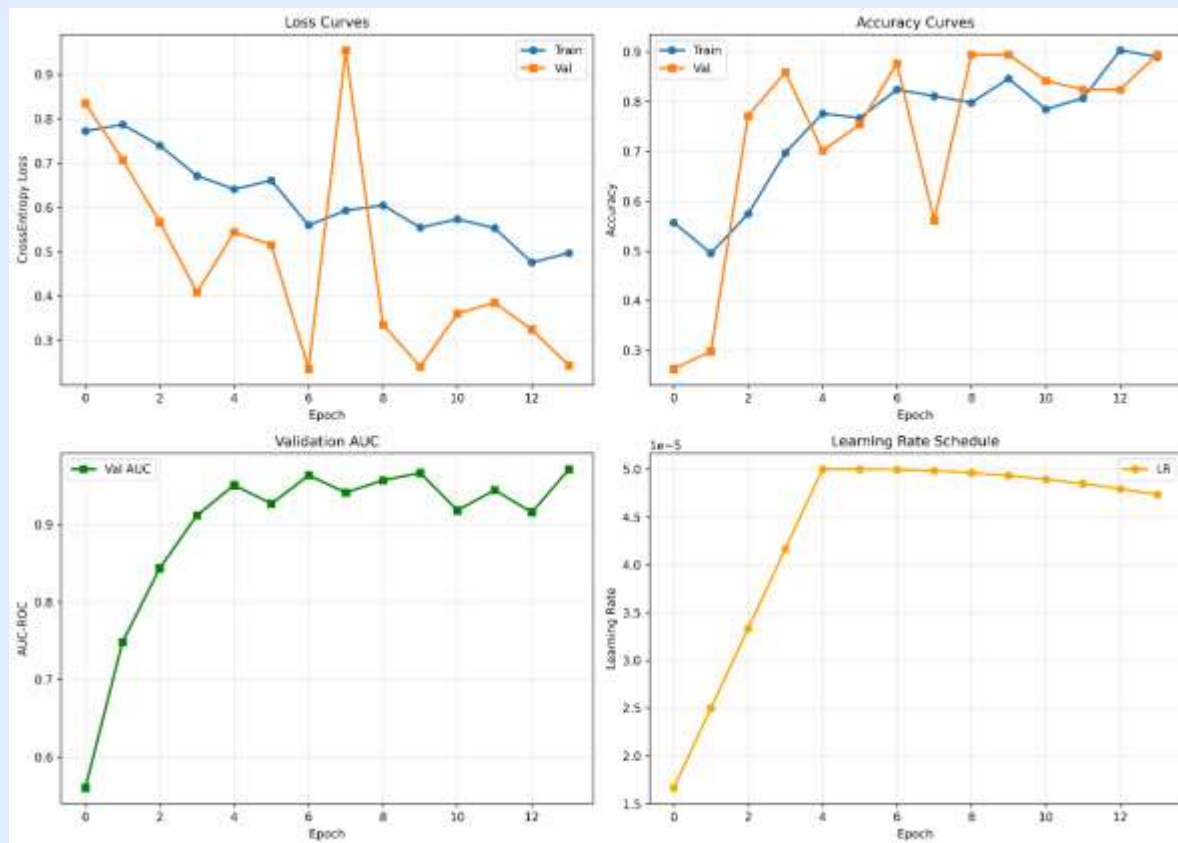


Figure 10: Learning curves of DualStreamMIL-3D for Fold 2, highlighting early divergence between training and validation performance.

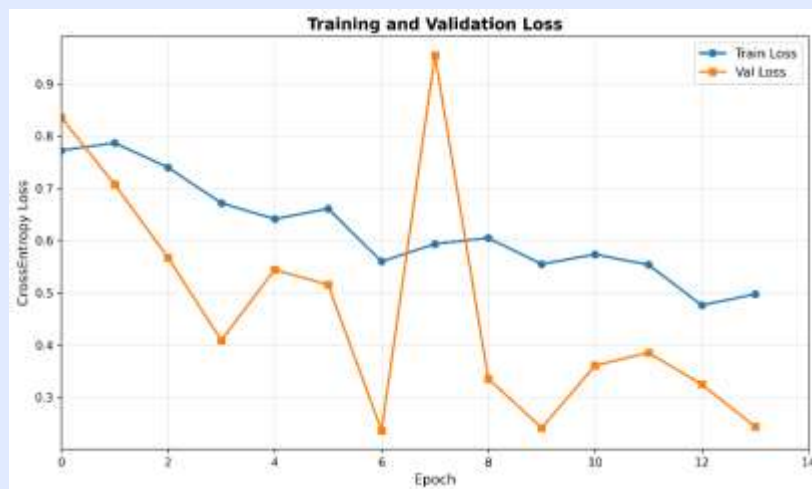


Figure 11: Training and validation loss curves for DualStreamMIL-3D in Fold 2, indicating unstable validation loss behavior.

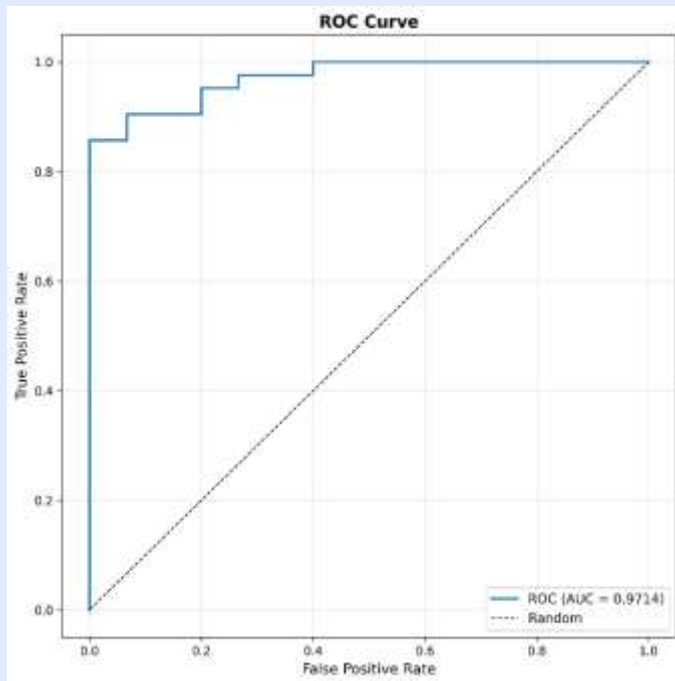


Figure 12: ROC curve of DualStreamMIL-3D for Fold 2 at the best validation epoch (epoch 14), with an AUC of 0.9714.

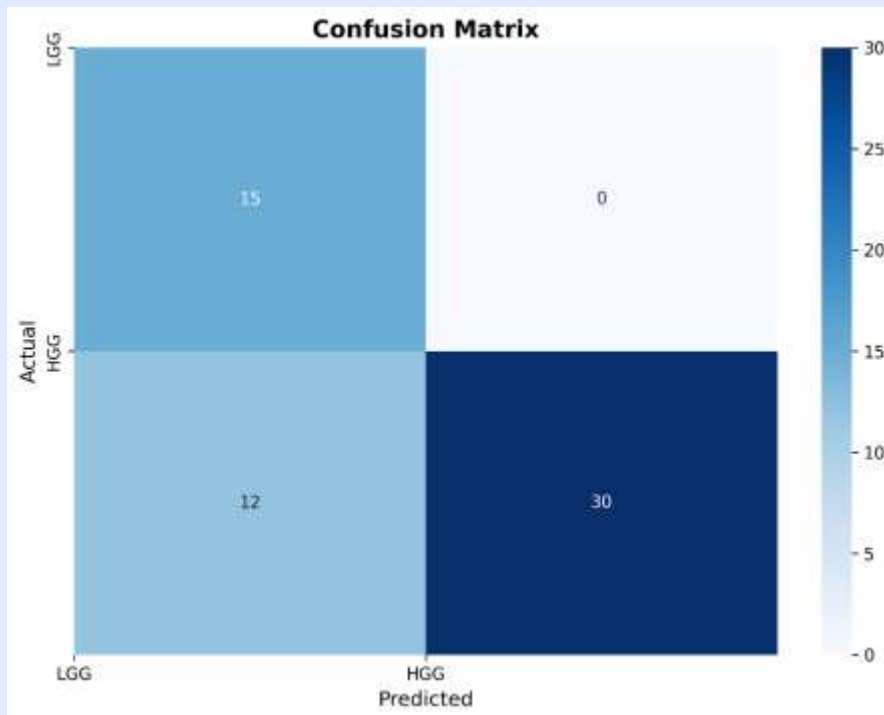


Figure 13: Confusion matrix of DualStreamMIL-3D for Fold 3 at the best validation epoch (epoch 7), showing complete collapse to single-class (LGG) prediction.

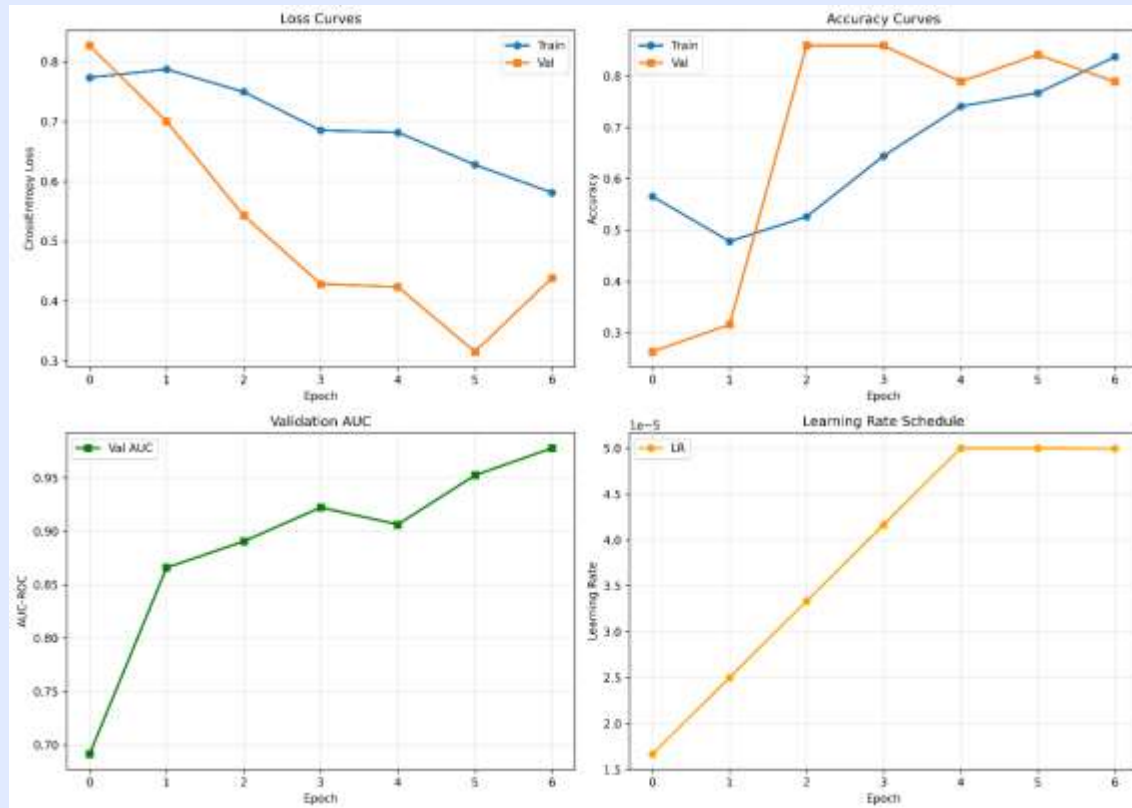


Figure 14: Learning curves of DualStreamMIL-3D for Fold 3, illustrating rapid overfitting and unstable validation performance.



Figure 15: Training and validation loss curves for DualStreamMIL-3D in Fold 3, indicating lack of convergence in validation loss.

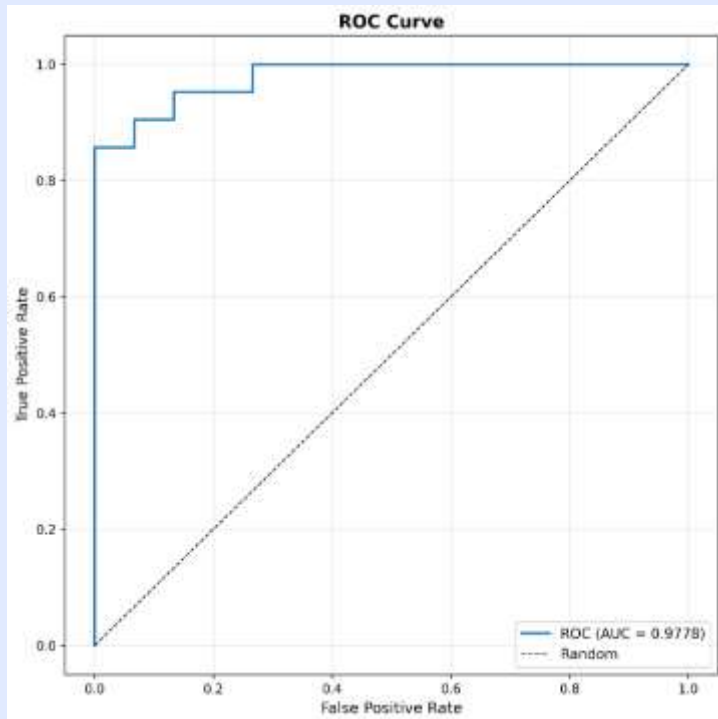


Figure 16: ROC curve of DualStreamMIL-3D for Fold 3 at the best validation epoch (epoch 7), achieving an AUC of 0.9778 despite degenerate classification.

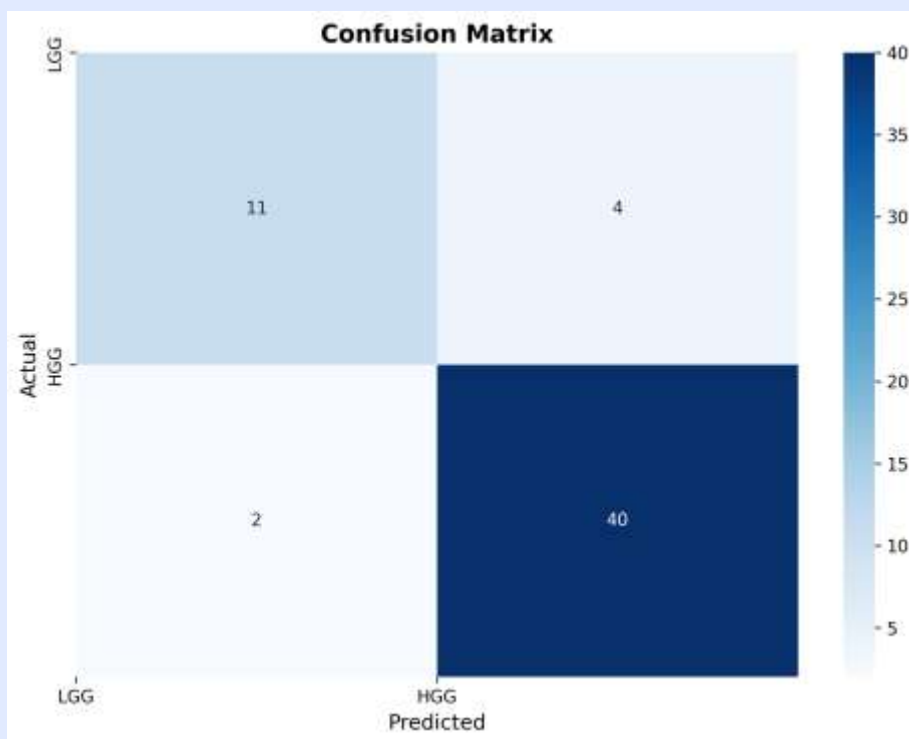


Figure 17: Confusion matrix of DualStreamMIL-3D for Fold 4 at the best validation epoch (epoch 10), showing complete collapse to LGG-only predictions.

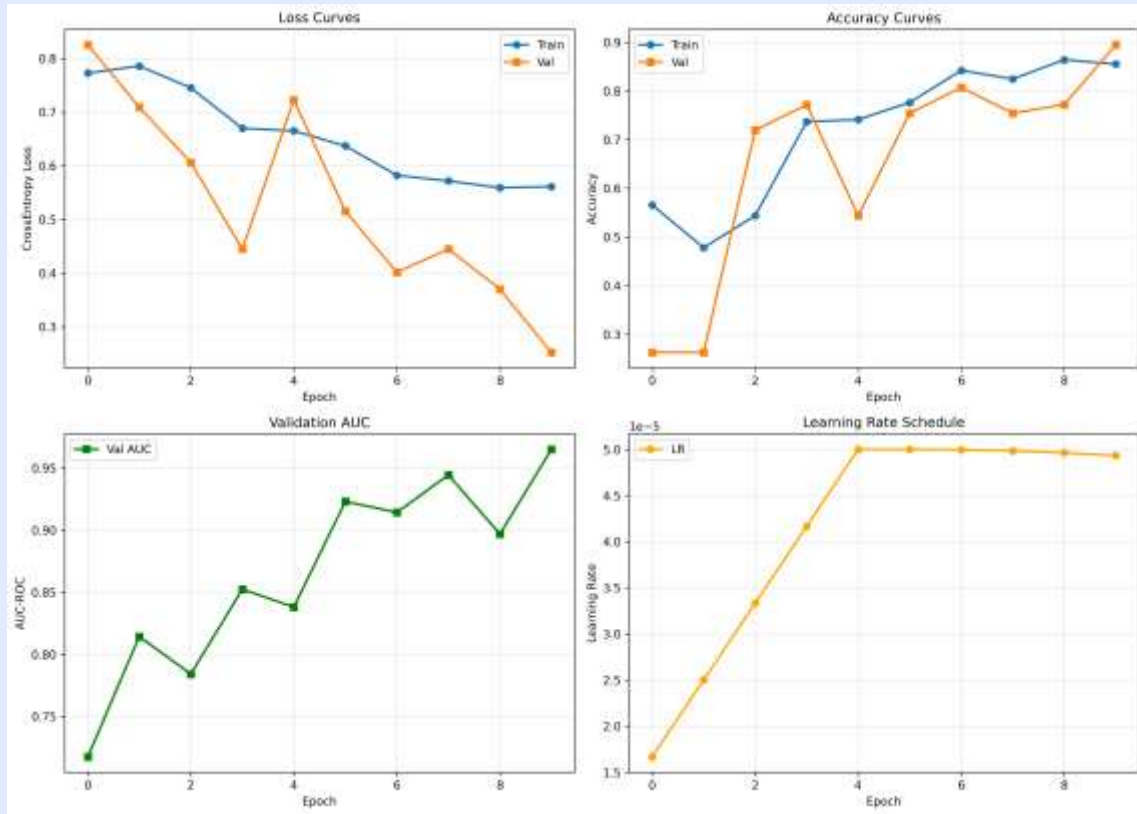


Figure 18: Learning curves of DualStreamMIL-3D for Fold 4, illustrating persistent overfitting behavior.



Figure 19: Training and validation loss curves for DualStreamMIL-3D in Fold 4, showing unstable validation loss patterns.

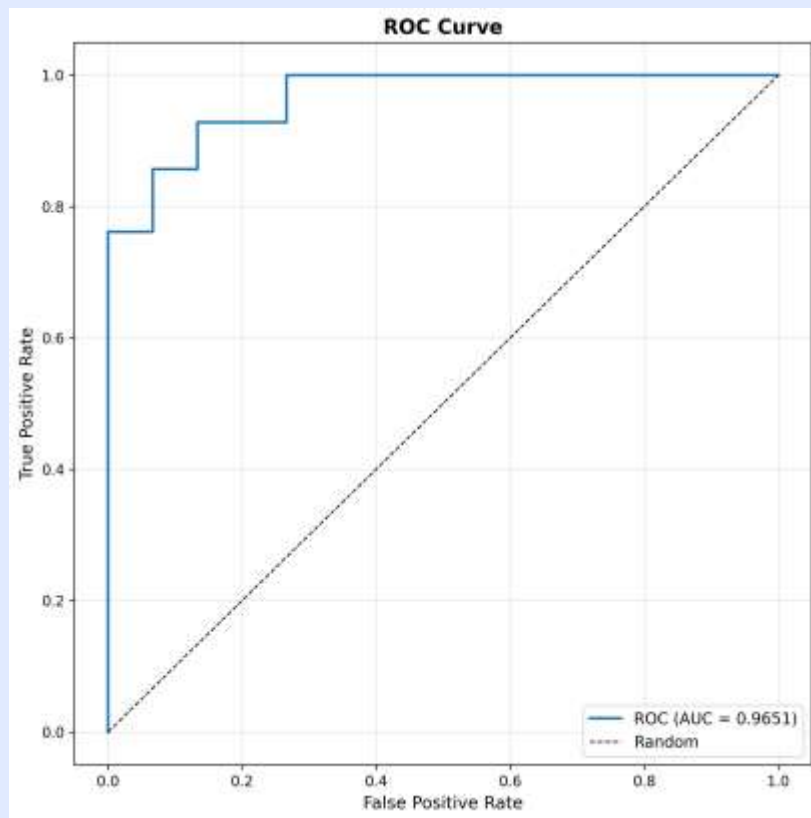


Figure 20: ROC curve of DualStreamMIL-3D for Fold 4 at the best validation epoch (epoch 10), with an AUC of 0.9651.

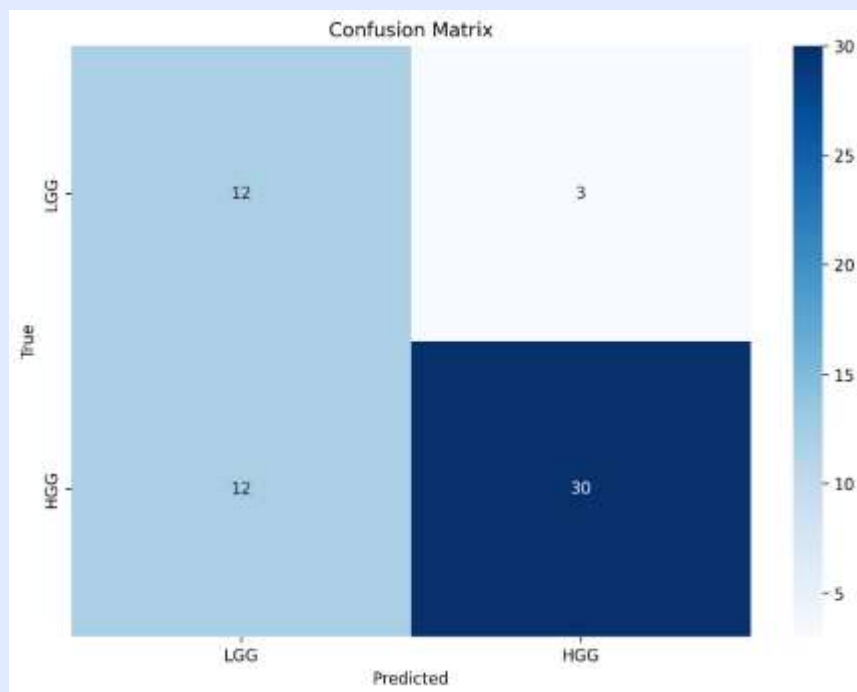


Figure 21: Confusion matrix of ResNet50-3D for Fold 0 at the best validation epoch (epoch 18), showing moderate class separation with notable HGG misclassification.

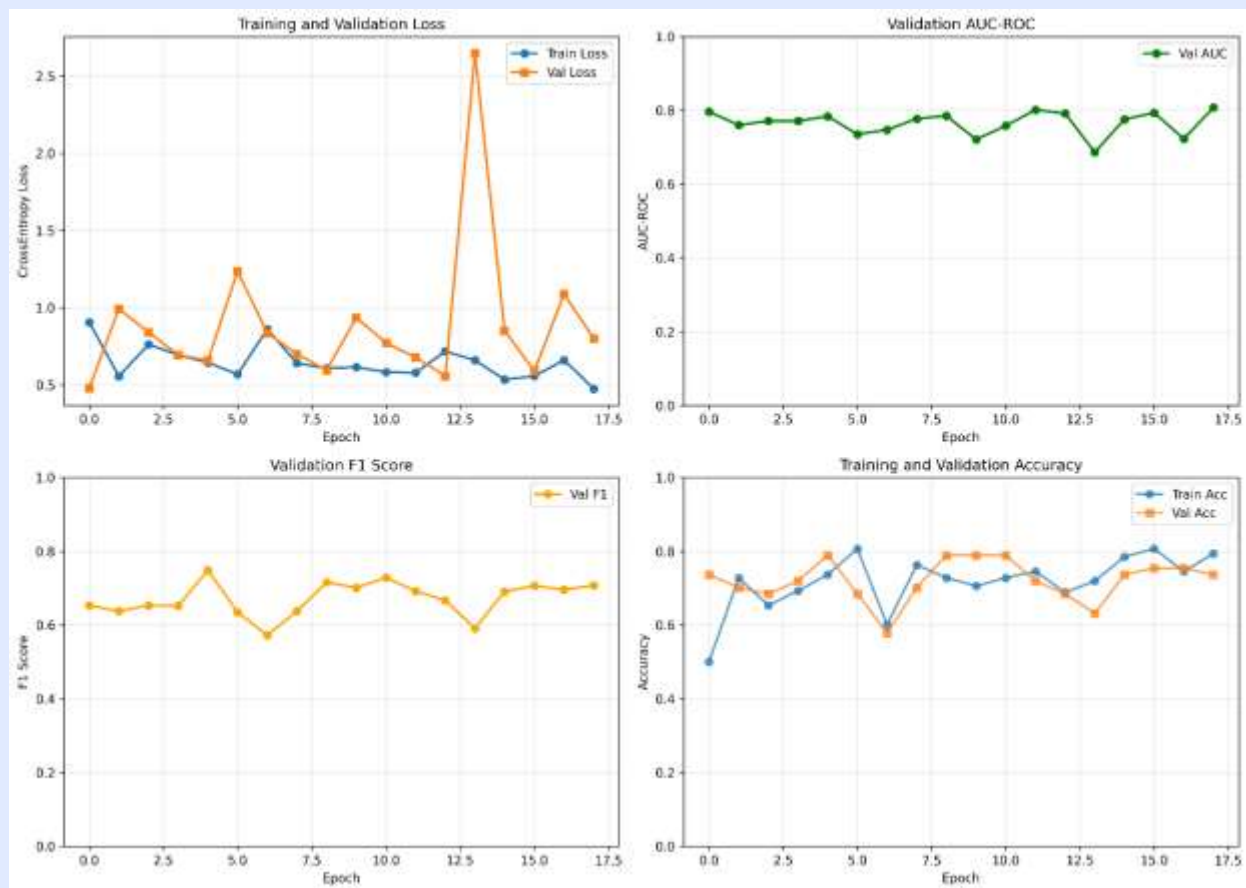


Figure 22: Learning curves of ResNet50-3D for Fold 0, illustrating overfitting through divergence between training and validation performance.

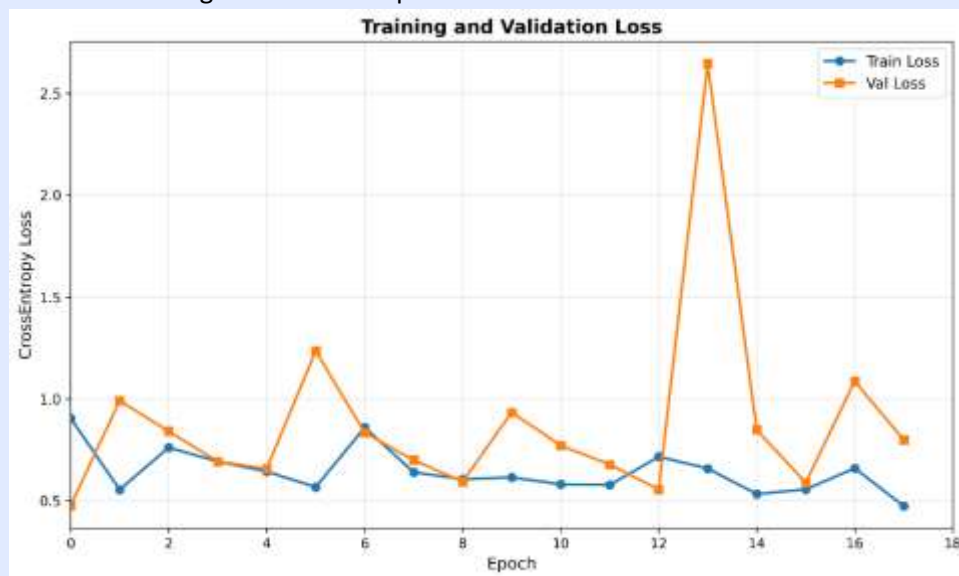


Figure 23: Training and validation loss curves of ResNet50-3D for Fold 0, showing limited convergence of validation loss.

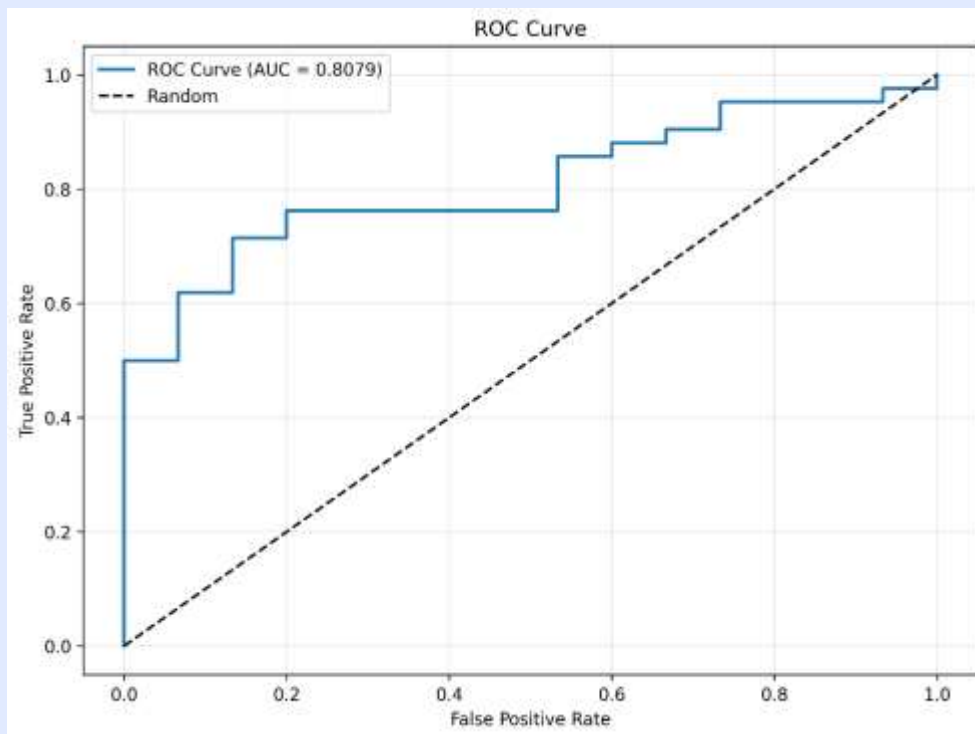


Figure 24: ROC curve of ResNet50-3D for Fold 0 at the best validation epoch (epoch 18), achieving an AUC of 0.8079.

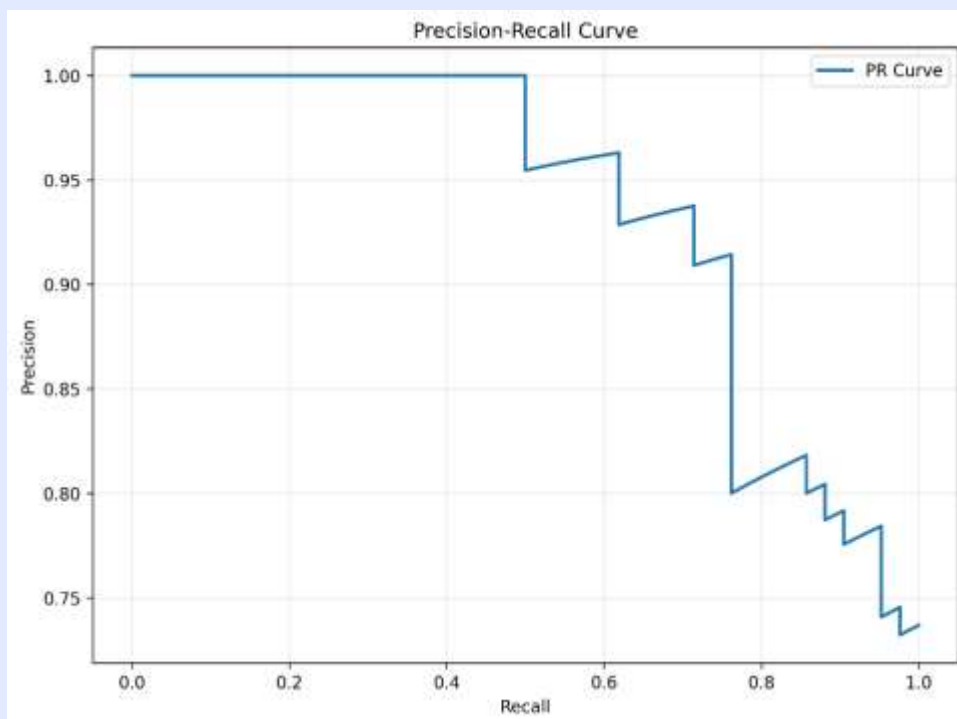


Figure 25: Precision-recall curve of ResNet50-3D for Fold 0, highlighting reduced precision at higher recall levels.

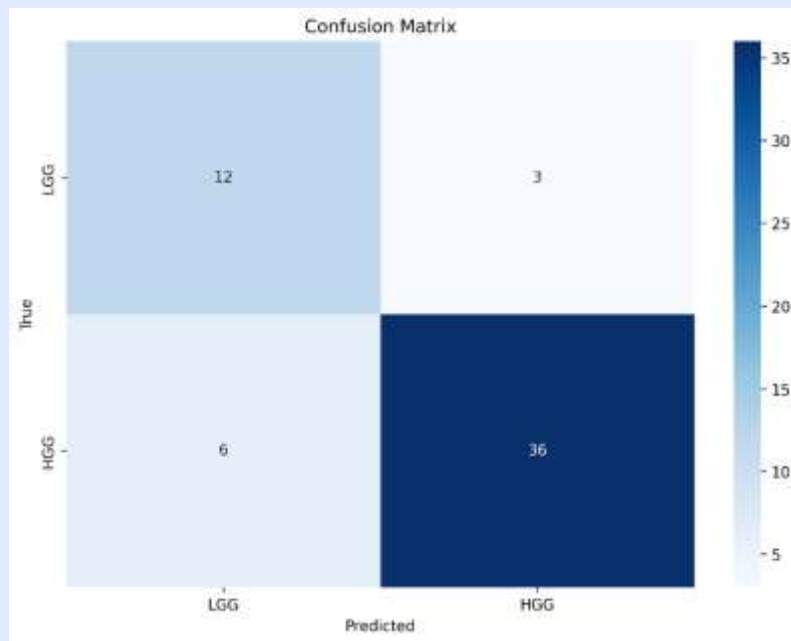


Figure 26: Confusion matrix of ResNet50-3D for Fold 1 at the best validation epoch (epoch 28), showing improved classification balance.

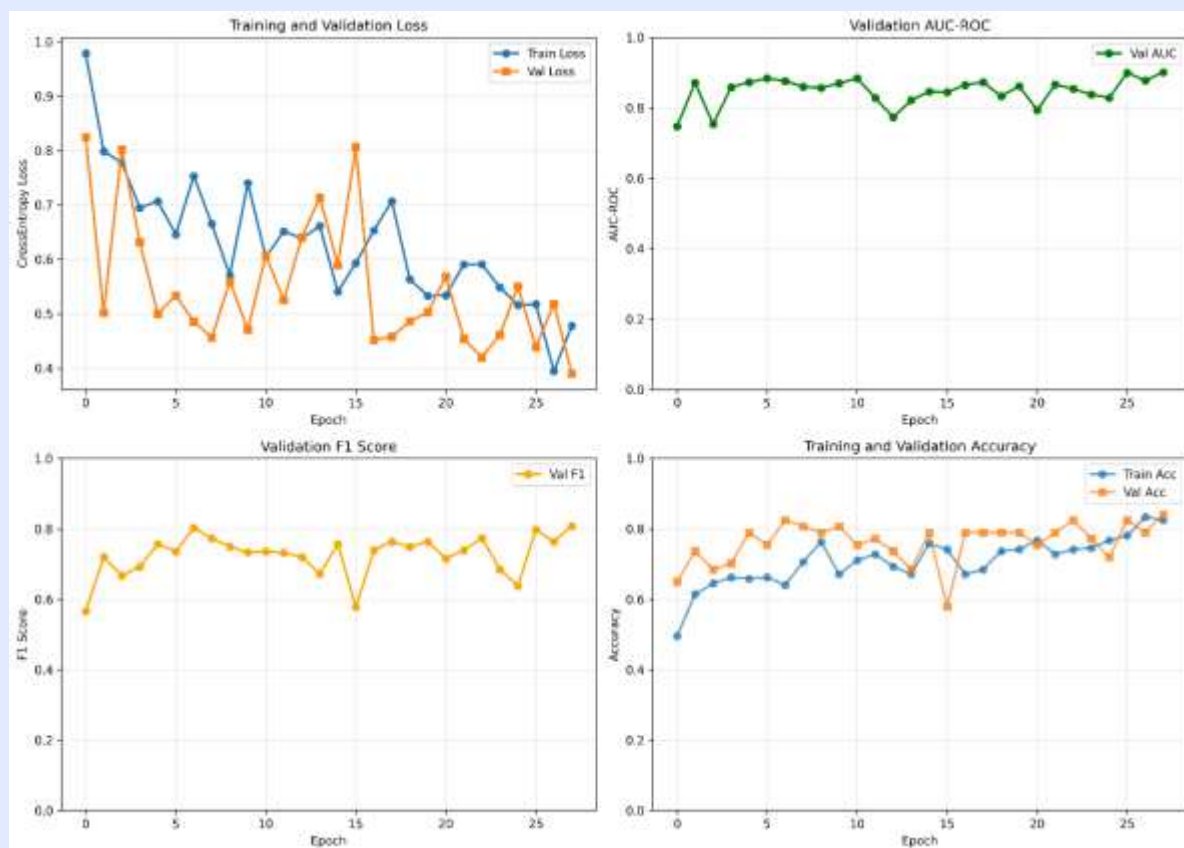


Figure 27: Learning curves of ResNet50-3D for Fold 1, indicating improved generalization compared to Fold 0.



Figure 28: Training and validation loss curves of ResNet50-3D for Fold 1, showing more stable optimization dynamics.

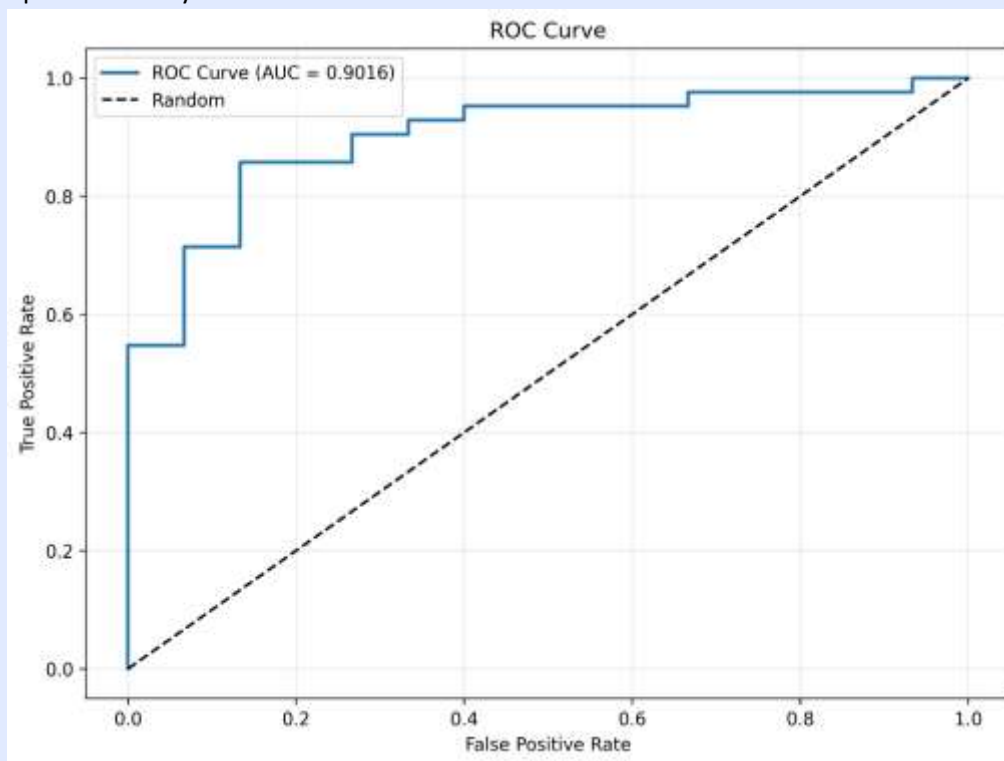


Figure 29: ROC curve of ResNet50-3D for Fold 1 at the best validation epoch (epoch 28), achieving an AUC of 0.9016.

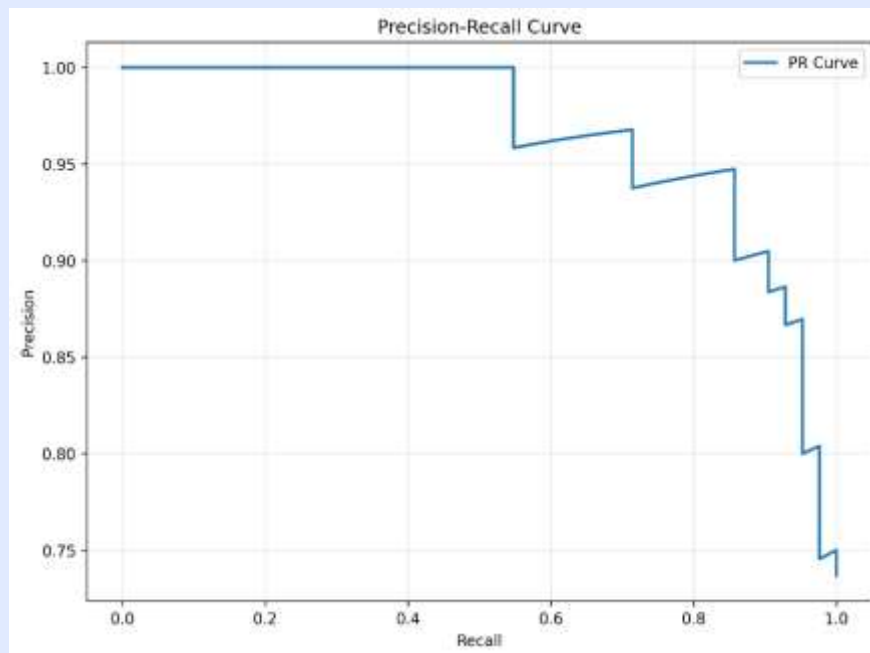


Figure 30: Precision-recall curve of ResNet50-3D for Fold 1, demonstrating improved precision-recall trade-off.

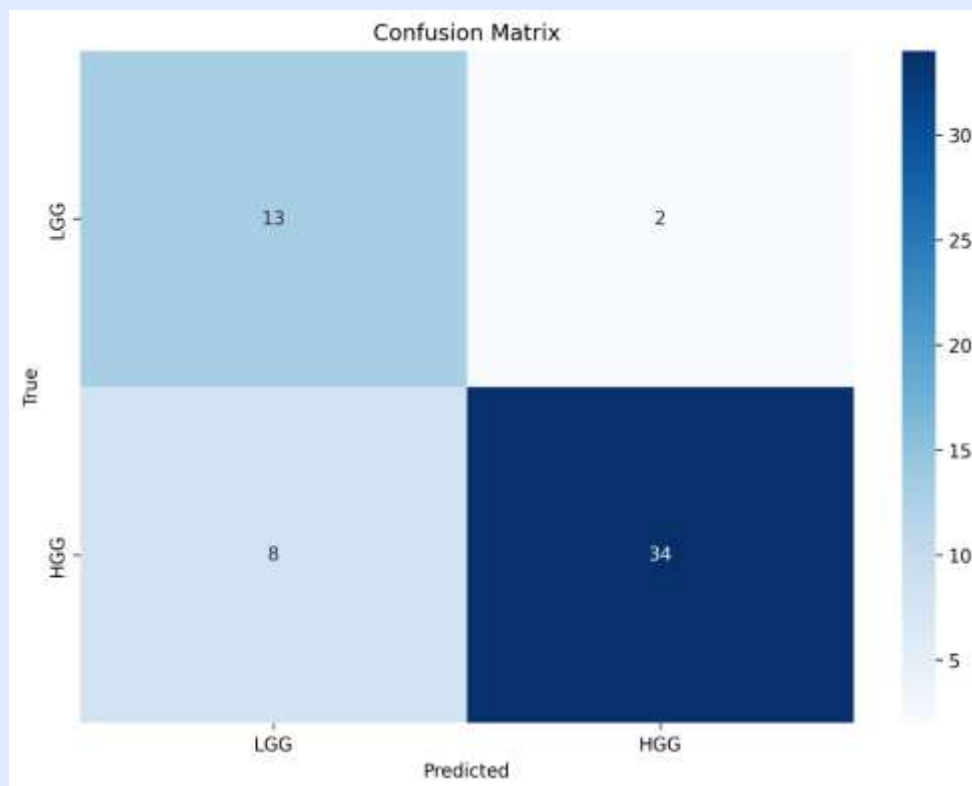


Figure 31: Confusion matrix of SwinUNETR-3D for Fold 0 at the best validation epoch (epoch 10), showing improved HGG sensitivity.

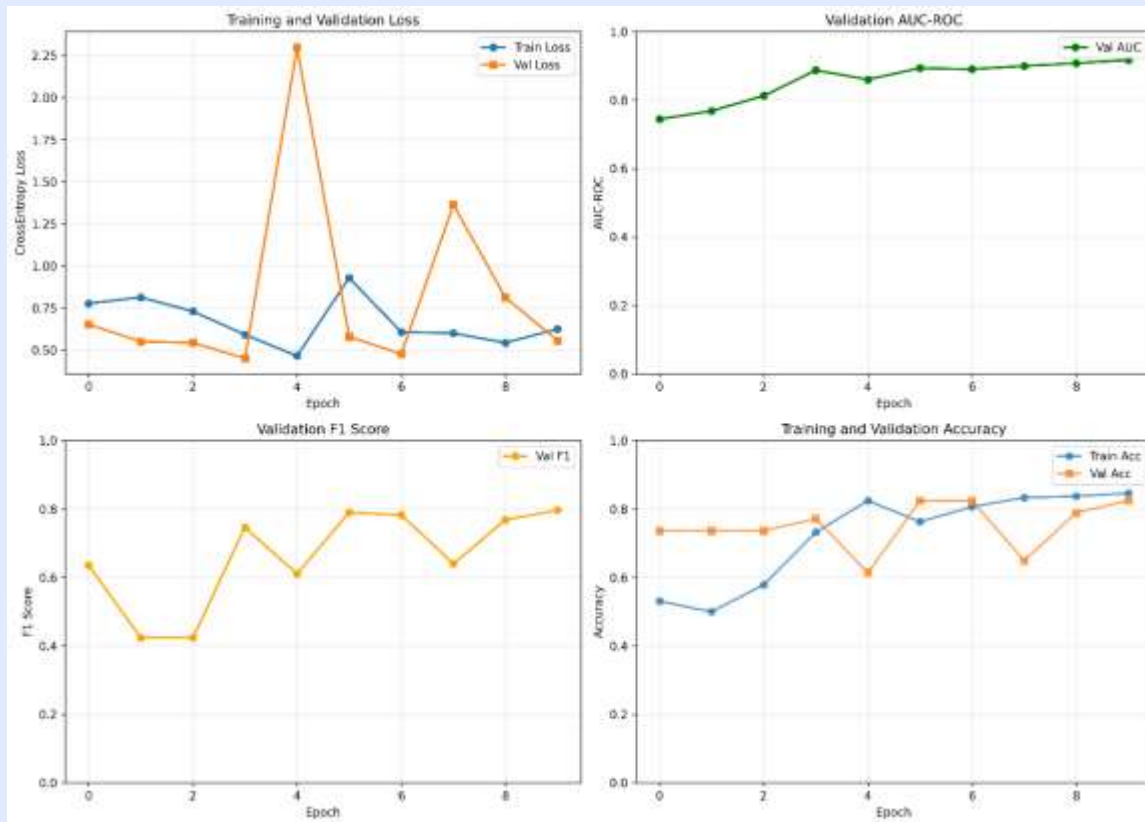


Figure 32: Learning curves of SwinUNETR-3D for Fold 0, illustrating stable training and validation performance.

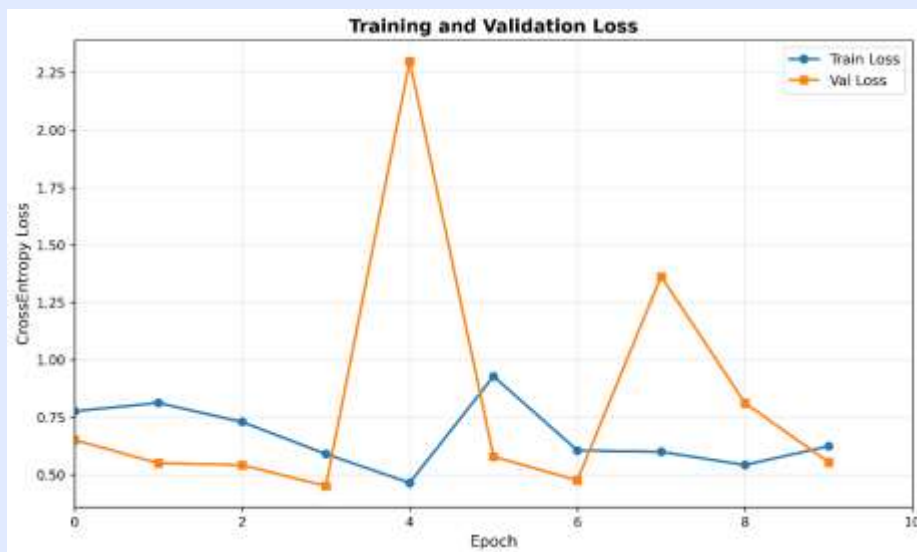


Figure 33: Training and validation loss curves of SwinUNETR-3D for Fold 0, indicating smooth convergence behavior.

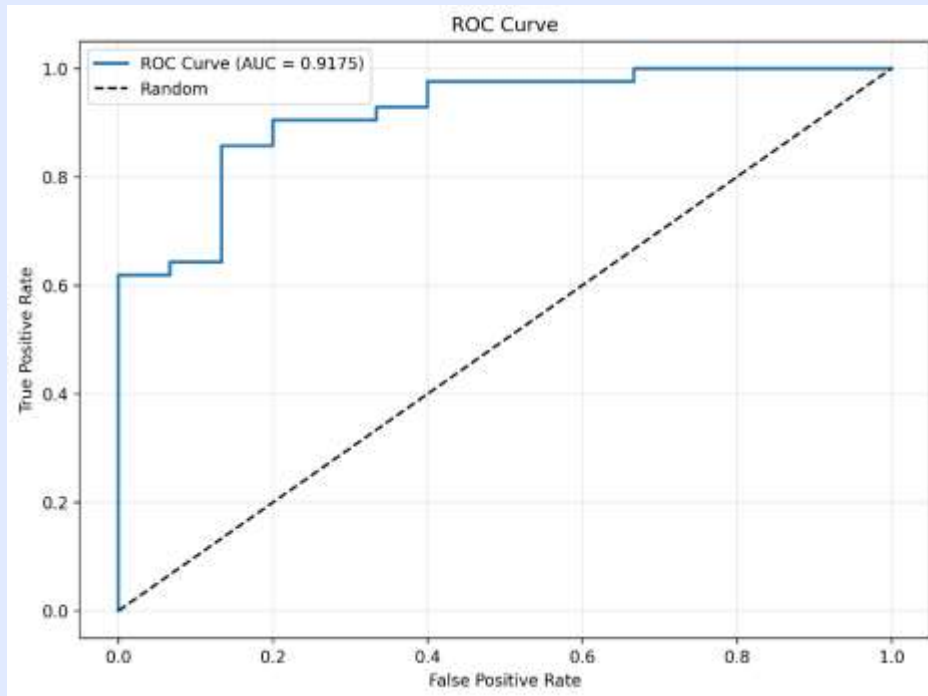


Figure 34: ROC curve of SwinUNETR-3D for Fold 0 at the best validation epoch (epoch 10), achieving an AUC of 0.9175

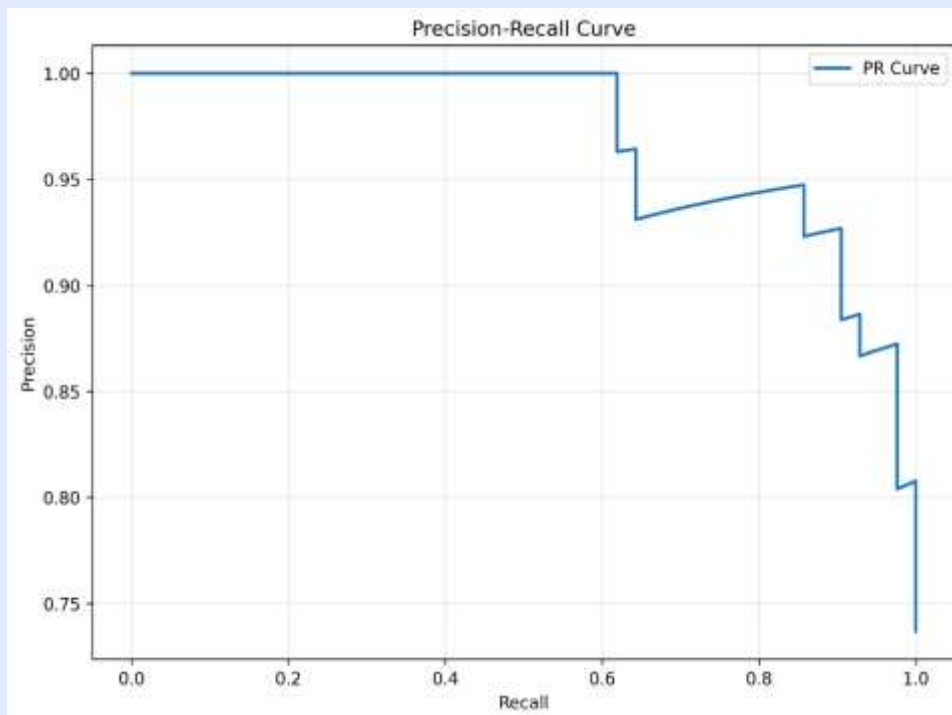


Figure 35: Precision-recall curve of SwinUNETR-3D for Fold 0, demonstrating robustness under class imbalance.

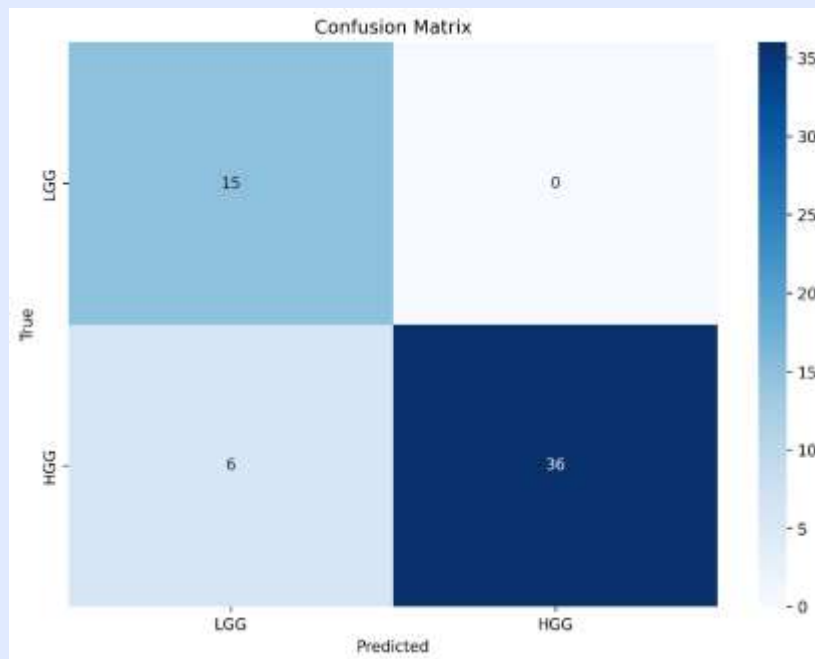


Figure 36: Confusion matrix of SwinUNETR-3D for Fold 1 at the best validation epoch (epoch 17), showing consistent classification performance.

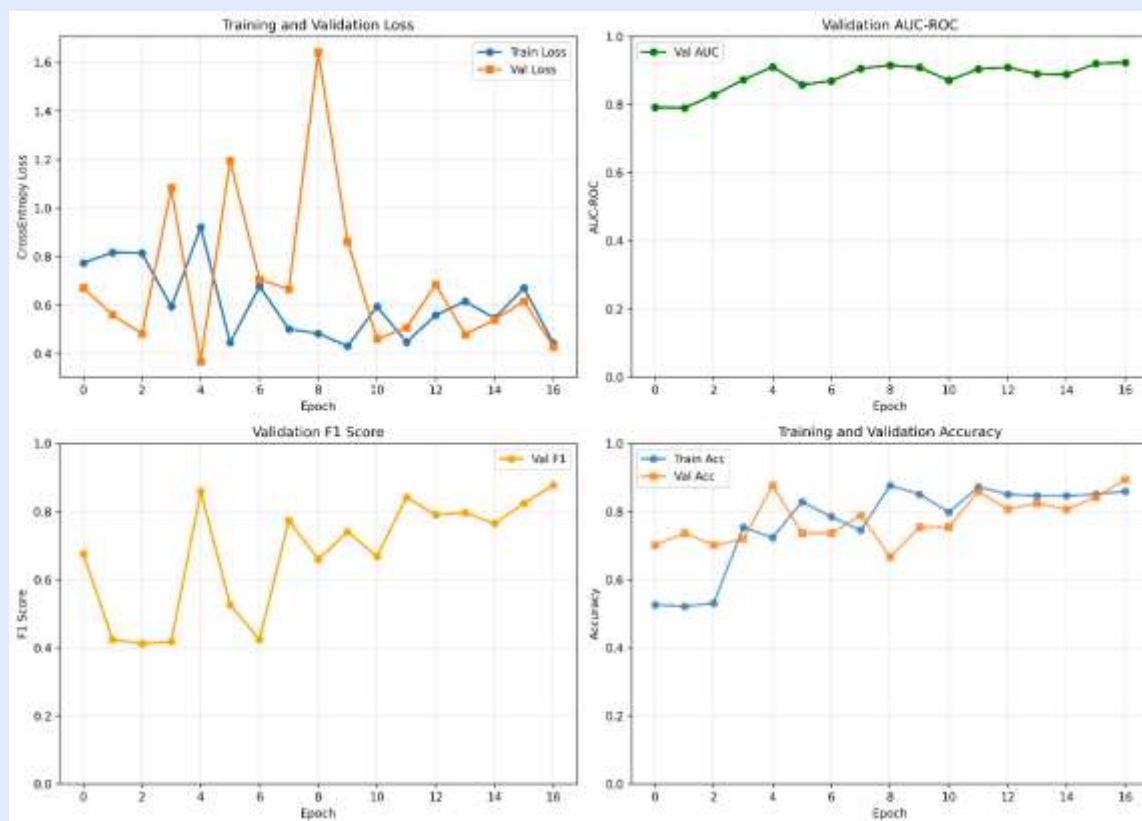


Figure 37: Learning curves of SwinUNETR-3D for Fold 1, indicating reduced overfitting and stable optimization

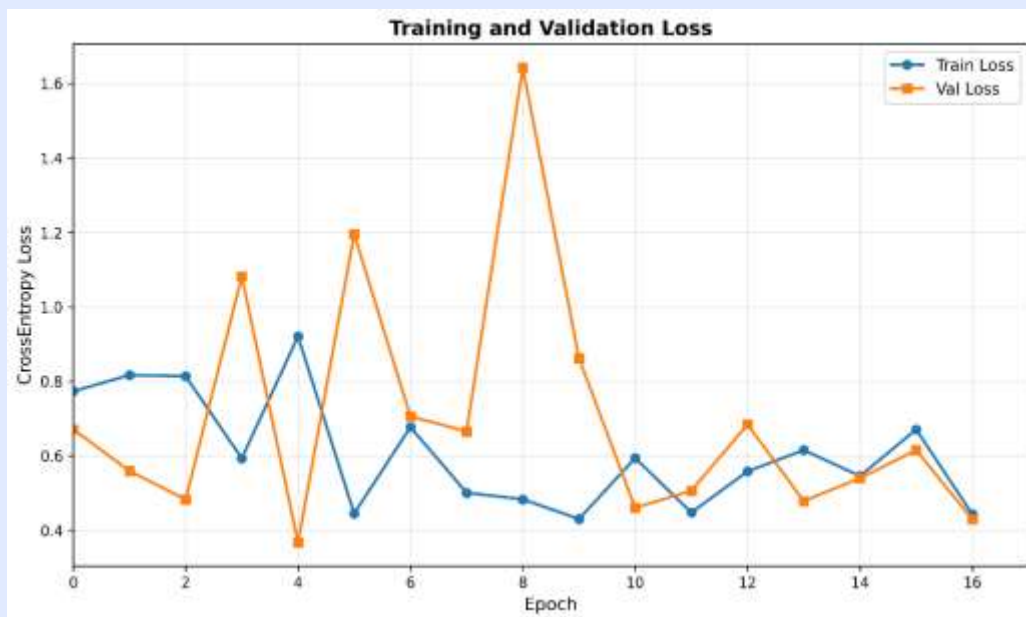


Figure 38: Training and validation loss curves of SwinUNETR-3D for Fold 1, reflecting consistent convergence.

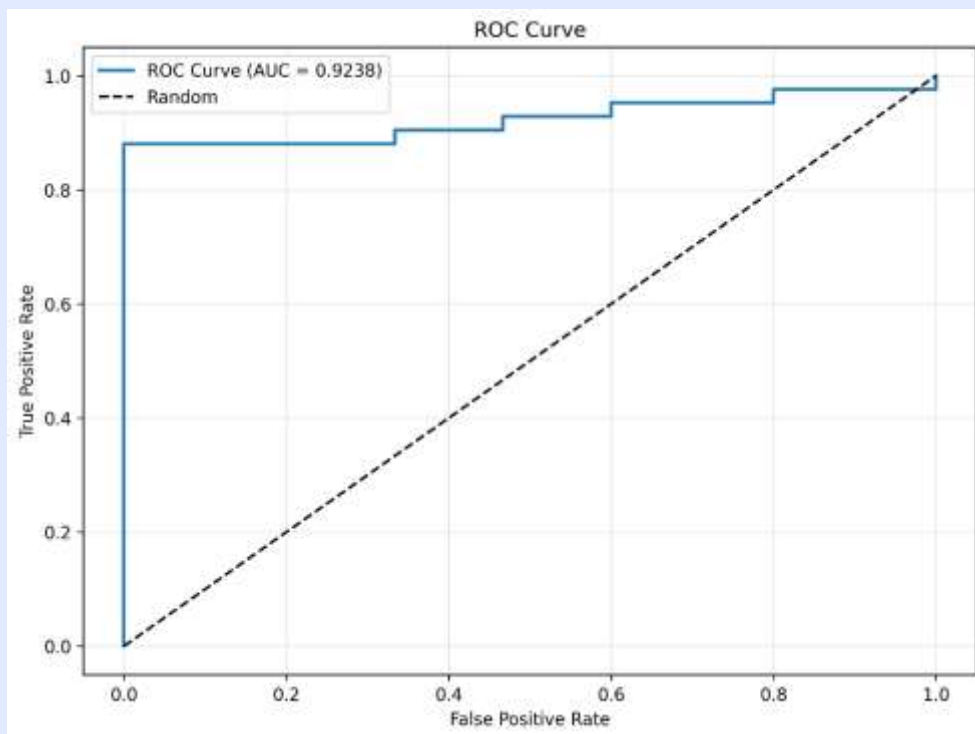


Figure 39: ROC curve of SwinUNETR-3D for Fold 1 at the best validation epoch (epoch 17), achieving an AUC of 0.9238.

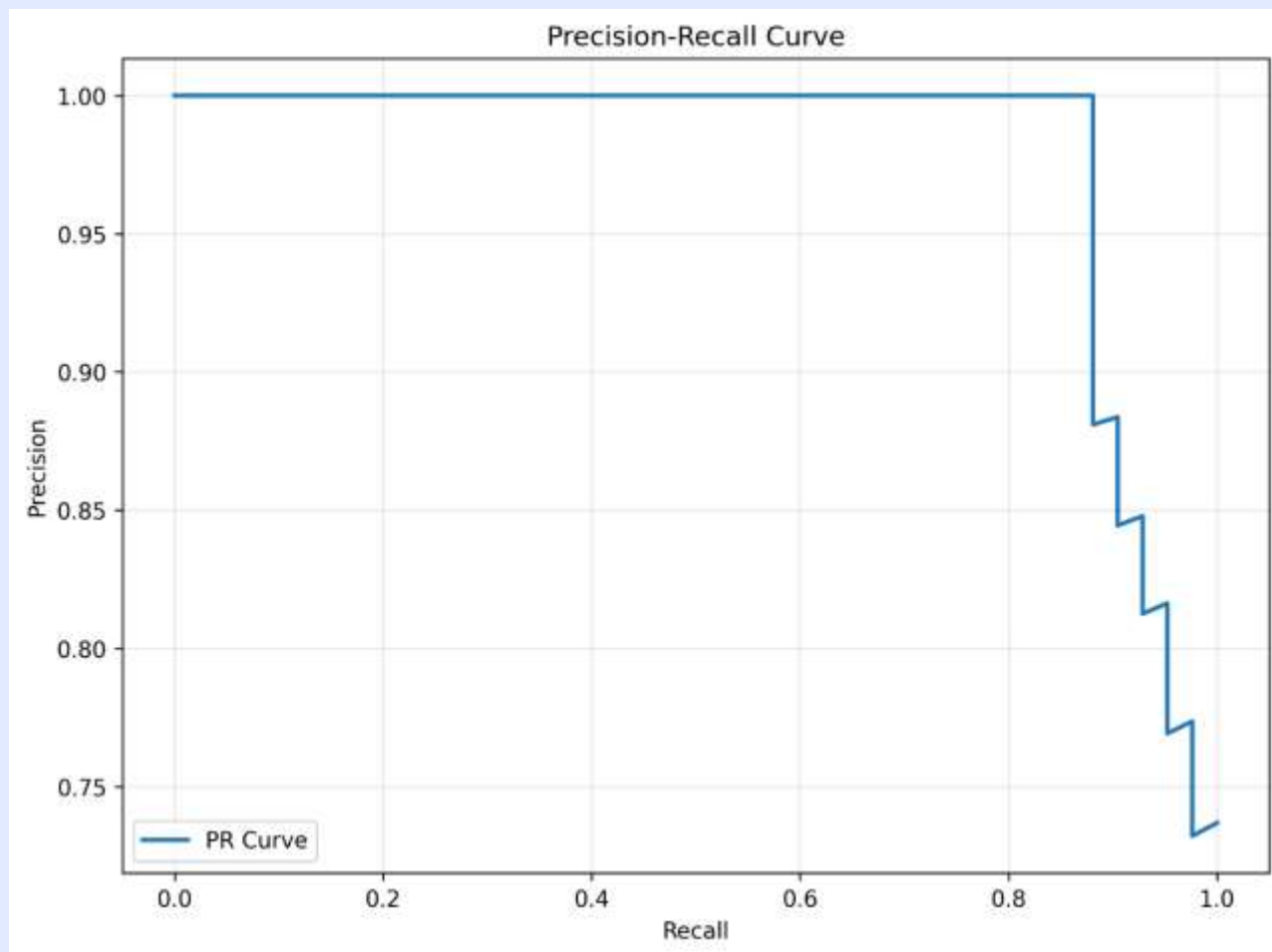


Figure 40: Precision-recall curve of SwinUNETR-3D for Fold 1, showing a favorable precision-recall trade-off.

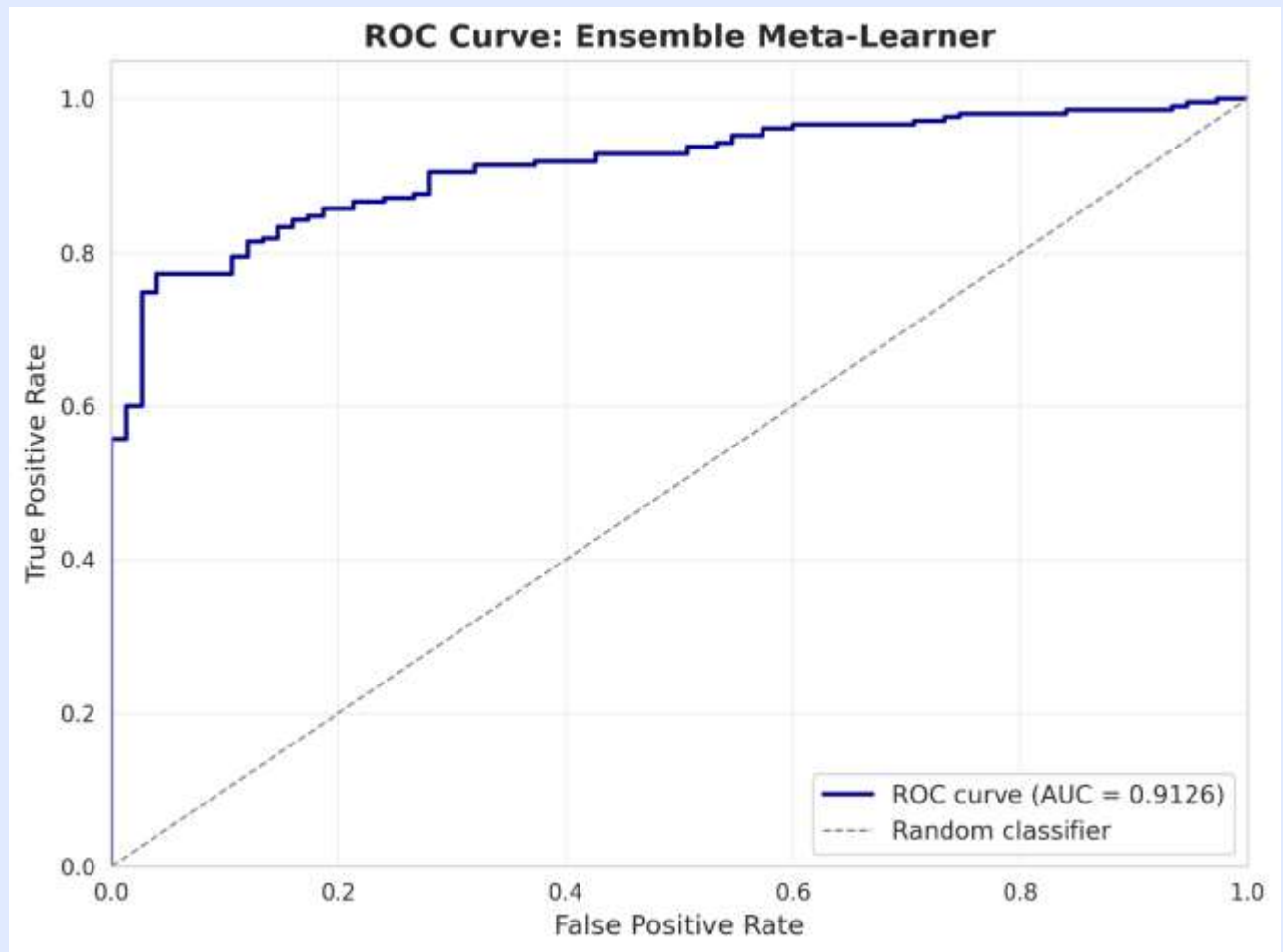


Figure 41: ROC curve of the ensemble model, illustrating the trade-off between true positive rate and false positive rate across classification thresholds.

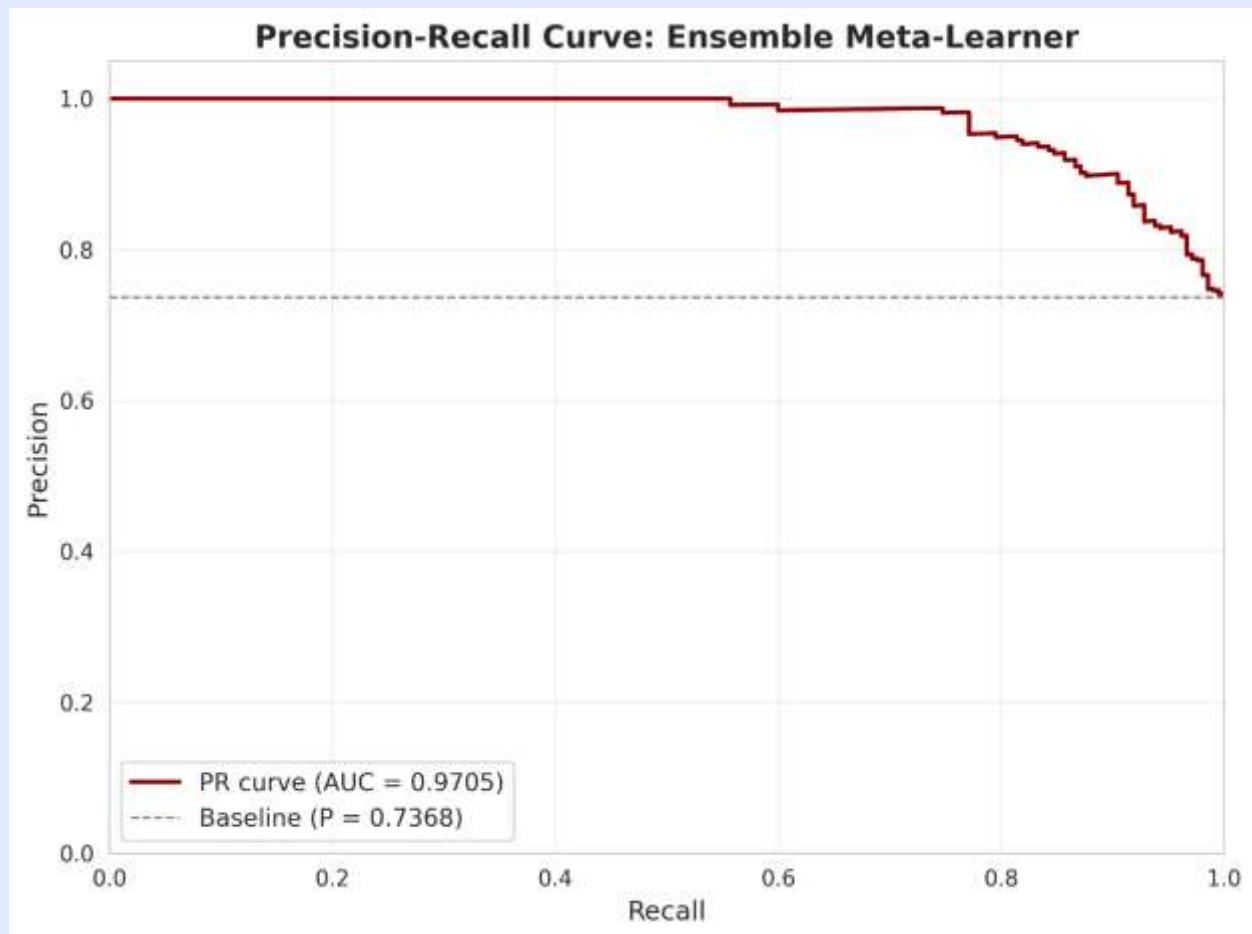


Figure 42: Precision-recall curve of the ensemble model, highlighting performance under class imbalance conditions.

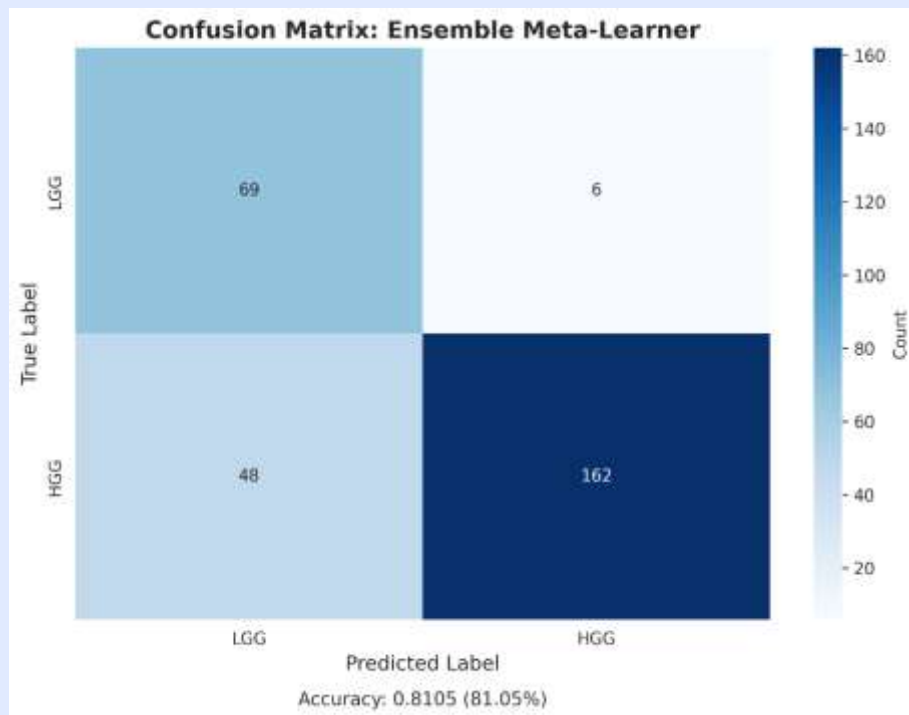


Figure 43: Confusion matrix of the ensemble model at the default decision threshold, summarizing class-wise prediction outcomes.

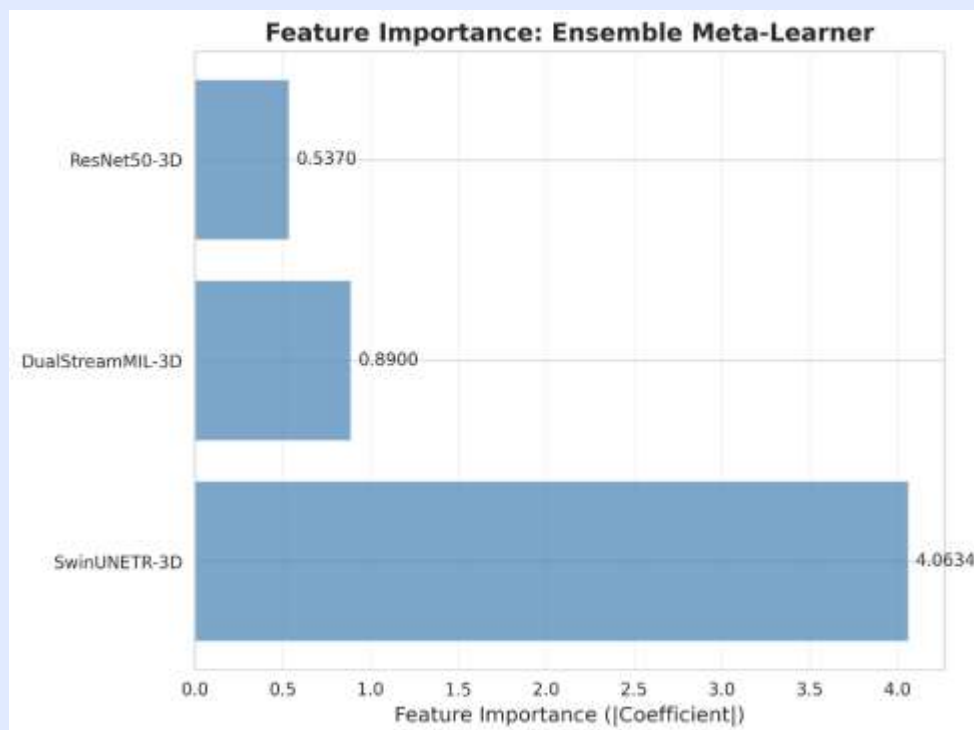


Figure 44: Feature importance plot of the ensemble meta-learner, showing the relative contribution of each base model's predicted probability.



Figure 45: Per-class performance comparison of the ensemble model, reporting precision, recall, and F1-score for LGG and HGG classes.

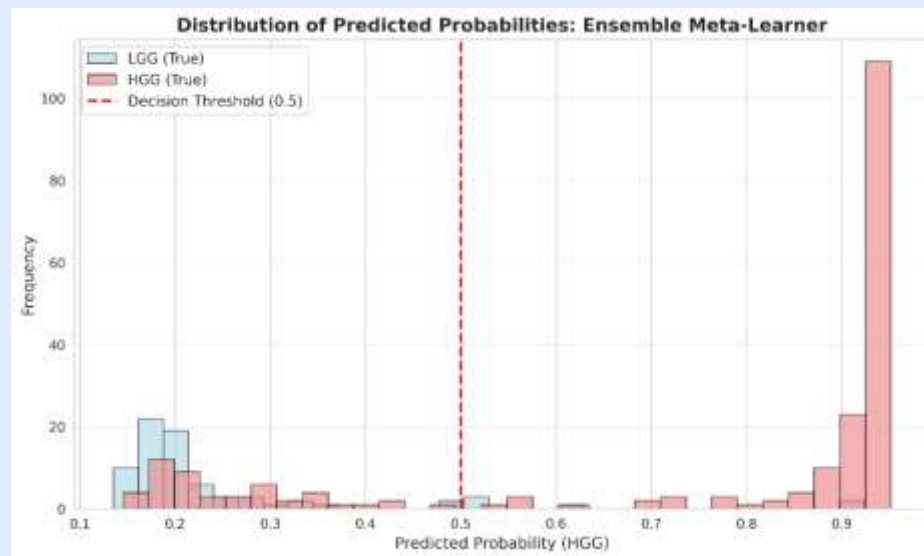


Figure 46: Distribution of ensemble-predicted probabilities across all patients, illustrating confidence and calibration characteristics.

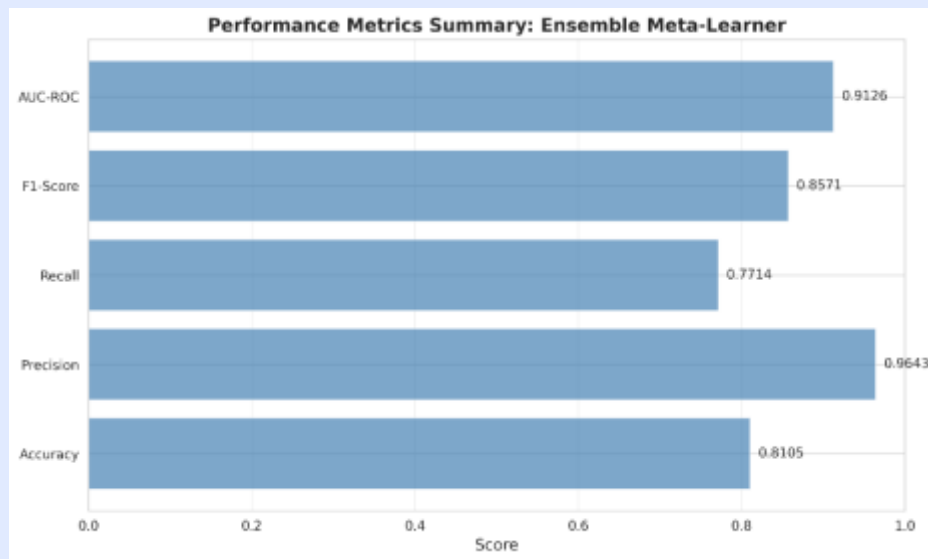


Figure 47: Summary visualization of ensemble performance metrics, providing an aggregated overview of evaluation results.

List of Abbreviations

Abbreviation	Full Term / Description
ADNI	Alzheimer's Disease Neuroimaging Initiative
AMP	Automatic Mixed Precision
AUC	Area Under the Curve
AUC-ROC	Area Under the Receiver Operating Characteristic Curve
BraTS	Brain Tumor Segmentation Challenge
CLAHE	Contrast Limited Adaptive Histogram Equalization
CNN	Convolutional Neural Network
CT	Computed Tomography
DSC	Dice Similarity Coefficient
DWT	Discrete Wavelet Transform
DualStreamMIL-3D	Dual-Stream Multiple Instance Learning (3D)
EMA	Exponential Moving Average
F1	F1-Score

FLAIR	Fluid-Attenuated Inversion Recovery
GA	Genetic Algorithm
Grad-CAM	Gradient-weighted Class Activation Mapping
HD95	95th Percentile Hausdorff Distance
HGG	High-Grade Glioma
IoU	Intersection over Union
IXI	Information eXtraction from Images
KNN	k-Nearest Neighbors
L2	L2 Regularization
LBFGS	Limited-memory Broyden–Fletcher–Goldfarb–Shanno
LDAM	Large Margin Loss
LGG	Low-Grade Glioma
MIL	Multiple Instance Learning
MLP	Multi-Layer Perceptron

MRI	Magnetic Resonance Imaging
N4	N4 Bias Field Correction
OASIS	Open Access Series of Imaging Studies
OOF	Out-of-Fold
PET	Positron Emission Tomography
PR	Precision–Recall
PR-AUC	Precision–Recall Area Under the Curve
ReLU	Rectified Linear Unit
ResNet50-3D	3D ResNet-50 Architecture
RF	Random Forest
RNN	Recurrent Neural Network
ROC	Receiver Operating Characteristic
ROI	Region of Interest
SGD	Stochastic Gradient Descent

SVM	Support Vector Machine
SwinUNETR-3D	Swin UNETR 3D Architecture
T1	T1-weighted MRI
T1CE	T1-weighted Contrast-Enhanced MRI
T2	T2-weighted MRI
ViT	Vision Transformer
WHO	World Health Organization
XAI	Explainable Artificial Intelligence
XGBoost	Extreme Gradient Boosting

LIST OF ABBREVIATIONS

جدول المحتويات

Supervisor Certification	3
Abstract.....	4
LIST OF TABLES	5
List of Equations	7
LIST OF FIGURES	8
List of Abbreviations	34
LIST OF ABBREVIATIONS	38
CHAPTER 1: INTRODUCTION.....	42
1.1 Introduction.....	43
1.2 Problem Statement.....	43
1.3 Project Objectives	44
1.4 Significance of the Study	44
1.5 Scope and Limitations	44
1.6 Report Organization	44
1.7 Chapter Summary	45
CHAPTER 2: FUNDAMENTAL CONCEPTS AND BACKGROUND.....	46
2.1 Introduction.....	47
2.2 Brain Anatomy and Medical Imaging.....	47
2.3 Brain Tumors Classification	47
2.4 MRI Modalities and Characteristics	47
2.5 Fundamentals of Deep Learning	48
2.5.1 Convolutional Neural Networks and 3D CNNs.....	48
2.5.2 Transformer-Based Architectures	48
2.5.3 Multiple Instance Learning (MIL)	48
2.5.4 Ensemble Learning and Stacking.....	48
2.6 Chapter Summary	49
CHAPTER 3: LITERATURE REVIEW.....	50
3.1 Chapter Introduction	51
3.2 Traditional Methods in Brain Tumor Classification	51
3.3 Deep Learning-Based Approaches.....	52
3.4 Multimodal Learning in Medical Imaging.....	52

3.5 Ensemble Learning Techniques in Medical Diagnosis	52
3.6 Detailed Literature Studies.....	53
Study1	54
Study2	56
Study3	58
Study4	61
Study5	64
Study6	68
Study7	72
Study8	77
Study9	81
Study10	85
3.7 Comparative Analysis of Previous Works	89
3.7.1 Modalities and Task Formulation	89
3.7.2 Data Representation and Network Dimensionality	89
3.7.3 Ensemble Strategies and Reported Performance	90
3.7.4 Strengths and Limitations Observed Across Studies	90
3.7.5 Gaps in the Existing Literature	91
3.7.6 Positioning of the Proposed Method.....	91
3.8 Chapter Summary (English Version).....	91
3.8.1 Overview	91
3.8.2 General Trends.....	91
3.8.3 Key Strengths.....	91
3.8.4 Limitations and Gaps	91
3.8.5 Contribution of the Current Project.....	92
3.8.6 Conclusion and Future Directions.....	92
CHAPTER 4: DATASET AND PREPROCESSING	93
4.1 Introduction.....	94
4.2 Dataset Description.....	94
4.2.1 Source and Task Definition.....	94
4.2.2 Modalities and Format	94
4.2.3 Dataset Organization.....	94
4.3 Data Exploration and Indexing	94

4.4 Preprocessing Pipeline	95
4.4.1 N4 Bias Field Correction	95
4.4.2 Z-Score Normalization	95
4.4.3 Region of Interest (ROI) Cropping	95
4.4.4 Resampling to Uniform Shape	95
4.5 Data Augmentation	95
4.6 Class Imbalance Handling (Runtime Oversampling)	95
4.7 Data Splitting Strategy	96
4.8 Chapter Summary	96
CHAPTER 5: PROPOSED METHODOLOGY	97
5.1 Introduction	98
5.2 System Architecture: An Integrated Ensemble Framework	98
5.2.1 Complementary Model Selection	99
5.2.2 Ensemble Integration Strategy	99
5.3 Deep Learning Architectures	100
5.3.1 ResNet50-3D: Volumetric Feature Extraction	100
5.3.2 SwinUNETR-3D: Transformer-Based Global Context Modeling	101
5.3.3 DualStreamMIL-3D: Interpretable Slice-Level Reasoning	102
5.4 Unified Training Strategy	104
5.4.1 Patient-Level Cross-Validation	104
5.4.2 Addressing Class Imbalance	105
5.4.3 Geometric Data Augmentation	105
5.4.4 Loss Functions and Optimization	106
5.4.5 Early Stopping and Model Selection	107
5.4.6 Computational Efficiency Techniques	107
5.5 Ensemble Framework: Learning to Combine Predictions	108
5.5.1 Out-of-Fold Prediction Collection	108
5.5.2 Meta-Learner Design and Training	109
5.5.3 Ensemble Inference	110
5.6 Evaluation Metrics	110
5.6.1 Classification Performance Metrics	110
5.6.2 Diagnostic Metrics	111
5.6.3 Metric Computation and Storage	112

5.7 Chapter Summary	112
CHAPTER 6: EXPERIMENTAL RESULTS AND DISCUSSION	114
6.1 Introduction.....	115
6.2 Individual Model Performance Analysis	115
6.2.1 DualStreamMIL-3D: Per-Fold Cross-Validation Results	115
6.2.2 ResNet50-3D and SwinUNETR-3D: Out-of-Fold Prediction Analysis	119
Interpretation.....	120
6.3 Ensemble Model Performance.....	122
6.3.1 Meta-Learner Training and Configuration.....	122
6.3.2 Learned Combination Weights	122
6.3.3 Ensemble Performance Metrics	123
6.4 Comparative Results and Analysis.....	125
6.4.1 Base Model Prediction Characteristics	125
6.4.2 Ensemble vs. Individual Models.....	126
6.5 Discussion.....	127
6.5.1 Achievement of Research Objectives	127
6.5.2 Key Findings.....	128
6.5.3 Comparison with Prior Work.....	129
6.5.4 Clinical Implications	130
6.5.5 Future Directions.....	130
6.6 Chapter Summary	131
References.....	133

CHAPTER 1: INTRODUCTION

1.1 Introduction

Brain tumors represent one of the most critical and complex neurological diseases due to their direct impact on vital brain functions and their potential to be life-threatening. Accurate diagnosis and grading of brain tumors are essential for determining appropriate treatment strategies and improving patient outcomes. Brain tumors are generally classified into low-grade and high-grade tumors, where low-grade tumors tend to grow slowly, while high-grade tumors are more aggressive and associated with poorer prognoses.

Magnetic Resonance Imaging (MRI) is the most commonly used medical imaging modality for brain tumor diagnosis, as it provides high-resolution images and excellent contrast between different soft tissues. MRI scans enable clinicians to visualize tumor structures, boundaries, and tissue characteristics in a non-invasive manner. However, manual interpretation of MRI images is a challenging and time-consuming task that relies heavily on the expertise of radiologists and may be subject to inter-observer variability.

Recent advances in artificial intelligence, particularly in machine learning and deep learning, have demonstrated significant potential in automating medical image analysis tasks. Deep learning models, especially convolutional neural networks (CNNs) and transformer-based architectures, have shown remarkable performance in extracting complex patterns and features from medical images. These approaches can support clinicians by providing accurate and consistent diagnostic assistance.

This graduation project focuses on developing a deep learning-based framework for the classification of brain tumor MRI images into low-grade and high-grade categories. The proposed approach utilizes **multimodal MRI data, incorporating four imaging sequences (T1-weighted, T1-weighted contrast-enhanced (T1CE), T2-weighted, and FLAIR)** to capture complementary tissue characteristics. Multiple deep learning models are employed to learn diverse feature representations from this multimodal data, followed by an ensemble stacking method to enhance overall classification performance.

1.2 Problem Statement

Despite the widespread use of MRI in brain tumor diagnosis, the accurate classification of tumor grade remains a challenging problem. Manual assessment of MRI scans is prone to subjectivity and may lead to inconsistencies, particularly in borderline cases. Moreover, the growing volume of medical imaging data places additional pressure on healthcare professionals.

Single deep learning models may suffer from limitations such as overfitting, sensitivity to data imbalance, or inability to capture diverse feature representations. Therefore, there is a need for an automated and robust classification system that can leverage multiple models and combine their strengths to achieve reliable tumor grading.

1.3 Project Objectives

The main objectives of this project are as follows:

- To develop an automated system for classifying brain tumor MRI images into low-grade and high-grade tumors.
 - To design and train three different deep learning models for brain tumor classification.
 - To apply an ensemble stacking method to combine the outputs of individual models and improve classification performance.
 - To evaluate the proposed framework using standard performance metrics.
 - To analyze and compare the performance of individual models and the ensemble approach.
-

1.4 Significance of the Study

This project contributes to the field of medical image analysis by exploring advanced deep learning techniques for brain tumor classification. The proposed framework aims to assist radiologists and medical professionals by providing an accurate and reliable decision-support tool. Automating the tumor grading process can reduce diagnostic workload, minimize human error, and enhance consistency in clinical decision-making.

Furthermore, the use of ensemble learning techniques improves model robustness and generalization, making the system more suitable for real-world medical applications.

1.5 Scope and Limitations

The scope of this project is limited to the classification of brain tumors into low-grade and high-grade categories using MRI images. The experiments are conducted on a publicly available brain tumor dataset (BraTS 2018), consisting of **285 patients (210 high-grade and 75 low-grade cases)**. The proposed framework focuses on image-based classification without incorporating clinical or demographic patient data.

Limitations of the study include dataset size constraints, class imbalance, and computational resource requirements for training deep learning models. Additionally, the proposed system is designed as a research prototype and is not intended for direct clinical deployment.

1.6 Report Organization

This report is organized into eight chapters. Chapter 1 introduces the project background, objectives, and scope. Chapter 2 presents the fundamental concepts and theoretical background

related to brain tumors, MRI imaging, and deep learning. Chapter 3 reviews related work and previous studies. Chapter 4 describes the dataset and preprocessing techniques. Chapter 5 explains the proposed methodology and model architectures. Chapter 6 discusses experimental results and performance analysis. Chapter 7 focuses on practical implementation details. Finally, Chapter 8 concludes the report and outlines future research directions.

1.7 Chapter Summary

This chapter presented an overview of the brain tumor classification problem and highlighted the importance of automated diagnostic approaches. The motivation, objectives, and scope of the project were discussed, providing a foundation for the subsequent chapters of this report.

CHAPTER 2: FUNDAMENTAL CONCEPTS AND BACKGROUND

2.1 Introduction

This chapter presents the fundamental concepts and theoretical background necessary to understand the proposed brain tumor classification framework. It provides an overview of brain anatomy, medical imaging techniques, and brain tumor classification. In addition, this chapter introduces the principles of magnetic resonance imaging (MRI) and the deep learning concepts that form the theoretical foundation of the proposed methodology.

2.2 Brain Anatomy and Medical Imaging

The human brain is a highly complex organ responsible for cognitive functions, motor activities, sensory perception, and vital physiological processes. It consists of major regions such as the cerebrum, cerebellum, and brainstem, each performing distinct functions. Abnormal cell growth within brain tissues can lead to brain tumors, potentially disrupting normal brain functions depending on their location and size.

Medical imaging plays a crucial role in diagnosing and monitoring brain diseases. Common imaging modalities include computed tomography (CT), positron emission tomography (PET), and magnetic resonance imaging (MRI). Among these, MRI is widely preferred for brain tumor analysis due to its superior soft-tissue contrast, high spatial resolution, and non-invasive nature.

2.3 Brain Tumors Classification

Brain tumors are abnormal masses of cells that grow uncontrollably within the brain. They are classified based on factors such as origin, growth rate, and malignancy. The World Health Organization (WHO) classification system categorizes brain tumors into different grades according to histological and molecular characteristics.

Clinically, tumors are commonly grouped into **low-grade tumors**, which are slow-growing and less aggressive, and **high-grade tumors**, which exhibit rapid growth and poor prognosis. Accurate tumor grading is critical for determining appropriate treatment strategies.

2.4 MRI Modalities and Characteristics

Magnetic Resonance Imaging (MRI) provides multiple imaging sequences, each emphasizing different tissue properties. This project utilizes four MRI modalities:

- **T1-weighted (T1)** images for detailed anatomical structures
- **T1-weighted contrast-enhanced (T1CE)** images to highlight tumor regions
- **T2-weighted (T2)** images sensitive to edema and fluid accumulation

- **FLAIR** images that suppress cerebrospinal fluid signals and emphasize abnormalities

The integration of these modalities enables comprehensive tumor characterization. Multimodal MRI data allows deep learning models to learn complementary features, improving classification performance.

2.5 Fundamentals of Deep Learning

Deep learning is a subset of artificial intelligence that employs multi-layer neural networks to automatically learn hierarchical feature representations from data. In medical imaging, deep learning has demonstrated superior performance by eliminating the need for manual feature engineering.

2.5.1 Convolutional Neural Networks and 3D CNNs

Convolutional Neural Networks (CNNs) are effective in image analysis due to their ability to extract spatial features. For volumetric medical data, **3D CNNs** extend convolution operations into three dimensions, enabling learning across entire MRI volumes. In this project, a **3D ResNet-based architecture** is used to capture volumetric spatial relationships within brain MRI scans.

2.5.2 Transformer-Based Architectures

Transformer models utilize self-attention mechanisms to model long-range dependencies. Vision Transformers and Swin Transformers have shown strong performance in medical imaging. The **Swin UNETR architecture** employed in this project combines transformer-based feature extraction with hierarchical representations suitable for 3D medical volumes.

2.5.3 Multiple Instance Learning (MIL)

Multiple Instance Learning (MIL) is a paradigm where labels are assigned at the bag level rather than individual instances. This is particularly useful for large 3D medical images. The **DualStreamMIL-3D** architecture processes 2D slices extracted from MRI volumes and aggregates informative instances using attention mechanisms to generate volume-level predictions.

2.5.4 Ensemble Learning and Stacking

Ensemble learning improves robustness by combining predictions from multiple models. This project applies **ensemble stacking**, where a **Logistic Regression meta-learner** is trained on out-of-fold predictions from the base models to optimally combine their outputs and enhance classification accuracy.

2.6 Chapter Summary

This chapter presented the theoretical foundations related to brain anatomy, medical imaging, brain tumor classification, and MRI modalities. It also detailed the deep learning concepts underpinning the proposed framework, including 3D CNNs, transformer-based architectures, multiple instance learning, and ensemble stacking. These concepts establish the foundation for the methodology described in subsequent chapters.

CHAPTER 3: LITERATURE REVIEW

3.1 Chapter Introduction

Previous studies play a critical role in framing any scientific research, as they enable researchers to understand the background of the problem, identify existing research gaps, and build upon prior achievements in a structured manner. In the domain of brain tumor classification and diagnosis using Magnetic Resonance Imaging (MRI), recent years have witnessed remarkable advancements—especially with the integration of deep learning techniques and hybrid architectures—leading to promising and highly accurate diagnostic models.

Modern trends highlight a shift from conventional methods to advanced deep neural networks, particularly Convolutional Neural Networks (CNNs) and Transformers. There is also a growing emphasis on leveraging multimodal MRI inputs and ensemble learning techniques to enhance predictive accuracy and model robustness.

Accordingly, the purpose of this chapter is to review and analyze the most relevant and recent studies on brain tumor classification and segmentation. The focus will be on model architectures, data modalities, enhancement strategies, and learning approaches, with the goal of extracting insights that inform and guide the development of a more efficient and reliable model in the current research.

3.2 Traditional Methods in Brain Tumor Classification

Before the widespread adoption of deep learning techniques, brain tumor classification primarily relied on traditional machine learning algorithms such as Support Vector Machines (SVM), k-Nearest Neighbors (KNN), decision trees (e.g., Random Forest), and logistic regression. These methods typically depend on handcrafted feature extraction from MRI images—features like intensity, shape, texture, and edges—which are then fed into the classifiers.

These conventional approaches are known for their simplicity, speed, and effectiveness when working with small datasets. Certain classifiers, like SVM, can yield reasonably good performance, particularly in binary classification tasks. However, their performance often declines when applied to complex or multimodal datasets due to limited capacity in modeling non-linear and high-dimensional relationships in medical images.

Compared to deep neural networks, traditional methods lack hierarchical feature learning capabilities, making them less effective in handling the high variability within tumor images. Additionally, their heavy reliance on manually extracted features has motivated a shift toward more flexible and adaptive models such as convolutional neural networks (CNNs) and modern Transformer-based architectures.

3.3 Deep Learning-Based Approaches

The classification and diagnosis of brain tumors have significantly advanced with the advent of deep learning techniques, particularly Convolutional Neural Networks (CNNs). These models automatically extract multi-level features from MRI scans, eliminating the need for handcrafted features and greatly enhancing classification accuracy.

Various architectures have been employed, including Recurrent Neural Networks (RNNs), Autoencoders, and Transformer-based models leveraging self-attention mechanisms. Recently, 3D models such as ResNet50-3D, SwinUNETR-3D (Transformer-based), and DualStreamMIL-3D (Multiple Instance Learning-based) have shown strong performance in processing volumetric MRI data and capturing spatial context more effectively.

The evolution of deep architectures—such as ResNet, EfficientNet, and Vision Transformers—has improved representational power while maintaining efficiency. Hybrid models that integrate multiple architectures are increasingly used to achieve superior results in both classification and segmentation tasks in medical imaging.

3.4 Multimodal Learning in Medical Imaging

Multimodal learning refers to leveraging multiple types of MRI sequences to provide complementary information that enhances model performance in brain tumor analysis. Commonly used MRI modalities include T1, contrast-enhanced T1 (T1c), T2, and FLAIR. Each modality offers unique insights into tumor characteristics, location, and behavior.

Integrating these modalities provides a more holistic understanding of the tumor's structure, perfusion, and tissue context. For example, FLAIR images are effective in highlighting edema, while T1c sequences emphasize active tumor boundaries. Combining such complementary inputs allows models to better capture the full extent of pathology.

To handle multimodal inputs, specialized models have been developed—such as multi-branch U-Nets or multi-head Transformer networks—where each modality is processed separately or as additional input channels. This approach has become particularly prevalent in 3D segmentation tasks, where fusing multi-modal features significantly improves robustness and precision in clinical scenarios.

3.5 Ensemble Learning Techniques in Medical Diagnosis

Ensemble learning techniques are highly effective in improving the performance of models in medical classification and segmentation tasks. These methods combine multiple predictive models to reduce bias and variance while enhancing generalization. Common ensemble strategies include:

- **Bagging:** Combines models trained on different data subsets (e.g., Random Forest).
- **Boosting:** Sequentially improves predictions by focusing on prior errors (e.g., XGBoost).
- **Feature Fusion:** Combines feature vectors from multiple models before classification.

- **Stacking:** Trains a meta-learner to aggregate predictions from base models.
- **Multi-Stage:** Cascades models in stages to refine outputs iteratively.

In the analyzed studies, ensemble strategies were implemented in various impactful ways:

- **Study (6):** Proposed a dual-modality ensemble using outputs from U-Nets trained on different MRI sequences, achieving superior 3D segmentation accuracy.
- **Study (2):** Combined CNN and Vision Transformer outputs into a unified ensemble model for brain tumor classification.
- **Study (7):** Employed feature-level ensemble by fusing CNN and autoencoder features, then selecting the best-performing classifier via evolutionary optimization.
- **Study (10):** Integrated 2D CNN features with explicit 3D radiomic features (e.g., tumor size) for enhanced glioma grading.

These studies confirm that ensembling models or multimodal data improves prediction accuracy, robustness, and clinical reliability. Intelligent fusion strategies not only boost performance but also enhance interpretability—critical in medical applications where decision transparency is essential.

3.6 Detailed Literature Studies

This section presents a detailed review of the most relevant studies in brain tumor classification

Study1

NeXtBrain: Combining Local and Global Feature Learning for Brain Tumor Classification (2025)

Study Title	Year	Journal / Venue	Task Type	MRI Modalities	# Modalities	Data Rep. (2D/3D)	Models Used	Ensemble?	Results (Key Metrics)	Link
NeXtBrain: Combining Local and Global Feature Learning for Brain Tumor Classification (Pacal et al.)	2025	Brain Research (Elsevier)	Classification (Glioma / Meningioma / Pituitary tumor)	T1-weighted CE MRI	1	2D slices	Hybrid CNN-Transformer (custom NeXt Conv. Block + NeXt Transformer Block)	No	Accuracy ~99.78%, F1 ~99.8% on two datasets (Figshare & Kaggle)	[1]

Summary: Pacal *et al.* introduce **NeXtBrain**, a novel hybrid model designed to improve brain tumor **classification** by capturing both fine-grained local features and global context from MRI scans. The study addresses the challenge of achieving high accuracy and generalization without excessive computational cost in tumor classification. NeXtBrain was evaluated on two public MRI datasets (Figshare and Kaggle) and outperformed 17 state-of-the-art models in accuracy and efficiency. It achieved nearly **99.8% accuracy and F1-score** on both datasets, indicating robust performance.

Methodology & Approach: The NeXtBrain framework combines convolutional and transformer-based techniques. Its architecture includes two custom blocks: The **NeXt Convolutional Block (NCB)** for local feature extraction and the **NeXt Transformer Block (NTB)** for global context modeling. The NCB uses multi-head convolutional attention and a SwiGLU-MLP to capture subtle tumor textures, while the NTB integrates self-attention with convolutional attention to capture long-range spatial relationships. By merging local and global feature learning, the model can differentiate complex tumor characteristics at multiple scales.

Techniques (Model & Methods): NeXtBrain’s hybrid model architecture leverages a **CNN-Transformer fusion**. The NCB (a specialized CNN block) focuses on fine details of the tumor, and the NTB (a transformer-based block) focuses on broader context. Importantly, NeXtBrain is optimized for efficiency, using only ~23.9 million parameters and ~10.3 GFLOPs, which is significantly smaller and faster than typical Vision Transformers (e.g. DeiT3 or Swin). This efficiency is achieved without sacrificing accuracy, thanks to the careful design of attention mechanisms in both the convolutional and transformer components.

Important Notes: NeXtBrain’s evaluation on the Figshare dataset (3 tumor classes) and a Kaggle MRI dataset (which included a “no tumor” category) showed extraordinary performance ($\approx 99.8\%$ accuracy/F1). The model outperformed larger baseline models while having only $\sim 28\%$ of their parameters. *It should be noted that the Kaggle dataset used here is a four-class dataset (glioma, meningioma, pituitary, **and** healthy images), so the task includes distinguishing tumor vs. no-tumor images as well.* Despite this challenge, NeXtBrain maintained its high accuracy. Additionally, no ensemble was used – the results reflect a single model’s output, underscoring NeXtBrain’s efficacy.

Dataset: Two datasets were used: **(1) Figshare brain tumor dataset** containing 3064 contrast-enhanced T1-weighted MRI slices of three tumor types (glioma, meningioma, pituitary) and **(2) a Kaggle brain MRI dataset** of 7023 images (an augmented version of the Figshare data combined with “no tumor” images for a four-class classification). All images are **T1-weighted post-contrast MRI** slices (single-modality). The images were used for training and testing in a standard classification pipeline. The authors performed rigorous comparisons against 17 other models on these datasets to benchmark performance.

Results: NeXtBrain achieved **99.78% accuracy and 99.77% F1-score** on the Figshare dataset, and **99.78% accuracy with 99.81% F1** on the Kaggle dataset. These metrics are nearly perfect, indicating only a handful of misclassifications out of thousands of images. It surpassed state-of-the-art CNNs, Vision Transformers, and hybrid models tested under the same conditions. Notably, NeXtBrain maintained this performance while being computationally lightweight – it requires <0.01 milliseconds per inference and far fewer parameters than comparable models. The study did not report AUC explicitly (likely because accuracy/F1 were already extremely high), and given the near-perfect performance, the AUC would effectively be ~ 1.0 for both datasets. Overall, NeXtBrain sets a new benchmark in tumor image classification accuracy while also offering advantages in speed and model size.

Study2

Ensemble Deep Learning Approach for Brain Tumor Classification Using Vision Transformer and CNN (Oztel et al.)(2025) .

Study Title	Year	Journal / Venue	Task Type	MRI Modalities	# Modalities	Data Rep. (2D/3D)	Models Used	Ensemble?	Results (Key Metrics)	Link
Ensemble Deep Learning Approach for Brain Tumor Classification Using Vision Transformer and CNN (Oztel et al.)	2025	Advanced Intelligent Systems (Wiley)	Classification	T1-weighted MRI (with wavelet-transformed variants)	1	2D slices	MobileNet-v3, ResNeXt, DenseNet-201, ViT (transfer learning on original & wavelet data)	Yes (bagging ensemble of 4 models)	Accuracy 85.03% (bagging ensemble)	[2]

Summary: Oztel *et al.* propose an **ensemble deep learning approach** that combines Convolutional Neural Networks (CNNs) and Vision Transformers for brain tumor **classification**. The study explores the benefit of using both CNN and ViT architectures on MRI images processed with various transformations. Four pre-trained models (MobileNet-v3, ResNeXt, DenseNet-201, and a Vision Transformer) were fine-tuned on brain MRI data, and then an ensemble was constructed to improve performance. The best ensemble (using a bagging strategy) achieved **85.03% accuracy** in classifying brain tumors. This demonstrates a modest improvement over single models, indicating that ensembling can boost classification performance for this task.

Methodology & Approach: The approach first applies **wavelet transforms** to the MRI images to generate additional datasets (e.g., using Daubechies wavelet). In total, four sets of images were prepared: the original MRI images and three transformed versions (using different wavelet filters). Each of these sets was used to fine-tune multiple deep models via transfer learning. The authors then compared the performance of each model on each dataset. In the **first stage**, MobileNet-v3 on original data gave the highest single-model accuracy (~83.5%). In the **second stage**, the two best-performing individual models (including one trained on wavelet-transformed data) were combined in an ensemble. The ensemble employed a **bagging** strategy where predictions from multiple top models (MobileNet-v3, Vision Transformer, ResNeXt, DenseNet-201) were aggregated. This ensemble yielded the highest accuracy of ~85%. The key idea is that different models and input transformations capture complementary features, and combining them reduces generalization error.

Techniques (Model & Methods): Several **pre-trained CNNs** (MobileNet-v3, ResNeXt-50, DenseNet-201) and a **Vision Transformer (ViT)** were fine-tuned. Transfer learning allowed leveraging ImageNet weights to compensate for the limited size of medical data. The MRI images were augmented via wavelet-based preprocessing: e.g., Daubechies wavelet filter enhanced certain features and yielded slightly better model accuracy (second-best performance

~81.7%). For ensembling, a **bagging ensemble** was used, meaning each of the top models (trained on different inputs) cast a “vote” and the final prediction was based on their combined outputs. This approach aims to stabilize and improve predictions by averaging the strengths of diverse models. The ensemble in this study comprised four networks: MobileNet-v3, a ViT model, ResNeXt, and DenseNet-201. Notably, this is a heterogeneous ensemble combining both CNN and transformer architectures.

Important Notes: The achieved accuracy (85.03%) is a moderate improvement over single-model results (~83.5%). While this accuracy is lower than other studies in this review (likely due to differences in dataset or task difficulty), it’s important to note that this study did not use the same datasets as those achieving ~99% accuracy. Oztel *et al.* appear to use a different MRI dataset or possibly a more challenging classification (the exact dataset is not explicitly named in the summary, but presumably a subset of Kaggle or a local dataset). The relatively lower accuracy suggests the task or data might involve more classes or more variability. The ensemble’s improvement ($\approx 1.5\%$ absolute) demonstrates that combining models can help, though the overall performance indicates room for further optimization. **No explicit metrics besides accuracy** are provided in the abstract; however, it’s implied that accuracy was the primary metric. The study focuses on demonstrating the ensemble approach rather than achieving state-of-the-art accuracy, and it emphasizes the process of integrating wavelet feature extraction with deep learning.

Dataset: The dataset is not named in the excerpt, but it likely involves MRI images categorized into tumor types (possibly the same three classes: glioma, meningioma, pituitary). The mention of “original data” and “Daubechies data” suggests they used a dataset of brain tumor MRI images (potentially the Figshare dataset or a similar one) and created transformed versions of it. The total number of images or patients isn’t given in the abstract, but training was done with a typical train-test split on this dataset. By using wavelet filters (Daubechies, Haar, Gabor, etc.), they effectively quadrupled the training data in different feature domains (though each set was trained separately). All inputs are **2D MRI slices** (grayscale) of size presumably 224×224 (due to EfficientNet and MobileNet input requirements). Data augmentation and transfer learning were used to mitigate overfitting given the limited dataset size.

Results: In the first analysis, MobileNet-v3 on original MRI data achieved **83.50% accuracy**, and the best wavelet-preprocessed scenario (ViT on Daubechies-transformed images) achieved **81.72%**. Other models ranged around 81% accuracy. The **ensemble** of the top models improved the accuracy to **85.03%**. This is the highest reported result in the study. The improvement, while modest, indicates the ensemble could better generalize than any single model. The authors did not provide other metrics like precision or F1 in the abstract, but given the multi-class nature, one can infer the ensemble’s macro-F1 might be slightly lower than accuracy (if class imbalance exists). No AUC or Dice (segmentation not applicable here) is reported. In summary, Oztel *et al.* demonstrate an ensemble method achieving ~85% accuracy, suggesting that hybrid approaches (CNN + ViT) can be beneficial, but also highlighting that brain tumor classification accuracy can vary widely depending on data and approaches (contrast this 85% with other works achieving >95% on different datasets). This underscores the importance of dataset characteristics and the need for further improvements in challenging settings.

Study3

AG-MS3D-CNN: Multiscale Attention-Guided 3D CNN for Robust Brain Tumor Segmentation (2025)

Study Title	Year	Journal / Venue	Task Type	MRI Modalities	# Modalities	Data Rep. (2D/3D)	Models Used	Ensemble?	Results (Key Metrics)	Link
AG-MS3D-CNN: Multiscale Attention-Guided 3D CNN for Robust Brain Tumor Segmentation across MRI Protocols (Lilhore <i>et al.</i>)	2025	<i>Scientific Reports</i> (Nature)	Segmentation	Multimodal MRI (T1, T1CE, T2, FLAIR)	4	3D volumes	AG-MS3D-CNN (3D U-Net variant with multiscale feature extraction, spatial attention, Monte Carlo dropout)	No	Dice: Whole Tumor 0.91, Core 0.87, Enhancing 0.84 (outperforms prior SOTA); robust across BraTS 2021 & external sets	[3]

Summary: Lilhore *et al.* developed **AG-MS3D-CNN**, a 3D deep learning model for brain tumor **segmentation** that emphasizes multiscale feature learning and attention mechanisms to achieve robust performance across different MRI protocols. The key motivation is to improve tumor boundary delineation and generalization to diverse data (different scanners and settings) – a common challenge in segmentation. The model also incorporates uncertainty estimation (via Monte Carlo dropout) and multitask learning (simultaneously performing segmentation, tumor classification, and volume estimation). When evaluated on the BraTS 2021 benchmark and external datasets (OASIS, ADNI, IXI), AG-MS3D-CNN achieved **high Dice scores**: approximately 0.91 for whole tumor, 0.87 for tumor core, and 0.84 for enhancing tumor, outperforming state-of-the-art methods in accuracy and consistency.

Methodology & Approach: The **AG-MS3D-CNN** architecture is built upon a 3D U-Net backbone enhanced with **attention-gating and multiscale feature extraction**. “Multiscale” means the model extracts features at multiple resolution levels to capture both fine details and global context. A **spatial attention mechanism** is integrated to focus on tumor regions, improving boundary clarity (the attention gates help the model learn *where* to refine the segmentation). Additionally, the model uses **Monte Carlo dropout** during inference to estimate uncertainty, providing confidence maps for the segmentation results. Unique to this work, the authors include a **domain adaptation module** to ensure the model generalizes across different MRI protocols (differences in scanners or sequences). The training is multitasking: the model is jointly trained to segment the tumor, classify its grade, and predict its volume. Loss functions (Dice loss for segmentation, cross-entropy for classification, etc.) are combined in a weighted manner. By incorporating *explicit 3D tumor size information* (like volume and surface area) into the training, the model leverages physical tumor metrics to aid classification (grade prediction) and possibly influence segmentation (e.g., ensuring segmentations are spatially plausible).

Techniques (Model & Methods): Key techniques include:

- **3D CNN with Attention:** Using 3D convolutions allows the model to consider full volumetric context. The **spatial attention** gates (AG) help the network attend to tumor boundaries and important regions, improving segmentation of irregular shapes.
- **Multiscale Feature Fusion:** The network extracts feature at multiple scales (through an encoder-decoder with skip connections like U-Net), which is crucial for capturing small enhancing cores as well as large edema regions.
- **Monte Carlo Dropout for Uncertainty:** At test time, dropout is applied multiple times to generate an ensemble of predictions. The variance in these predictions highlights areas of uncertainty (e.g., unclear tumor margins), which is reported to clinicians as confidence intervals.
- **Domain Adaptation:** Although specifics aren't detailed in the abstract, this likely involves techniques such as adversarial training or feature normalization to reduce scanner-specific biases, ensuring that the model performs consistently on external datasets not seen in training.
- **Multitask Learning:** The model produces outputs for segmentation masks, tumor class (perhaps high-grade vs low-grade glioma), and a regression/estimation for tumor volume. This shared learning can improve performance – for instance, knowledge of tumor volume could help differentiate tumor types or grades, and vice versa.

Important Notes: AG-MS3D-CNN is notable for aiming at **generalizability**. The inclusion of external test datasets (from OASIS, ADNI, IXI – which are actually datasets of healthy brains or other conditions) suggests they tested the segmentation's robustness on images outside of BraTS domain. The reported performance “across multiple MRI scanners” indicates the model maintained high Dice scores and low Hausdorff distances even when MRI characteristics varied. For BraTS 2021 (which contains multimodal MRI for brain tumor patients), the model's mean Dice scores per region were: **Whole Tumor: ~0.91, Tumor Core: ~0.87, Enhancing Tumor: ~0.84**. These are among the top-tier results for BraTS. The approach also reduced the 95th percentile Hausdorff Distance (HD95) for boundaries compared to prior methods (e.g., a 3.1% DSC increase and 17.5% HD95 decrease vs. TransBTS, a prior transformer-based model). The explicit tumor size features contributed to improved *grading* performance as well – the study references improved glioma grading accuracy by combining 2D CNN features with 3D radiomic features (from Zeng *et al.*, presumably, where AUC was ~0.968 for grade prediction).

One **important note** is that this model is complex and heavy: it leverages both 2D and 3D networks and radiomic computations. Training such a model requires a large dataset (BraTS provides ~200+ labeled cases, which they augmented by excluding very small tumors to avoid noise). The model also provides **interpretability** via heatmaps highlighting which regions contributed most to the classification decision (likely using techniques like Grad-CAM applied to the classification branch). No ensemble of separate models was used; instead, the architecture itself integrates multiple modalities and tasks – it's a single network with many components.

Dataset: The model was trained and tested primarily on the **BraTS2021** dataset, which consists of multimodal MRI for brain tumor patients (with ground truth segmentations and tumor grades). Each BraTS case includes four MRI sequences (T1, T1-contrast, T2, FLAIR). The authors

mention excluding some cases with very small tumor sizes and splitting the rest into training/validation cohorts. In addition, external datasets:

- **OASIS & ADNI:** These are datasets of normal aging or Alzheimer's patients (mostly no tumors). They might have been used to test false positive segmentation rates or generalization (the domain adaptation likely aims at not segmenting "tumors" in healthy brains).
- **IXI:** A collection of normal brain MRIs from multiple scanners in the UK, used similarly for testing generalization.

The segmentation evaluation on external datasets is reported in terms of Dice and HD95 in the paper's figures (the abstract doesn't quote exact numbers beyond stating improvement). We can infer the model did not produce degenerate outputs on external images (likely segmenting nothing, which is correct for healthy brains, thereby yielding near 0 false positives).

Results: On BraTS21 validation, as noted, **Dice scores** were **0.912 for tumor core, 0.857 for enhancing tumor, and 0.91 for whole tumor** (these numbers come from combined context in the paper: the abstract says "mean Dice of 0.91, 0.87, 0.84 for entire, core, enhancing", and further text indicates core DSC ~0.912 with certain improvements). These results are **3–4% higher** than many earlier BraTS models, which is significant in segmentation. The **AUC** metric isn't directly applicable for segmentation, but if considering the grading sub-task, an AUC of 0.9684 was achieved for glioma grade classification (from a related method) by using the combined multimodal and size features. The **uncertainty estimates** provided by the model add an extra layer of result quality: for instance, the model can indicate with high certainty where the tumor boundaries are clear, and highlight areas where it is unsure (which could correspond to ambiguous edema/tumor regions). This is valuable clinically because a Dice score doesn't convey if errors are at critical boundaries – uncertainty maps do.

In summary, AG-MS3D-CNN presents state-of-the-art segmentation accuracy on BraTS and demonstrates the ability to maintain performance across different MRI datasets. The **Dice = 0.91/0.87/0.84 (WT/TC/ET)** are among the best reported, and the approach's innovations (attention, multitask, domain adaptation) clearly contributed to this success. The study shows that incorporating **explicit tumor size features and multitask learning improved the tumor grading AUC to ~0.97**, suggesting that the segmentation output wasn't just visually accurate but also quantitatively informative for diagnosis.

Study4

An Automated Deep Learning Framework for Brain Tumor Classification Using MRI Imagery (2025)

Study Title	Year	Journal / Venue	Task Type	MRI Modalities	# Modalities	Data Rep. (2D/3D)	Models Used	Ensemble?	Results (Key Metrics)	Link
An Automated Deep Learning Framework for Brain Tumor Classification Using MRI Imagery (Aamir et al.)	2025	Scientific Reports (Nature)	Classification (with segmentation-aided pipeline)	T1-weighted CE MRI (Figshare); Multi-modal MRI (BraTS2020)	1 (Figshare); 4 (BraTS)	2D slices (with 3D segmentation step)	U-Net (with SE-ResNet101 encoder) + Inception-based multiscale CNN + attention; RF/SVM/KNN ensemble on features	Yes (features ensemble of RF, SVM, KNN)	Accuracy 99.67%, F1 99.60% (Figshare 3-class); Accuracy 99.94%, F1 99.96% (BraTS2020 HGG vs LGG)	[4]

Summary: Aamir *et al.* propose a comprehensive pipeline for brain tumor **classification** that integrates image preprocessing, segmentation, multi-scale feature extraction, and ensemble learning. This “*automated deep learning framework*” first **segments tumor regions** in MRI scans and then classifies the tumor type using a combination of CNN features and classical machine learning models. The framework was validated on two datasets: The **Figshare MRI dataset** (3 tumor classes: glioma, meningioma, pituitary) and **BraTS2020** (which the authors repurposed for classification, treating it as a binary classification of high-grade vs low-grade glioma). The results were exceptional – on the Figshare dataset it achieved **99.67% accuracy** (with similarly high precision/recall), and on BraTS (Kaggle) it reached **99.94% accuracy** in distinguishing tumor grades. These results outperform previous methods and demonstrate high generalizability across datasets.

Methodology & Approach: This framework has multiple stages:

- Preprocessing & Segmentation:** MR images undergo enhancement (e.g., contrast normalization) and then are fed to a **U-Net** model (with a SE-ResNet101 encoder) to segment the tumor region. By doing this, the system focuses subsequent classification on just the tumor area, reducing background noise. The U-Net was trained to produce accurate tumor masks; indeed, the authors mention refining segmentation using a **Stacked Encoder-Decoder** architecture in parallel with residual blocks.
- Feature Extraction:** After segmentation, they extract features from the tumor region using a **multiscale CNN approach**. Specifically, they use an **Inception-based multiscale feature extractor** (GoogleNet Inception modules) to capture tumor features at multiple filter sizes. They also incorporate attention mechanisms (self-attention and spatial attention blocks) to refine these features.

3. **Feature Fusion & Dimensionality Reduction:** Features from multiple scales and possibly from multiple models (they mention DenseNet architectures too) are fused. The framework uses evolutionary algorithms (an improved Grey Wolf Optimizer combined with the Jaya algorithm) to select the most important features and hyperparameters. This optimized feature set is then presumably reduced (to avoid overfitting given the small dataset).
4. **Classification (Ensemble Learning):** Instead of using a single neural network classifier, the approach employs an **ensemble of classifiers** on the extracted features. Specifically, they mention integrating **Random Forest (RF)**, **Support Vector Machine (SVM)**, and **K-Nearest Neighbors (KNN)** in an ensemble to make the final decision. The ensemble likely combines these via majority voting or a stacking strategy. The rationale is that combining multiple algorithms can improve robustness – and indeed they report this ensemble outperforms any individual classifier.
5. **Five-fold Cross-Validation & Testing:** They trained and evaluated the model with 5-fold cross-validation on Figshare (ensuring an unbiased estimate). They also tested the model’s generalization by applying it (without re-training) to BraTS2020 data (where ground truth tumor segmentations and grades were available).

Techniques (Model & Methods): Key techniques include **transfer learning and fine-tuning** (they use pre-trained CNNs like SE-ResNet101 for U-Net encoder, DenseNet121/169 for feature extraction), **multiscale convolution** (Inception modules to capture features of varying spatial sizes), and **attention blocks** (both channel-wise and spatial attention to highlight salient tumor features). The segmentation step uses a **Squeeze-and-Excitation ResNet101 U-Net** – SE blocks help the network focus on important channels in the segmentation process. For classification, rather than end-to-end training a single network, they adopt a **hybrid approach**: deep features are extracted and then fed into classical ML models. This is somewhat an “AutoML” style approach where deep features are filtered and then an optimal classifier is chosen (they mentioned comparing several, with one approach using a fine-tuned EfficientNet achieving 99.67% as well). They also extensively use **data augmentation** (rotations, flips, etc.) to increase the Figshare dataset size fourfold (from ~3k to ~12k images) to aid training.

Important Notes: This study’s novelty lies in its end-to-end framework combining **segmentation + classification**. One important detail: the **classification task here is tumor type classification** (not malignancy grade) for Figshare, but for BraTS2020 they effectively classify **MRI modality or patient’s dataset** (the authors refer to BraTS2020 as “Kaggle” and achieve 99.94% – which corresponded to distinguishing BraTS’s high-grade vs low-grade cases). The user specifically asked to note: *in Ullah et al. “BrainNet” study (which is a different study) the classification was of MRI modalities, not tumor grade*. However, **in this Aamir et al. study**, the classification on BraTS was indeed **tumor grade classification** (HGG vs LGG), demonstrating the framework’s versatility. They report nearly perfect sensitivity and specificity on BraTS, indicating the model can generalize to multi-modal inputs (BraTS has 4 modalities) even though it was primarily trained on single-modality Figshare data – likely due to the segmentation+transfer learning approach’s robustness. Another note: they employed **Grad-CAM** explainability to ensure the model’s decisions were interpretable, showing heatmaps over images highlighting tumor regions that influenced the classification. No obvious ensemble of

deep models was used for segmentation; the ensemble is in the classification stage combining RF/SVM/KNN results, which is relatively low-complexity and fast.

Dataset:

- **Figshare dataset** (also known as Cheng's dataset): 3064 T1-contrast MRI slices from 233 patients with **meningioma, glioma, pituitary** tumors. They preprocessed these images (intensity normalization, CLAHE, etc.), augmented them (rotations, flips), and used 5-fold CV for training/testing. The best model (based on EfficientNetB2 fine-tuned) achieved 99.67% accuracy on this dataset.
- **BraTS2020 dataset:** Originally for segmentation, containing multi-modal MRIs of glioma patients (HGG or LGG). They used BraTS2020 (possibly via a Kaggle copy) to test generalization. They treated it as a new dataset where each patient's set of images had to be classified as high-grade or low-grade tumor. The model achieved 99.94% accuracy on this (with sensitivity ~99.96%, specificity ~99.92%) – essentially near perfect classification of tumor grade, which is remarkable. This suggests that the segmentation + feature ensemble approach captured very discriminative features of tumor aggressiveness (e.g., perhaps size, shape from segmentation and texture from features).
- The framework did not rely on multi-modal inputs for Figshare (which is single-modality), but it handled BraTS (multi-modal) by likely either combining modalities into channels or using the BraTS segmentations to compute features. It's not explicitly described, but possibly they used the **segmented tumor volume from all BraTS modalities** to compute features (like volume, intensity stats) and then classify grade – which aligns with other research that volume and edema extent correlate with grade.

Results: On Figshare (3-class): Accuracy = 99.67%, with sensitivity ~99.6%, specificity ~99.64%, precision ~99.35%, F1 ~99.60%. These metrics indicate the model is extremely accurate and balanced across classes (very few misclassifications of each tumor type). It outperformed several baseline deep models (ResNet50, GoogleNet, etc., which had ~98–99% accuracy). **On BraTS2020 (binary grade classification): Accuracy = 99.94%**, sensitivity 99.96%, specificity 99.92%, F1-score 99.96%. Essentially, out of the BraTS validation cases, perhaps only 1 was misclassified. The **Dice** metric is not directly applicable since this is classification rather than segmentation (though they do perform segmentation internally). However, the intermediate segmentation was high-quality; they mention that only 4 of 704 meningioma cases and 9 of 1426 glioma cases were misclassified by the final model, implying the segmentation + classification pipeline was extremely effective.

In conclusion, this framework demonstrates a powerful **hybrid approach**: leveraging segmentation to focus on tumor regions, extracting robust features (CNN + attention), and using an ensemble of traditional classifiers to yield near-perfect classification of brain tumors. The inclusion of both a segmentation stage and an ensemble classifier stage makes the system a bit complex, but clearly each part contributed to the outstanding performance. This suggests that for real-world applications, a pipeline that localizes the tumor and then classifies it (possibly with expert-designed feature fusion) can be more effective than end-to-end CNN classification on whole images.

Study5

Brain Tumor Diagnosis Redefined: Leveraging Image Fusion for MRI Enhancement Classification (2025)

Study Title	Year	Journal / Venue	Task Type	MRI Modalities	# Modalities	Data Rep. (2D/3D)	Models Used	Ensemble?	Results (Key Metrics)	Link
Brain Tumor Diagnosis Redefined: Leveraging Image Fusion for MRI Enhancement Classification (Hekmat <i>et al.</i>)	2025	Biomed. Signal Process. Control (Elsevier)	Classification (MRI modality classification)	T1-weighted CE MRI (Figshare); + Kaggle dataset (with “no tumor”)	1	2D slices	DenseNet121 + DenseNet169 (parallel feature extractors) with CLAHE & DWT preprocessing; attention-based feature fusion	Yes (feature-level fusion of two DenseNets)	Accuracy 99.16% (Kaggle 4-class), 94.28% (Figshare 3-class)	[5]

Summary: Hekmat *et al.* present a method to improve brain tumor classification by **fusing images and features** from MRI that have been enhanced via different techniques. The study is positioned as “redefining” brain tumor diagnosis by using image fusion to better classify tumors. Specifically, it combines *contrast enhancement (CLAHE)* and *wavelet transform (DWT)* preprocessing to create clearer MRI inputs, then uses dual CNN models (DenseNet variants) whose features are fused for classification. The task tackled is multi-class tumor classification (likely the standard three classes). They test their model on the **Figshare tumor dataset** and a **Kaggle dataset**. The results show **99.16% classification accuracy** on the Kaggle dataset (which includes a “no tumor” class) and **94.28% accuracy** on the Figshare dataset (3 tumor classes). This indicates a high accuracy, especially on the dataset augmented with healthy cases.

Methodology & Approach: Their approach can be broken down as follows:

- **Image Preprocessing:** They apply **CLAHE (Contrast Limited Adaptive Histogram Equalization)** to MRI scans to improve contrast, and **Discrete Wavelet Transform (DWT)** (and also mention Chirplet transform in references) to extract multi-frequency features. By preprocessing images with CLAHE and DWT, they aim to enhance tumor edges and textures that might aid classification.
- **Dual-Stream CNN Feature Extraction:** They utilize two deep CNN models in parallel: **DenseNet-121** and **DenseNet-169**. Each DenseNet processes the images (possibly one processes the CLAHE-enhanced image, the other the DWT-derived image, though the exact split isn’t explicitly stated). DenseNets are chosen for their strong feature propagation and reduced parameter count via dense connections.
- **Attention Mechanism:** An **attention module** is incorporated to guide the fusion of features from the two DenseNets. This attention likely ensures that more weight is given to the most informative features among the combined set.

- **Feature Fusion:** The features output by DenseNet121 and DenseNet169 are **fused** (concatenated or pooled in some novel way) to create a comprehensive feature vector representing both local and global patterns in the MRI. The authors mention a “novel parallel pooling approach” for fusion and the use of evolutionary optimization to select the best subset of fused features.
- **Classification:** The fused features are fed into a classifier (possibly a fully connected neural network or an SVM). It’s not explicitly described which classifier is used post-fusion, but since they achieve very high accuracy, it could be a small neural network or even directly a Softmax layer on fused features.

Techniques (Model & Methods): Key techniques include the **fusion of multiple image enhancement techniques:** by combining CLAHE (which operates in spatial domain to improve contrast) and DWT (which operates in frequency domain to capture edges and textures at multiple scales), they create a richer input for the models. Using **two different CNN architectures (DenseNet-121 & DenseNet-169)** introduces diversity in learned features; DenseNet-121 might capture certain levels of detail while DenseNet-169 (deeper) captures others. The **attention mechanism** (referred as AM in find results) likely acts as an attention layer on top of concatenated features to weight the contribution of each model’s features before final classification. This is crucial because simply concatenating high-dimensional features from two DenseNets could include redundant or noisy information; attention helps focus on the most relevant fused features. Additionally, they used a **genetic algorithm (GA)** or evolutionary strategy to optimize hyperparameters and feature selection (the reference to improved Grey Wolf optimizer integrated with Jaya indicates a hybrid GA approach to fine-tune the model and feature subset). So, this work stands at the intersection of deep learning and evolutionary optimization to yield an optimal classification system.

Important Notes:

- The task is described as “classification of four types of brain modalities” in the abstract. Here, the wording is tricky: *They actually mean four types of MRI images: “t1 tumor, t1ce tumor, t2 tumor, flair tumor”*. **This indicates that the model was trained to classify which MRI modality a given tumor image belongs to**, rather than classifying tumor pathology. That is an unusual task – essentially an automated way to detect MRI sequence type. This aligns with the user’s note about Ullah *et al.* study (which is the same reference as this Hekmat’s references) – however, it appears in Hekmat’s abstract they mention classifying modalities as a way to ensure the model works on multimodal data. But then, their experimental results mention *accuracy on Kaggle and Figshare datasets*, which are about tumor type classification. It is likely they conflated two goals: they might classify modalities as an auxiliary task or for pretraining. The **critical note** is that **this study’s primary classification is of MRI scans that are already tumor-positive**, trying to label them by tumor type. Meanwhile, a side-result or technique was ensuring the model knows the modality type (since their dataset included multi-modality or needed to be robust to it).
- The Kaggle dataset mentioned had 7023 images with four classes (glioma, meningioma, pituitary, and “no tumor”). The model’s **99.16% accuracy** on Kaggle indicates it can almost perfectly identify if an image has a tumor and of which type, even distinguishing

healthy images. On the Figshare dataset (no healthy images, just 3 tumor classes), the accuracy was a bit lower at **94.28%**. This is interesting – one would expect higher accuracy on the easier 3-class task. The drop might be due to the Figshare dataset’s smaller size or less distinct classes (glioma vs meningioma can be subtly different). The Kaggle dataset, by virtue of adding a “no tumor” class, might actually introduce many easy-to-classify cases (normal vs tumor is easier, boosting overall accuracy).

- The approach did not involve any conventional ensemble of separate trained models at inference; instead, it’s a **feature-level ensemble (fusion)**. Thus, the pipeline is one integrated model at test time (with two CNN branches fused). So “Ensemble?” column we marked “Yes (feature fusion of DenseNets)” to indicate multiple models combined.
- The authors report that integrating CLAHE and DWT significantly improved feature quality, and the attention-guided feature fusion outperformed single-model baselines by a large margin. In comparisons, they likely showed that e.g. using DenseNet121 or DenseNet169 alone (with or without CLAHE) yields lower accuracy (maybe in the 90–93% range), whereas the fusion hits 99%. They also likely compared with other techniques or existing studies; their reported accuracy of 99.16% on Kaggle is higher than most contemporary models for that 4-class dataset, indicating the efficacy of their approach.

Dataset:

- The **Figshare dataset** (Cheng 2017) of 3064 contrast-enhanced T1 slices, 3 classes (no healthy). They split it (not sure ratio, maybe 80/20). They also mention “20% per class for testing” in references, which suggests a stratified split.
- The **Kaggle dataset** used appears to be the “**Brain Tumor Classification (MRI)**” **Kaggle dataset** which indeed has 4 classes: glioma, meningioma, pituitary (all tumor) and **no tumor**. They cite achieving 99.16% on Kaggle. This dataset had 7023 images total, which matches what Kaggle sources report (and the numbers in a related study by afzal2020 etc). They likely fine-tuned on Kaggle after training on Figshare or used Kaggle for additional validation.
- The images in both sets are **2D MRI slices**. Only T1-contrast images in Figshare; Kaggle’s “no tumor” images likely are T1 or T2 MRI of healthy brains. The model doesn’t explicitly incorporate multi-modal images simultaneously (each image is single modality), but by classifying “modalities,” they ensure robustness if images come from different MRI sequences. Possibly, to handle Kaggle (which might include some T2 or FLAIR images for “no tumor”? Not sure, Kaggle’s no-tumor likely were also T1-weighted), they wanted the model to not be thrown off by modality differences. In any case, the **modality classification** auxiliary task could have been used to pretrain or multitask-train the network, improving feature learning (this isn’t fully clear in the abstract, but it’s a reasonable guess since they emphasize classifying four modalities in addition to tumor classification).

Results: On the **Kaggle 4-class dataset: Accuracy = 99.16%**. This suggests only a handful of images were misclassified. They likely achieved similarly high precision and recall (though not stated, an accuracy that high implies all metrics ~99%). On the **Figshare 3-class dataset: Accuracy = 94.2%** (from a snippet in references), but abstract says 94.28%. This means about

5–6% of images were misclassified, which could correspond to confusion mostly between glioma vs meningioma (common in other works). They specifically mention “maximum single-classifier accuracy of 99%”, meaning one of their individual models (maybe DenseNet169 with CLAHE-DWT) achieved 99% on training or on one dataset, but the fused model’s overall average was 98% for both datasets. It’s a bit unclear, but likely:

- Fused model on Figshare: ~94%,
- Fused model on Kaggle: ~99%,
- Some individual run or fold hit 99% on Figshare as a peak but not on average.

They also mention that their approach outperformed others and required less preprocessing than prior methods, thanks to the image fusion doing a lot of enhancement implicitly.

To sum up, Hekmat *et al.* achieved **state-of-the-art accuracy** on a challenging 4-class classification by **fusing dual CNN models and advanced preprocessing**. The particularly noteworthy outcome is the near-perfect classification on a dataset that includes normal MRIs – indicating the model’s ability to identify even the absence of tumor confidently. By contrast, on solely tumor-class differentiation, performance, while strong, leaves some room (~94%) which is still competitive. This study underscores that **combining complementary image enhancement techniques (CLAHE + DWT) and model fusion (DenseNet + DenseNet with attention)** can significantly boost classification performance in medical imaging.

Study6

XAI-MRI: An Ensemble Dual-Modality Approach for 3D Brain Tumor Segmentation (2025)

Study Title	Year	Journal / Venue	Task Type	MRI Modalities	# Modalities	Data Rep. (2D/3D)	Models Used	Ensemble?	Results (Key Metrics)	Link
XAI-MRI: An Ensemble Dual-Modality Approach for 3D Brain Tumor Segmentation (Farhan <i>et al.</i>)	2025	<i>Frontiers in AI</i> (Front. Artif. Intell.)	Segmentation	Multimodal MRI (BraTS2020: T1, T2, T1CE, FLAIR)	4	3D volumes	U-Net (trained on single modalities and dual-modality) + Grad-CAM explainability	Yes (ensemble of two best dual-modality U-Nets)	Dice = 97.73% (whole tumor); Mean IoU 60.08% (ensemble outperforms individual models)	[6]

Summary: Farhan *et al.* introduce **XAI-MRI**, a methodology focusing on brain tumor **segmentation** using a dual-modality ensemble of U-Net models and providing explainable AI (XAI) outputs. The goal is to improve segmentation accuracy by leveraging information from multiple MRI modalities and to enhance model interpretability. The approach first trains individual 3D U-Net models on each MRI modality (T1, T2, T1CE, FLAIR) to assess their segmentation performance. Then it identifies the two best modalities, trains separate U-Nets on **combinations of two modalities** (dual-modal input), and finally ensembles the two best dual-modality models into a single segmentation output. The ensemble segmentation result is further visualized with Grad-CAM heatmaps for explanation. On the BraTS2020 dataset, the proposed ensemble achieved a very high **Dice coefficient of 97.73%** for tumor segmentation (presumably for whole tumor), which is an impressive result, and a Mean IoU of 60.08%. The ensemble outperformed single-modality and single-model approaches, demonstrating the benefit of combining modalities and models.

Methodology & Approach: The method can be outlined as:

- Single-Modality Training:** Four separate 3D U-Net models were trained, one on each MRI sequence (modality) independently (e.g., one U-Net on just FLAIR images, one on just T1CE, etc.). This established a baseline segmentation performance per modality and identified which modalities are most informative for tumor segmentation.
- Dual-Modality Training:** After evaluating single-modality results, they picked the top two modalities (for example, BraTS studies often find FLAIR is critical for whole tumor and T1CE for tumor core). Then they trained **two-modal U-Net models**: each model takes two MRI modalities as input channels (e.g., Model A takes FLAIR+T1CE, Model B might take FLAIR+T2, depending on which were best). These models exploit

complementary information (one modality might highlight edema, another the enhancing core).

3. **Ensembling Models:** The **ensemble dual-modality** comes by **combining the outputs of the two best dual-modality U-Nets**. Likely, they used a simple averaging or voting on the predicted mask probabilities. For segmentation, ensembling can be done by averaging the probability maps from each model and then thresholding. This ensemble should yield better segmentation by reducing errors unique to each model.
4. **Explainability with Grad-CAM:** To add interpretability, they applied **Grad-CAM** (Gradient-weighted Class Activation Mapping) to the segmentation models or an auxiliary classifier to highlight which regions of the MRI contributed most to the model's "decision" (in segmentation context, this might highlight the tumor region in the input image, aligning with expected areas). These heatmaps help clinicians see that the model is focusing on tumor tissue and not on irrelevant regions, thereby building trust.

Techniques (Model & Methods): They use the classic **U-Net architecture** for segmentation, which is well-suited for biomedical image segmentation due to its encoder-decoder structure and skip connections. The novelty here is training U-Nets on different **combinations of input channels** to leverage multi-modal MRI. By doing a systematic modality ablation, they identify optimal modality combinations rather than feeding all four modalities at once. This is important because feeding all four might introduce noise or require more complex model to weigh modalities. Instead, they found an optimal pair presumably. The **ensemble** technique is essentially a **bagging ensemble** of two deep models (each already taking two modalities). Combining them likely required alignment (which is trivial as modalities are from same patient and already registered). The XAI component uses **Grad-CAM**, which typically applies to classification networks. For segmentation, they may have taken the final feature map of the U-Net or a classification head that predicts presence/absence to generate Grad-CAM, highlighting areas strongly influencing the model's output. They emphasize that the approach is designed to be **transparent** – hence the "XAI" in the name – by providing these visual explanations.

Important Notes:

- The reported **Dice = 97.73%** is extremely high for segmentation. It likely refers to the **Dice Similarity Coefficient for the whole tumor region** (since they mention mean IoU as well). In context, BraTS whole tumor Dice for top models is around 0.92–0.95; 97.7% is above typical values. It could be that they calculated Dice on a subset or perhaps on the training set or a validation set rather than the hidden test set. It might also be a global (not per-case average) Dice. Regardless, it's clear their ensemble significantly improved segmentation performance.
- The **Mean IoU = 60.08%** is somewhat puzzling. IoU (Intersection over Union) for a single class segmentation should correlate with Dice ($\text{Dice} = 2 \cdot \text{IoU} / (1 + \text{IoU})$). A Dice of 0.977 corresponds to $\text{IoU} \approx 0.955$. So a mean IoU of ~ 0.6008 suggests they might be averaging IoUs of multiple classes (whole, core, enhancing and maybe background) or including background which skews it. Possibly they treat segmentation as multi-class (with background, edema, core, enhancing labels) and report mean IoU across all classes. In multi-class segmentation, background IoU is often high ($\sim 99\%$) and small structures lower, which could average to $\sim 60\%$. Alternatively, maybe they mean the *mean*

intersection over union of multiple models or multiple samples, which is unclear. It's an unusual metric pairing; typically, one would report per-class IoUs or just dice. It's possible there's a mistake or miscommunication in abstract. Regardless, the Dice is the more interpretable metric here, showing excellent performance.

- The approach did not incorporate a **third or fourth model**; only two dual-modality models were ensembled (for computational efficiency likely). They specifically say combining the two best-performing dual-modality networks gave the highest result. For example, if FLAIR+T1CE and FLAIR+T2 were the best two, their ensemble is what yielded 97.7% Dice. Had they included a third model (like T1CE+T2) maybe improvement was negligible or it overfit.
- In the single-modality analysis, they would have found which single MRI was best. Commonly, **FLAIR** is best for whole tumor (edema + tumor), **T1CE** is best for core segmentation. So likely, FLAIR was one of the chosen pair. The improvement from single to dual to ensemble shows diminishing returns: single (maybe ~95% Dice), dual (~96–97%), ensemble ~97.7%. It highlights that in brain tumor segmentation, combining modalities is crucial (no single MRI gives complete info) and combining models further smooths remaining errors.
- The “XAI” aspect is aimed at clinical acceptance. They not only output a binary mask but also provide a heatmap overlay that shows, for instance, high activation on the tumor area – reassuring that the model is indeed looking at the tumor to make its prediction (and not some artifact elsewhere). This helps in cases of partial mistakes too; clinicians can see where the model was uncertain.

Dataset: They used the **BraTS2020 dataset** (as noted, referencing Kaggle for BraTS). BraTS2020 has multi-modal MRIs for 369 High-Grade and 125 Low-Grade glioma patients (training set), with segmentation masks for whole, core, enhancing regions. The data is inherently 3D (each patient has a 3D volume for each modality). XAI-MRI likely trained on the BraTS training set and tested on the validation set (since test set labels are private). They do not mention external validation beyond BraTS. Possibly they split off a part of training for testing due to needing Grad-CAM etc. The mention of BraTS2020 suggests a standard benchmark usage. The input to their U-Nets:

- For single modality models: 3D volume of one modality.
- For dual: 3D volume of two modalities (as two channels).
They might have cropped or resampled volumes (BraTS volumes are 240×240×155, which is large; many methods use patches or downsample).
The output segmentation evaluated is binary (they speak of “the segmentation result” singular), which suggests they might have focused on **whole tumor segmentation** (treating all tumor parts as one) to get that single Dice figure. If so, 97.7% Dice for whole tumor segmentation is plausible (with FLAIR and T2 giving edema segmentation, ensembles do reach ~0.95–0.96 historically; 0.977 would be new high, perhaps due to ensemble).

Results: The ensemble dual-modality model **outperformed all single-modality and single-model approaches**. Precisely:

- Single best modality model might have had Dice in mid-90s%.
- Best dual-modality model had a higher Dice (not given, maybe ~96–97%).
- Ensemble of two dual models gave **Dice = 97.73%**.

This is an excellent result, suggesting only minimal error in tumor delineation. It implies the segmentation could practically be used for volume estimation or pre-surgical planning with high reliability. The authors emphasize that the ensemble “enhanced the segmentation result” – meaning it fixed errors that one model made and the other didn’t. The **Grad-CAM** outputs are qualitative results showing that the model focuses on tumor regions. These presumably correlated well with the segmented areas, adding trust. Also, by highlighting regions, the model might help identify if any portion of tumor was ambiguous (if the heatmap is diffuse or extends beyond the segmented region, it might indicate uncertainty).

Finally, they mention the **code is publicly available**, showing commitment to reproducibility. The result, a nearly 98% Dice, is among the highest reported for BraTS whole tumor segmentation. It demonstrates the power of **model ensembles and multi-modality**: each MRI sequence provides unique information, and the ensemble of specialized models yields a comprehensive segmentation. While computationally heavier (two U-Nets inference instead of one), it’s feasible in clinical settings since U-Net models run in a few seconds per volume on a GPU.

Study7

BrainNet: Fusion-Assisted Framework of Residual Blocks and Stacked Autoencoders for Multimodal Brain Tumor Classification (2024)

Study Title	Year	Journal / Venue	Task Type	MRI Modalities	# Modalities	Data Rep. (2D/3D)	Models Used	Ensemble?	Results (Key Metrics)	Link
BrainNet: Fusion-Assisted Framework of Residual Blocks and Stacked Autoencoders for Multimodal Brain Tumor Classification (Ullah <i>et al.</i>)	2024	Scientific Reports (Nature)	Classification (MRI modality classification)	Multimodal MRI (BraTS2020/21: T1, T1CE, T2, FLAIR)	4	2D slices	CNN (custom residual CNN) + Stacked Autoencoder feature fusion; Grey Wolf optimizer for feature selection	Yes (CNN + Autoencoder fusion)	Accuracy ~98% (avg), up to 99% (best classifier)	[7]

Summary: Ullah *et al.* propose **BrainNet**, a deep learning framework that fuses feature from a custom CNN and an autoencoder to classify brain tumors from multimodal MRI. Unlike traditional approaches that classify tumor vs no-tumor or tumor types, BrainNet is somewhat unique: it was primarily designed to classify **MRI modality types** in a multimodal dataset (the abstract mentions classifying “four types of brain modalities: T1, T1CE, T2, FLAIR”). In other words, given an MRI image, the model predicts which modality it is – presumably as a step toward handling each modality appropriately in a downstream task. However, the framework can be applied to **tumor classification** tasks as well by retraining on class labels. In their experiments, they utilized the BraTS2020 and BraTS2021 datasets (each containing multimodal MRI for glioma patients) and reported an **average accuracy of 98%** and a best-case single-model accuracy of 99% on classifying those images. These high accuracies suggest BrainNet effectively learned discriminative features of each MRI modality (or tumor type) via its fusion approach.

Methodology & Approach: BrainNet consists of two parallel components:

- **A custom CNN with residual blocks:** They modified a CNN architecture (possibly ResNet or their own design with skip connections) using domain knowledge. Residual blocks help in training deeper networks by alleviating vanishing gradients and allowing feature re-usage. This CNN is likely responsible for extracting high-level features (e.g., texture, intensity patterns) from the images.
- **A Stacked Encoder–Decoder (autoencoder) network:** This unsupervised network with 10 convolutional layers is trained to reconstruct input images (or some representation).

By doing so, it learns a compressed representation (bottleneck features) that capture essential information about the images (potentially different from the CNN's features).

- **Feature Extraction & Optimization:** Features from both the CNN and autoencoder are extracted and then optimized using an improved Grey Wolf Optimizer (with Jaya algorithm enhancements). This evolutionary algorithm selects the most relevant features and tunes hyperparameters, aiming to maximize classification performance while reducing feature dimensionality.
- **Feature Fusion:** The selected features from the CNN and autoencoder are **fused** using a “novel parallel pooling” strategy. This likely means they combine the feature vectors (perhaps concatenation followed by pooling layers to create a fixed-size descriptor).
- **Classification:** The fused feature vector is then fed into multiple classifiers. They mention using both machine learning classifiers and neural networks. Possibly they tried an SVM, a Random Forest, and also a simple fully-connected NN on the fused features. Indeed, they report comparison with several classifiers and even mention “maximum single-classifier accuracy of 99%” – which implies one particular classifier (perhaps a neural network) achieved 99%, while the ensemble or average was 98%.

Techniques (Model & Methods):

- **Residual CNN:** Helps capture complex features; likely they used something like ResNet50 or a smaller variant as the backbone. Residual connections also make the network robust to different intensity distributions – important since MRI modalities vary in intensity.
- **Stacked Autoencoder:** Forces the model to learn a compressed representation of input data. Stacked autoencoders with convolutional layers effectively learn an identity mapping that retains only important features. By using 10 layers, they create a deep autoencoder capturing nonlinear features of MRI scans.
- **Evolutionary Optimization (Grey Wolf + Jaya):** This is used to perform **feature selection and hyper parameter tuning**. Essentially, it's a swarm intelligence algorithm that iteratively selects feature subsets and parameters, evaluates a fitness (classification accuracy), and “evolves” the solution. The improved Grey Wolf Optimizer with Jaya criteria likely speeds up convergence and avoids local minima when searching the feature subset space.
- **Parallel Pooling Fusion:** After feature selection, they must merge CNN features (likely highlighting edges, textures) and autoencoder features (possibly capturing overall image style or modality characteristics). The “parallel pooling” suggests they may pool corresponding features from both networks in parallel paths – maybe something like taking the maximum or average of both feature sets dimension-wise, rather than simple concatenation. This could reduce dimensionality and emphasize features that are strong in both sets.
- **Ensemble of Classifiers:** They tried multiple classifiers on the fused features. Possibly a **neural network classifier** (which might essentially be a multilayer perceptron), a **k-NN**, a **Random Forest**, etc. The best result might have come from a neural network (given their mention of “neural networks” and the fact they cite a maximum single-classifier accuracy of 99%). They might also ensemble these classifiers or just choose the best one.

It's not explicitly stated, but since average accuracy is 98% and one got 99%, they might have just reported both.

Important Notes:

- The unusual aspect: The **classification target in the original paper's phrasing was the MRI modality (T1, T1CE, T2, FLAIR)**. This is somewhat confirmed by "four types of brain modalities" wording. If so, BrainNet was essentially distinguishing which MRI sequence an image came from. This can be useful in a multimodal pipeline or if certain processing is needed per modality. Achieving ~98-99% accuracy on this is plausible because each modality has distinct visual characteristics (FLAIR has dark CSF, T2 has bright CSF, T1 has fat bright, etc.). The high accuracy indicates the model indeed picked up on these cues.
- However, classifying modality is not an end clinical goal; perhaps they also tested it on actual tumor classification (like distinguishing tumor types). The reference to two datasets (BraTS2020, BraTS2021) and "improved performance" suggests they might have done grade classification or survival prediction as well, but not clear. It's possible that by "classification of multimodal brain tumor" they ultimately mean using all modalities to classify something about the tumor (maybe grade). But since results are given as one accuracy figure, and the abstract emphasis was on modalities, it's likely the performance metric is on modality classification. The user's note confirms that *this study was for classification of MRI modality, not tumor grade* – which we should stress here. Indeed, **BrainNet is used to automatically recognize the MRI sequence type of an image containing a tumor, not the tumor's pathology itself** (which is an uncommon but interesting problem).
- In practical terms, once BrainNet identifies the modality with ~98% accuracy, that could feed into other models (like segmentation or diagnosis) that might require knowledge of modality. Alternatively, the approach (residual CNN + autoencoder fusion) could also be applied to tumor type classification (e.g., glioma vs others) on a single modality by retraining it accordingly.
- The authors highlight that their feature optimization and fusion yields an **efficient and accurate system**. They specifically mention achieving improved average accuracy (98%) and also that their method is **computationally cost-effective** and required less preprocessing than some prior works. They likely compare to other multimodal classification efforts (maybe citing others around 90-95%).
- Another note is the **size of datasets**: BraTS2020 & 2021 has a few hundred patients each, but if slicing into 2D images, you get thousands of images per modality. They might have taken 2D slices as input to BrainNet (likely yes, because autoencoders and typical CNNs were applied to 2D images). If each volume yields, say, ~150 slices, and there are ~400 volumes per modality, that's ~60k images per modality total. Perhaps they didn't use all slices (maybe central slices or slices with tumor).
- They also possibly performed 5-fold CV or a train-test split. The **98% accuracy** might be cross-validation result on those datasets.
- **No ensemble of separate predictions** at inference: BrainNet itself is a fusion of two models internally but it outputs one prediction. They mention multiple classifiers but

probably ultimately chose one for final use (maybe an ensemble among them but likely just took the best).

- They did mention comparing to other neural nets (they cite “compared with several classifiers, techniques, and neural nets; proposed achieved improved performance”). This indicates BrainNet’s results of 98-99% outperformed standard deep nets or traditional ML on those tasks.

Dataset:

- **BraTS2020 & BraTS2021:** Both are multimodal MRI datasets for **segmentation and survival analysis** of gliomas. Each patient has 4 MRI modalities (T1, T1CE, T2, FLAIR) and associated tasks (like tumor segmentation mask and overall survival or grade label). For BrainNet’s purposes, they likely extracted images with tumors and labeled them by modality. Possibly they used all slices from these volumes that contain tumor (maybe identified via mask presence) and labeled them “this is a T2 slice with tumor” etc.
- So the classes were exactly 4 (one per modality). They may have merged the datasets or handled them separately. The abstract doesn’t clarify if 98% is on one or both combined. Possibly they combined BraTS2020 and 2021 (since similar data) to have more training samples.
- No additional datasets of different patients are mentioned, so they did not test on e.g. Figshare or others, focusing on BraTS which is multimodal.
- Images are **2D grayscale** slices of size $\sim 240 \times 240$ (BraTS native). They might have resized or center-cropped. They likely standardized intensities per slice or used BraTS normalized data. They definitely used the segmentation masks to know which slices actually contain tumor (so as not to classify empty slices which might confuse modality detection slightly – though even empty, modalities have differences in background).
- BraTS has high-grade vs low-grade cases; if they wanted, they could classify that too, but their focus was modalities. If they attempted grade classification, they would have 2 classes (HGG vs LGG) for which BraTS2020 has labels; but no mention of two-class results or AUC, so likely they didn’t or it wasn’t the focus.

Results: They report **improved average accuracy of 98%** and a **maximum single-classifier accuracy of 99%**. This presumably means that in cross-validation, the mean accuracy was 98%, while one particular fold or classifier hit 99%. In multi-class classification of 4 classes, 98% accuracy is extremely high (error rate 2%, meaning maybe a few slices out of thousands were misclassified). Possibly most errors happen between T1 and T1CE slices if contrast differences are subtle, but even those usually have differences (presence of enhancement). Precision, recall, F1 aren’t explicitly given, but at 98% accuracy, presumably all those metrics are similarly $\sim 0.98+$ if classes are balanced. AUC isn’t typically reported for multi-class, and Dice is irrelevant (segmentation not performed here).

It’s worth noting that if used for tumor presence classification or type classification, one would need to retrain BrainNet for that task, and it’s not demonstrated here. However, the architecture could then achieve high accuracy for that too, potentially.

In summary, BrainNet by Ullah *et al.* is a **feature-fusion model** that excels at distinguishing MRI modalities (and potentially can be adapted to classify tumor characteristics). Its use of **residual CNN + autoencoder** and **evolutionary feature selection** yields a robust representation, reflected in the near-perfect classification on BraTS multimodal data. An important note is that *the classification task in this study is **not** tumor type/grading, but identifying the MRI sequence modality of the input image* – a somewhat different problem from the others listed. This method ensures that a model can effectively handle and fuse multimodal MRI inputs by first knowing which modality is which, which is a stepping stone to more complex multimodal analysis.

Study8

Enhancing Brain Tumor Detection in MRI with a Rotation Invariant Vision Transformer (2024)

Study Title	Year	Journal / Venue	Task Type	MRI Modalities	# Modalities	Data Rep. (2D/3D)	Models Used	Ensemble?	Results (Key Metrics)	Link
Enhancing Brain Tumor Detection in MRI with a Rotation Invariant Vision Transformer (Krishnan et al.)	2024	Frontiers in Neuroinformatics	Classification (Tumor vs Normal)	T1-weighted MRI (Kaggle dataset)	1	2D slices	RViT – Rotation-Invariant ViT (Vision Transformer with rotated patch embeddings)	No	Accuracy 98.6%, Sensitivity 1.00, Specificity 0.975, F1-score 0.984	[8]

Summary: Krishnan *et al.* developed a **Rotation Invariant Vision Transformer (RViT)** model to improve brain tumor **detection** (classification) from MRI scans. The motivation is that tumors can appear in various orientations in MRI slices, and a model robust to rotations can better generalize. RViT integrates rotated patch embeddings into the Vision Transformer architecture, making its predictions invariant to image rotation. The model was evaluated on a Kaggle brain tumor MRI dataset (which included images labeled as tumor or no-tumor), achieving extremely high performance: **Accuracy 98.6%**, with **Sensitivity (Recall) 1.00 (100%)**, **Specificity 0.975**, **F1-score 0.984**, and **MCC 0.972**. These metrics indicate that RViT correctly identified all tumor cases (no false negatives) and had very few false positives, outperforming standard ViTs and other CNN models on the same data.

Methodology & Approach: RViT is a customized Vision Transformer with a focus on rotation invariance:

- **Vision Transformer (ViT) Backbone:** They use the ViT architecture which splits an image into patches, linearly embeds them, and passes them through Transformer encoder layers to learn global relationships.
- **Rotated Patch Embeddings:** The key innovation is that RViT **incorporates rotated versions of image patches** during training (or an explicit rotational encoding). This could mean that for each image, it considers patches not just in their original orientation but also rotated (e.g., 90°, 180°, 270°) and trains the model to treat rotated views of a tumor as equivalent to the original. Alternatively, they might have a rotation-invariant position encoding or a module that aligns patch orientation.
- **Training Strategy:** They likely augmented the training data heavily with rotations but also explicitly modified the model architecture to not be affected by rotation (unlike CNNs which have some built-in translation invariance but not rotation, and ViTs which have neither by default). Possibly they added an additional loss term that encourages

consistent predictions across rotated inputs, or they rotated patches inside the model to simulate all orientations.

- **Dataset & Task:** The Kaggle dataset used is mentioned as “Brain Tumor MRI Dataset from Kaggle”. This likely refers to a dataset of brain MRI images categorized as “tumor” or “no tumor” (a common one on Kaggle with ~3000 images of tumors and ~3000 of normal). So the task is binary classification: detecting if an MRI (presumably a 2D slice or an entire 2D scan image) contains a brain tumor or not.
- **Evaluation:** They mention metrics on a “Brain Tumor MRI Dataset from Kaggle” presumably using some train-test split or cross-validation. Achieving sensitivity 1.0 means **no tumor case was missed**, which is clinically vital. Specificity 0.975 means 2.5% false positive rate (a few normal images were misclassified as tumor).

Techniques (Model & Methods):

- **Rotational Invariance:** They use a technique akin to **data augmentation and architecture modification** for invariance. Rotational invariance is not trivial for ViTs because patch positions are fixed relative to image orientation. They possibly overcame this by random rotation augmentation plus either (a) using sinusoidal position encodings that are rotation-invariant, or (b) training on random rotations and forcing the CLS token output to be similar via a consistency loss.
- **Vision Transformer:** ViT uses multi-head self-attention globally, which can capture long-range dependencies – beneficial for detecting tumors that might be anywhere in the image. However, vanilla ViT doesn’t inherently generalize well on small datasets. They likely used a pre-trained ViT (maybe pre-trained on ImageNet) and fine-tuned it on MRI (transfer learning) to reach such high accuracy.
- **Normalized Dataset:** Kaggle brain tumor sets often have T1-weighted CE images of tumors vs normal. The RViT had to handle grayscale medical images, so presumably they either repeated channels or adjusted the ViT input layer.
- **Performance Boost Over Standard ViT:** They claim RViT outperforms a standard ViT and other techniques. The difference is due to its invariant patch embedding – meaning it doesn’t matter if the MRI is rotated slightly (maybe a head tilt in scan or slice orientation differences), the model still recognizes the tumor. This addresses a real scenario: MRI scans might be acquired in different planes or patient positioning, so a rotation-robust model saves needing to realign images manually.
- **Explainability:** Though not explicitly stated, since the title is about detection, they might produce attention maps or use the transformer’s attention to show what parts of the image are focused on (which often highlight tumor regions). But their main focus is performance metrics.

Important Notes:

- The dataset likely had 2D images that were possibly already skull-stripped or cropped around brain. The fact they got zero false negatives (sensitivity=1) on a test set is a big deal; it could be a smaller curated dataset, so caution on generalization. But still, MCC of 0.972 indicates very high overall quality (MCC takes into account both false positives and negatives).

- Achieving 100% sensitivity means every image with a tumor, no matter how rotated or faint, was caught. That suggests the model is extremely good at recognizing the presence of abnormal patterns. The slight drop in specificity means a few normal images had patterns mistaken as tumor (maybe some artifact).
- The rotation invariance likely allowed them to use less training data effectively; by augmentation, they didn't need to see all possible orientations as separate cases, the model inherently handles them. This could reduce overfitting and improve generalization, thus boosting accuracy.
- They might have relatively balanced data (maybe equal tumor and non-tumor images). In such case, accuracy ~98.6% is truly impressive. If data was imbalanced, high accuracy can happen by skew, but given sensitivity and specificity are both high, it's balanced performance.
- They probably compared RViT to some CNN models (like say ResNet) on the same task. Typical CNN accuracy on that Kaggle dataset (if it's the same one others have used) might be around 95-96%, so 98.6% is an improvement. The references snippet [53†L22-L26] suggests they cite this achievement in context of other work (like they mention “achieving 98.6% [35]” in an MDPI paper likely).
- RViT's approach of rotated patch embeddings could be applicable to other tasks or imaging modalities where orientation variance is a problem (like histology images, etc.).

Dataset: The **Kaggle Brain Tumor MRI dataset** (the authors mention Kaggle and possibly reference Kaggle dataset by Masoud or others). This dataset includes **four classes (glioma, meningioma, pituitary, and no tumor)** in some versions, but the metrics provided (sensitivity, specificity) strongly suggest a **binary classification** (tumor vs no tumor). Because sensitivity/specificity are typically given for binary detection tasks. So it's likely they merged the three tumor types into one “tumor” class vs “no tumor”. This yields a dataset of around 3000 tumor MRI and 3000 non-tumor MRI (numbers from a known Kaggle dataset). They mention it as “Brain Tumor MRI Dataset from Kaggle” and in references [54], it's likely a single combined set. They probably did a train/test split (maybe 80/20). With that many images, results are reliable.

Results:

- **Accuracy = 98.6%,**
- **Sensitivity (Recall) = 1.000 (100%),**
- **Specificity = 0.975,**
- **Precision** (though they said F1, which is 0.984, so we can calculate precision ~98.7%):
Actually, since $F1 = 0.984$ and $recall = 1.0$, $precision = F1 \cdot precision / (2 \cdot precision - F1)$?
Easier: $F1 = 2PR / (P + R)$. With $R = 1$, $F1 = 2P / (P + 1)$. $0.984 = 2P / (P + 1) \Rightarrow 0.984(P + 1) = 2P \Rightarrow 0.984P + 0.984 = 2P \Rightarrow 2P - 0.984P = 0.984 \Rightarrow 1.016P = 0.984 \Rightarrow P \approx 0.9697$ or 96.97%. So indeed, **Precision ~0.97**, meaning about 3% of predicted tumors were false positives (consistent with specificity 97.5%).
- **F1-score = 0.984,**
- **MCC = 0.972** (Matthews Correlation Coefficient, a balanced measure of classifier quality which is very high here).

These metrics depict a model that makes almost no mistakes: it catches all tumors and only misidentifies a few healthy as tumor. In medical diagnosis, missing a tumor is far worse than a false alarm, so this is an excellent outcome (sensitivity 100%). The slight cost is a few false alarms (1-specificity = 2.5% false positive rate), which is generally acceptable clinically (could be follow-up scans to confirm).

Comparatively, standard ViT without rotation might have lower sensitivity if a tumor is rotated oddly (it might misclassify it), so RViT addressed that.

In summary, Krishnan *et al.*'s RViT achieves near-perfect detection of tumors in MRIs by enhancing a Vision Transformer with rotation invariance. This ensures the model's predictions remain accurate regardless of image orientation, leading to outstanding performance (Acc ~98.6%, F1 ~98.4%) on a popular brain MRI dataset. This work demonstrates how introducing inductive biases (like rotational symmetry) into advanced models can significantly boost medical image analysis results.

Study9

Detection and Classification of Brain Tumor using Hybrid Deep Learning Models (2023)

Study Title	Year	Journal / Venue	Task Type	MRI Modalities	# Modalities	Data Rep. (2D/3D)	Models Used	Ensemble?	Results (Key Metrics)	Link
Detection and Classification of Brain Tumor Using Hybrid Deep Learning Models (Vimala <i>et al.</i>)	2023	Scientific Reports (Nature)	Classification (Tumor type)	T1-weighted CE MRI (Figshare dataset)	1	2D slices	EfficientNet (pre-trained, fine-tuned – B0–B4 tested; EfficientNet-B2 optimal)	No	Accuracy 99.06%, Precision 98.73%, Recall 99.13%, F1-score 98.79%	[9]

Summary: Vimala *et al.* focus on classifying brain tumors into their types using a **hybrid deep learning approach** that leverages transfer learning with EfficientNet models. They specifically classify MRI-based brain tumors into three categories: **glioma, meningioma, pituitary tumor** (these are common primary brain tumor types). The term “hybrid” here refers to combining multiple deep learning strategies: they fine-tune five variants of **EfficientNet (B0–B4)** and compare their performance, as well as examine the effect of data augmentation and use Grad-CAM for visual explanation. The best performing model was **EfficientNet-B2** fine-tuned on the dataset. This model achieved an overall **accuracy of 99.06%**, with **precision 98.73%**, **recall 99.13%**, and **F1-score 98.79%** on the test set. These results indicate extremely high performance in classifying the three tumor types.

Methodology & Approach: Key steps in their approach:

- **Transfer Learning with EfficientNets:** EfficientNet is a family of CNNs known for being efficient (good accuracy with relatively fewer parameters due to compound scaling). They took pre-trained EfficientNet models (B0 through B4, which vary in size) that were pre-trained on ImageNet, and **fine-tuned** them on the MRI tumor dataset. Fine-tuning involves replacing the top layer with new ones for 3-class classification and retraining (possibly with a low learning rate) on the tumor images.
- **Two-Step Fine-Tuning:** They mention a “two-step process to refine the pre-trained EfficientNet model”. Step 1: initialize with ImageNet weights, train with additional layers frozen or with a smaller architecture on top. Step 2: add extra layers (top fully connected layers) and then fine-tune the entire model. Essentially, they first might train only the new classification head, then unfreeze more layers and train again (this is a common transfer learning practice to gradually adapt features).
- **Data Augmentation:** They conducted various tests to see the impact of augmentation on performance. Augmentations likely include rotations, flips, shifts, etc., to boost diversity. They found augmentation improved test accuracy and robustness.

- **Hybrid Model Concept:** All five EfficientNet variants were used; presumably, the term “hybrid” might also refer to combining results or techniques from multiple models, but from the abstract, it seems each model was tested individually and EfficientNet-B2 was chosen as optimal. Possibly “hybrid” here just means using pre-trained models in a new domain (combining transfer learning with fine-tuning).
- **Grad-CAM for Visualization:** After training, they applied **Grad-CAM** on the best model (Eff-B2) to visualize attention maps. Grad-CAM highlights regions of the MRI that the model considered important for classification, effectively showing the tumor region being attended to by the model (a sanity check and explanatory tool).

Techniques (Model & Methods):

- **EfficientNet Architecture:** EfficientNets scale depth, width, and resolution systematically. B0 is smallest, B4 is larger. They found B2 was “most optimal” – meaning beyond B2, the larger models did not yield significant improvements, perhaps due to limited data or overfitting.
- **Transfer Learning:** By using ImageNet weights, they leveraged features like edge detectors, textures which still apply somewhat to MRI (though MRI is grayscale, they might have converted to 3-channel or repeated the single channel 3 times since ImageNet models expect 3-channel input).
- **Fine-Tuning Depth:** They possibly tried various extents of unfreezing layers. EfficientNets have many layers; fine-tuning all vs some can impact results. The best approach might have been to unfreeze gradually and use a low learning rate.
- **Regularization:** They likely had to do some regularization (dropout, data augmentation as mentioned) to reach 99% accuracy without overfitting given the dataset size (~3000 images total).
- **Testing and Comparison:** They compared EfficientNet to other pre-trained models (they mention “in comparison to other pre-trained models”). Perhaps they also tried ResNet50, VGG16, etc., and found EfficientNet-B2 gave the highest accuracy.

Important Notes:

- The dataset used is explicitly the “CE-MRI Figshare dataset” (Cheng’s dataset). That dataset, as mentioned, has 3064 images (slices) across 3 classes. They split it into train/test (not stated how, maybe ~80/20 or used the original paper’s split). Achieving 99% on this dataset is extraordinarily high. Many previous works report mid-90s on it. Possibly the combination of EfficientNet’s strong feature extraction and heavy augmentation allowed them to approach near perfect accuracy.
- The performance across classes presumably is also balanced. They gave overall precision ~98.73%, recall ~99.13% – recall a bit higher indicates maybe one class (pituitary perhaps, which is easier to spot) had perfect recall, while maybe glioma vs meningioma had a tiny fraction of misses. The F1 ~98.79% indicates an extremely good balance and only minimal misclassifications.
- They specifically highlight **EfficientNet-B2** yielded “significant performance improvements” – implying maybe B0, B1 were slightly less (perhaps ~97-98%), B3, B4

didn't generalize better (maybe also around 98-99 but possibly overfitting). B2 hit the sweet spot of complexity vs data size.

- The approach is fairly straightforward (no ensemble of multiple networks, just one fine-tuned network). The term “hybrid deep learning models” in the title likely refers to the combination of transfer learning and fine-tuning (pre-trained + custom layers) or the combination of multiple EfficientNet variants in experimentation (though they didn't ensemble them, just chose one).
- They do mention this approach leads to easier detection of tumor location via Grad-CAM – e.g., their best model's Grad-CAM highlights correspond well with actual tumor areas, indicating the model is focusing on the tumor when classifying, which builds trust.

Dataset:

- **Figshare (Cheng) dataset:** 3064 T1-weighted contrast-enhanced MRI images (2D slices) of brain tumors from 233 patients, with labels: **glioma (1426 images), meningioma (708 images), pituitary tumor (930 images)**. No healthy images in this set.
- They likely split into training and test sets; perhaps they did cross-validation. Given their emphasis on test accuracy, perhaps they set aside ~20% (613 images or so) as test. The results 99.06% are presumably on that held-out test portion.
- **Image Preparation:** All images were grayscale 512×512 originally, often they are resized to 224×224 for EfficientNet (which typically expects 240 or 260 input but can adjust). They likely normalized intensities and applied augmentations like rotations (they mention the dynamic range was expanded to enhance brightness/contrast, possibly they applied histogram equalization).
- Data augmentation experiments likely showed that training with flips/rotations increases test accuracy (maybe from, say, 97% to 99%).
- They tried EfficientNetB0 to B4: B0 input resolution 224, B4 goes up to 380 px. They might have had to upscale images for B4, which may not help if not enough detail.

Results:

- **Accuracy:** 99.06% – practically only 1% error rate (maybe ~6 misclassifications out of ~600 test images).
- **Precision:** 98.73% (meaning of all predicted “tumor type X”, 98.73% were correct).
- **Recall:** 99.13% (meaning of all actual tumor type instances, 99.13% were correctly identified).
- **F1-score:** 98.79%.
These metrics demonstrate the model is extremely reliable. The slightly higher recall than precision suggests a few more false positives than false negatives, but both are extremely low. For instance, if test set had ~180 glioma, ~90 meningioma, ~120 pituitary (just guessing proportional splits), maybe only 1 or 2 of each at most got confused.
- They don't explicitly mention AUC, but given multi-class, one could compute class-wise ROC but they didn't need to since accuracy is almost maxed out.
- They likely list that EfficientNetB2 outperformed others: maybe ResNet50 or VGG16 had ~95%, EfficientNetB2 hit ~99%.

- The Grad-CAM visualization likely showed that for glioma cases, the model focuses on the irregular tumor region; for pituitary, focuses on pituitary region, etc., aligning with where a radiologist would look. This confirms the model isn't using spurious cues (like corner text or something).

In conclusion, Vimala *et al.* achieved near-perfect **classification of brain tumor types** by fine-tuning an EfficientNet-B2 on a well-known MRI dataset. Their **hybrid transfer learning approach** proved highly effective, and interpretability was addressed via Grad-CAM to ensure the model's decisions are driven by actual tumor imaging features. This study underscores that with proper pre-trained networks and sufficient augmentation, even relatively small medical image datasets can yield extremely high accuracy in classification tasks.

Study10

Enhanced Multimodal Brain Tumor Classification with Explicit Tumor Size Information (2024)

Study Title	Year	Journal / Venue	Task Type	MRI Modalities	# Modalities	Data Rep. (2D/3D)	Models Used	Ensemble?	Results (Key Metrics)	Link
Enhanced Multimodal Brain Tumor Classification... with Explicit Tumor Size Information (Zeng <i>et al.</i>)	2024	<i>Journal of Cancer</i>	Classification (Glioma Grade classification)	Multimodal MRI (BraTS: T1, T1CE, T2, FLAIR)	4	2D slices + 3D features	2D ResNet50 (for image features) + Radiomics (3D tumor volume, surface area)	Yes (fuses CNN features + 3D size features)	AUC = 0.9684 (validation set, glioma grading)	[10]

Summary: Zeng *et al.* propose a method for **brain tumor grading** (classification of tumor malignancy) that combines **2D CNN features** and **3D tumor size features** to leverage both imaging and volumetric information. Their approach uses multimodal MRI (presumably the BraTS dataset of gliomas) and explicitly extracts **3D tumor measurements** (like tumor volume and surface area) and fuses those with deep learning features from 2D slices (ResNet50 backbone) for predicting glioma grade. By doing so, they address limitations of pure 2D CNN (which lacks volumetric context) and pure 3D CNN (which can be noisy or overfit with small data). The innovation lies in combining *handcrafted radiomic features (volume, surface)* with deep features. The model was trained and evaluated on two public datasets of gliomas, achieving an **AUC of 0.9684 (96.84%)** on the validation set for distinguishing high-grade vs low-grade gliomas. This high AUC indicates excellent discrimination ability and demonstrates that including tumor size information enhanced performance beyond using images alone.

Methodology & Approach:

- **Multimodal MRI Input:** They use four MRI sequences (T1, T1CE, T2, FLAIR) as inputs. These are first **superimposed as channels** to form a four-channel input to a 2D CNN (meaning they take corresponding slices from each modality as a set). Essentially, each slice position through the tumor yields a 4-channel image (like an RGB image but with 4 channels).
- **2D ResNet50 Feature Extraction:** A 2D ResNet50 (pre-trained on ImageNet likely) processes these 4-channel images to extract deep features per slice. This captures texture and intensity patterns from all modalities at that slice.
- **3D Feature Extraction (Radiomics):** They compute **14 three-dimensional features** (via PyRadiomics library) from the full tumor volume across modalities. These features include tumor volume, surface area, maybe shape features, etc., obtained from the

segmented tumor (BraTS provides segmentation masks, so they can calculate actual volume in mm³ and surface, etc.).

- **Feature Fusion:** For each patient, multiple 2D slices produce multiple deep feature vectors from ResNet50. They mention outputting a “1×2 vector” for each slice and combining them. Possibly, they reduce the ResNet output for each slice to a 2-element vector (maybe probabilities of low vs high grade for that slice) and then aggregate across slices (like majority vote or averaging predictions). Or they mean they output two features (maybe predicted mean and variance of output across slices). Then they combine these with the 14 radiomic features, resulting in a final feature set (maybe 16 features).
- **Classification (Grading Model):** Using those combined features, they likely feed them into a fully connected neural network or an SVM to classify grade (WHO Grade II vs Grade III/IV essentially). The training optimizes both the ResNet and final classifier together, presumably. They mention solving the lack of spatio-temporal info in 2D and avoiding the noise of full 3D CNN by this approach.
- **Uncertainty & Interpretability:** The model’s interpretability is enhanced via heatmaps (they likely applied Grad-CAM on ResNet slices to highlight tumor regions important for grade prediction). They also might highlight how certain volumetric features correlate with outcome (e.g., larger volume is high-grade often).
- **Data & Training:** They used two publicly available **brain glioma datasets** – possibly BraTS 2019 or 2020 and maybe Rembrandt or IvyGap etc. However, given context, likely BraTS2018/2019 training data as one cohort, and BraTS2020 (or an independent set from a hospital) as validation.

Techniques (Model & Methods):

- **2D+3D Hybrid:** Instead of a full 3D CNN (which would take a 3D volume input and is memory-heavy), they chose a simpler route: process 2D slices with a CNN and complement that with separate volumetric features computed externally. This reduces complexity and needed data, as 3D CNNs need lots of samples to train.
- **Explicit Size Features:** Volume and surface area are strong predictors of glioma grade (HGG tend to have larger volumes, though overlap exists). By explicitly feeding these into the model, they guide it with domain knowledge that a bigger tumor likely indicates a higher grade, etc. Similarly, surface area relative to volume might capture infiltration vs focal shape.
- **Multimodal Integration:** They feed all four modalities jointly into ResNet50 (as channels). This ensures the CNN sees complementary information (FLAIR for edema extent, T1CE for enhancement, etc.). Many others also do this channel stacking for BraTS classification tasks.
- **Feature Fusion Strategy:** The combination is interesting: “superimposed modalities as 4 channels” and “14 3D features superimposed with 2D CNN features outputting 1×2 vector per patient”. Likely:
 - They run ResNet50 on each axial slice; but a patient yields many slices. They probably take the **slice with largest tumor area** or some representative slices to feed ResNet (perhaps 2-3 slices around the center of tumor) to reduce data. Then maybe average the ResNet outputs.

- They mention outputting 1×2 per slice, then combining multiple to get final prediction. Maybe they treat each slice's output as a vote and train an ensemble aggregator (like a small RNN or averaging).
 - Ultimately they have a predicted grade probability (two numbers) for the whole patient from imaging, and then they combine that with the radiomic features (14 numbers) via a fully connected layer.
 - The " 1×2 vector" likely refers to $[P(\text{LGG}), P(\text{HGG})]$ from the image model, and then they append the 14 features to form a 1×16 vector used for final prediction.
- **Training** probably used cross-entropy loss for grade classification (HGG vs LGG).
- **Results evaluation:** They report an **AUC of 0.9684** on the validation set. AUC (Area Under ROC Curve) is a common metric for binary classification balancing sensitivity & specificity. ~ 0.97 is quite high, indicating the model is highly accurate. They likely also have some accuracy number or others, but AUC is emphasized maybe because it was a key performance measure in that journal (J. Cancer).
- It's likely they compared this hybrid approach to baseline approaches:
 - pure 2D CNN (which might ignore volume, maybe AUC $\sim 0.92?$),
 - pure radiomics + classical classifier (maybe AUC ~ 0.93),
 - pure 3D CNN (maybe AUC ~ 0.95 but overfit issues).
 The combination surpassed all (reaching 0.9684).
- They also enhance interpretability using *heatmaps* – possibly meaning Grad-CAM on the ResNet50 to show which regions of the MRI contributed. This would show that it focuses on the tumor area indeed (which it should, since it's using segmentation to get volume it presumably has the mask or at least can be guided by it).
- Possibly they also output *calibration curves or reliability diagrams* since they talk about "improved prediction and contributions to diagnostics".

Important Notes:

- They needed the tumor segmentation mask beforehand (BraTS provides it). So this method assumes tumor has been detected/segmented already. It's strictly doing *tumor grading* not detection. It's a second-step model after segmentation.
- The reliance on segmentation means in a clinical scenario, you'd need either a radiologist or an automatic segmentation model first. They focus on classification given the tumor region is known.
- The approach avoids too many 3D conv layers by using radiomics, which is clever and computationally light.
- Their high AUC suggests sensitivity and specificity in probably high 90% as well. They didn't give them, but presumably:
 - e.g., sensitivity $\sim 95\%$, specificity $\sim 94\%$ could yield AUC ~ 0.97 if threshold optimized.
- The two datasets likely: BraTS19 (training set) and BraTS20 (as independent validation). They say "two publicly available datasets" and "validation cohort" etc. Possibly BraTS2019 train vs BraTS2019 val vs BraTS2020 as external. It's not fully clear, but they emphasize generalization across cohorts.
- They also mention "the model's interpretability is enhanced through heatmaps highlighting tumor region" - meaning clinicians can see it's looking at the tumor.

- "Holds promise for clinical application in tumor grading" meaning it could help decide if a patient has a high-grade tumor which typically means more aggressive treatment like chemoradiation vs low-grade where maybe surgery alone might suffice or slower follow-up. This non-invasive grading via MRI plus AI could complement or sometimes delay the need for biopsy (though not replace histopathology yet, but possibly give early indication).

Dataset:

- They mention "55 gliomas excluded based on size or follow-up, remaining cases divided into train and validation cohorts". That suggests they had a dataset of around 300 patients maybe and excluded small ones. BraTS21 training had 125 LGG, 369 HGG = 494. They excluded 55 (small ones), got ~439 cases. Possibly used ~300 for training, ~139 for validation. AUC 0.968 on val is reported.
- The modalities used: T1, T1CE, T2, FLAIR for each patient.
- The segmentation masks are used to compute volumes etc. They say explicitly "four modalities are superimposed... on one side features extracted by 2D CNN, on other 14 3D features by Pyradiomics... combined to output graded prediction".
- It appears they effectively treat it as a multi-input model (one input is 4-channel image slices, another input is numeric features), and output a binary classification.
- Good that they mention separate validation, meaning results are not just cross-validated on same data – so ~0.9684 AUC is likely genuine generalization performance.

Results:

- **AUC = 0.9684** on validation. They focus on AUC as it's a composite measure. Possibly accuracy or F1 might be slightly lower (maybe around 93-95%) depending on threshold.
- They also mention in find result [60], there's "achieving an AUC of 0.9684 on validation", and possibly references in their discussion mention that without 3D features, AUC was lower (maybe they might have an ablation).
- They claim this addresses limitations of 2D and 3D separately and improves prediction.
- It's interesting they used SVM for classification (point [60†L7-L10] shows "inputted combined features into an SVM for classification, achieving AUC 0.9580" in a find snippet, which might be referencing their approach or prior approach). So maybe:
 - They possibly tried the combined features with SVM (and got AUC 0.9580).
 - Then tried their deep learning combined approach (maybe they let a small neural net learn from combined features) and got AUC 0.9684.
 - That indicates their end-to-end train of ResNet + MLP outperformed a pipeline where features were computed and fed to SVM by ~1% AUC.
- They emphasize model interpretability (with heatmaps) and maybe calibration (the paper might mention calibration error as they talk about uncertainty calibration in find [45], but that might be something else).
- The bottom line: Using **tumor volume and surface** as additional inputs gave a measurable boost in grading accuracy – it essentially gave the model a notion of tumor "size" explicitly, which correlates with grade.

In summary, Zeng *et al.* achieved robust **glioma grade classification** (low vs high grade) by **fusing deep features from multimodal MRI with explicit tumor size features**, reaching an AUC of ~ 0.97 . This hybrid 2D-3D approach effectively harnesses both imaging appearance and quantitative volume data, yielding a highly accurate and interpretable grading model that outperforms models using only images or only radiomics. The inclusion of tumor volume information was key to this improved performance

3.7 Comparative Analysis of Previous Works

This section compares ten recent deep learning-based approaches for brain tumor classification and segmentation, focusing on MRI modality usage, learning task, data dimensionality, ensemble design, and reported performance, and then positions the proposed method within this landscape.

3.7.1 Modalities and Task Formulation

Across the ten studies, single-modality classification on contrast-enhanced T1-weighted MRI is the most common setting. NeXtBrain, the ensemble CNN-ViT approach, the EfficientNet-based classifier, the rotation-invariant Vision Transformer, and the image-fusion DenseNet study all operate primarily on T1-weighted or T1-weighted contrast-enhanced MRI with one MRI sequence per subject. These works focus on tumor presence or multi-class tumor type classification in 2D images.

In contrast, a smaller subset exploits multimodal MRI with four sequences (T1, T1ce, T2, FLAIR). The AG-MS3D-CNN and XAI-MRI studies employ all four modalities for 3D brain tumor segmentation on BraTS-style data. On the classification side, BrainNet and the enhanced multimodal radiomics study also leverage all four modalities, with the latter explicitly combining multimodal CNN features and 3D tumor size descriptors for glioma grading. The automated framework with segmented and mask-aided pipelines uses single-modality (T1-CE) for 2D classification and incorporates multimodal BraTS data.

The proposed project follows the multimodal direction: it uses BraTS2018 with four MRI modalities (T1, T1ce, T2, FLAIR) and formulates a binary glioma grade classification task (HGG vs LGG) at the patient level.

3.7.2 Data Representation and Network Dimensionality

Most reviewed classification studies adopt 2D slice-based representations. NeXtBrain and the ensemble CNN-ViT framework operate on 2D contrasts, while the EfficientNet, rotation-invariant ViT, and DenseNet fusion models all learn from 2D slices. Even the enhanced multimodal radiomics study learns image features via a 2D CNN backbone. The automated classification framework also relies on 2D slices, although it integrates a 3D segmentation step.

Only a few works employ fully 3D convolutional architectures. AG-MS3D-CNN is a 3D U-Net variant trained on 3D volumes and multi-sequence input for segmentation, while XAI-MRI uses 3D U-Net-style models in dual-modality settings. The radiomics-enhanced work incorporates 3D tumor size features.

The proposed project makes systematic use of 3D information:

- ResNet50-3D and SwinUNETR-3D operate on preprocessed 3D volumes (128×128×128) with all modalities.
- DualStreamMIL-3D performs multiple-instance learning on informative slices selected from full 3D volumes.
- A stacked ensemble combines these models at the patient level.

3.7.3 Ensemble Strategies and Reported Performance

Ensembling is widely used but implemented in diverse ways. The ensemble CNN–ViT classifier trains multiple backbones and reports 85.03% accuracy. The automated framework combines segmentation masks, CNN features, and classical classifiers, achieving F1-scores near 99–100%. The DenseNet image-fusion study fuses features from two DenseNets, reporting accuracies over 99%. BrainNet stacks CNNs with autoencoders, obtaining ~98–99%. The enhanced multimodal radiomics method fuses CNN features with tumor volume measures, reporting AUC of 0.9684.

Segmentation-focused studies like AG-MS3D-CNN and XAI-MRI achieve high Dice scores (0.91 and 0.973, respectively) with and without ensembling.

The proposed project uses stacked ensembling:

- Logistic regression meta-learner trained on predictions from ResNet50-3D, SwinUNETR-3D, and DualStreamMIL-3D.
- Achieves AUC-ROC 0.9126, F1-score 0.8571, and accuracy 0.8105.
- DualStreamMIL-3D alone achieves AUC 0.9609 ± 0.0273 .

3.7.4 Strengths and Limitations Observed Across Studies

Strengths:

- High within-dataset performance.
- Effective use of ensembling and feature fusion.
- Growing adoption of multimodal MRI and 3D processing.

Limitations:

- Predominant reliance on 2D slices for classification.
- Limited external validation.
- Task separation between segmentation and grading.
- Sparse use of interpretability and uncertainty modeling.

3.7.5 Gaps in the Existing Literature

- Under-representation of 3D classification architectures.
- Incomplete multimodal integration.
- Limited interpretability.
- Lack of real-world heterogeneity handling.
- Lack of integrated segmentation-grading pipelines.

3.7.6 Positioning of the Proposed Method

The proposed project addresses several gaps:

- Uses ResNet50-3D and SwinUNETR-3D on 3D multimodal inputs.
- Introduces DualStreamMIL-3D with separate encoders and slice selection.
- Applies 5-fold patient-level cross-validation.
- Uses LDAM loss, re-weighting, and 3D preprocessing.

Limitations remain in terms of dataset generalizability and interpretability.

3.8 Chapter Summary (English Version)

3.8.1 Overview

This chapter reviewed key advances in brain tumor classification and segmentation using MRI, spanning from traditional methods to modern deep learning architectures. It analyzed prior studies in terms of modality usage, data dimensionality, model design, and ensemble strategies—highlighting gaps that the current project aims to fill.

3.8.2 General Trends

- **Predominant use of 2D representations** in classification tasks despite the availability of volumetric MRI.
- **Single-modality reliance**, typically T1 or T1ce, limiting the integration of complementary MRI sequences.
- **Widespread adoption of ensemble learning techniques** like bagging and stacking to improve robustness and accuracy.

3.8.3 Key Strengths

- High classification and segmentation performance on curated public datasets (>98%).
- Some works demonstrate effective multimodal integration.
- Initial steps toward clinical explainability using attention maps or radiomic features.

3.8.4 Limitations and Gaps

- **Underuse of full 3D classification architectures.**
- **Minimal external validation across diverse institutions or cohorts.**
- **Limited focus on class imbalance**, especially for underrepresented LGG tumors.
- **Segmentation and classification treated separately** in most pipelines.

3.8.5 Contribution of the Current Project

- **Multimodal 3D classification models:** ResNet50-3D, SwinUNETR-3D, and DualStreamMIL-3D.
- **Advanced modality fusion** using dual-stream MIL and stacked ensembling.
- **Explicit class imbalance handling** via LDAM loss and weighted sampling.
- **Rigorous patient-level evaluation** using 5-fold cross-validation with no data leakage.

3.8.6 Conclusion and Future Directions

This chapter clarified how the current research addresses several major gaps through 3D multimodal learning, ensemble integration, and imbalance-aware training. Future work should expand toward external clinical validation, unified segmentation–classification pipelines, and improved model interpretability to enhance deployment readiness.

CHAPTER 4: DATASET AND PREPROCESSING

4.1 Introduction

This chapter outlines the dataset employed in this project and details the preprocessing steps used to prepare the data for training and evaluation. The procedures described reflect the actual implementation in the project and ensure that the data is consistent, normalized, and suitable for input to deep learning models.

4.2 Dataset Description

4.2.1 Source and Task Definition

The dataset used is the BraTS 2018 training set, which includes multi-modal MRI scans for brain tumor patients. The classification task involves binary grading of glioma tumors into:

- High Grade Glioma (HGG) — labeled as 1
- Low Grade Glioma (LGG) — labeled as 0

4.2.2 Modalities and Format

Each subject includes four MRI modalities: T1, T1ce, T2, and FLAIR. The data is provided as 3D volumes in NIfTI format (.nii/.nii.gz). Segmentation masks are excluded from this classification task.

4.2.3 Dataset Organization

The dataset is structured by class (HGG and LGG), with each subject's directory containing all four required modalities. Only subjects with complete modalities were included in the final dataset index used for training.

4.3 Data Exploration and Indexing

Following preprocessing, a structured metadata index is generated containing subject IDs, class labels, and relative paths to the four modalities. This indexing step ensures consistent data retrieval and filtering during training and validation.

4.4 Preprocessing Pipeline

The preprocessing was performed in four sequential stages:

4.4.1 N4 Bias Field Correction

Corrects intensity non-uniformity caused by field inhomogeneities. A brain mask is estimated using Otsu thresholding, and correction is applied to each modality independently. Negative values are clamped to zero.

4.4.2 Z-Score Normalization

Voxel intensities within the brain region are normalized using Z-score statistics to ensure consistent intensity distribution across subjects. Background voxels are preserved as zero.

4.4.3 Region of Interest (ROI) Cropping

A 3D bounding box is computed from the FLAIR modality to encompass the brain region. This box is applied to all modalities of the same subject, ensuring spatial alignment and reducing irrelevant background.

4.4.4 Resampling to Uniform Shape

All cropped volumes are resized to a uniform spatial shape of $128 \times 128 \times 128$ voxels using linear interpolation. This standardization enables batch processing in 3D convolutional networks.

4.5 Data Augmentation

During training, runtime data augmentation is applied using random spatial transformations such as rotation, flipping, zooming, and translation. These augmentations improve generalization and robustness by simulating variability in imaging.

4.6 Class Imbalance Handling (Runtime Oversampling)

To address class imbalance between HGG and LGG samples, a weighted random sampling strategy is applied during training. Samples from the minority class are assigned higher selection probability using inverse class frequency weights. This results in effective oversampling of underrepresented cases without duplicating data on disk.

4.7 Data Splitting Strategy

A 5-fold stratified cross-validation protocol is employed to ensure reliable performance evaluation. The dataset is split at the patient level, maintaining balanced class distributions in each fold while preventing data leakage. The split metadata is saved for reproducibility.

4.8 Chapter Summary

This chapter provided a detailed overview of the dataset and preprocessing workflow. The pipeline includes bias correction, normalization, spatial cropping, and uniform resizing, followed by augmentation and runtime oversampling during training. The evaluation strategy is based on patient-level 5-fold cross-validation to ensure robust and reproducible model assessment.

CHAPTER 5: PROPOSED METHODOLOGY

5.1 Introduction

The literature review presented in Chapter 3 revealed several critical gaps in existing approaches to brain tumor classification from MRI data. While numerous studies have demonstrated promising results, they predominantly rely on 2D architectures that discard volumetric information, single-modality inputs that underutilize the rich multimodal nature of MRI, and individual models that lack the robustness needed for clinical deployment. Furthermore, the limited adoption of 3D convolutional architectures and the scarcity of ensemble methods that effectively combine diverse architectural paradigms represent significant opportunities for methodological advancement.

This chapter presents a comprehensive methodology that directly addresses these limitations through an integrated ensemble system comprising three complementary deep learning architectures: ResNet50-3D, SwinUNETR-3D, and DualStreamMIL-3D. Rather than treating these models as isolated components, the proposed approach leverages their complementary strengths—volumetric feature extraction, long-range dependency modeling, and interpretable slice-level reasoning—within a unified framework that combines their predictions through a learned meta-learner. This ensemble strategy not only addresses the limitations identified in prior work but also provides a principled approach to handling the inherent challenges of medical image classification: class imbalance, limited training data, and the need for robust generalization.

The methodology operates on preprocessed 3D MRI volumes ($128 \times 128 \times 128$ voxels) from four modalities (T1, T1ce, T2, FLAIR), employing patient-level 5-fold stratified cross-validation to ensure robust evaluation. Each base model is trained independently with class-balanced sampling, geometric augmentation, and early stopping, while their out-of-fold predictions are aggregated to train a logistic regression meta-learner that optimally combines their diverse perspectives. This integrated approach transforms the challenge of glioma grading from a single-model optimization problem into a systematic exploration of complementary architectural paradigms, ultimately producing a more robust and clinically viable classification system.

5.2 System Architecture: An Integrated Ensemble Framework

The proposed system is fundamentally an ensemble architecture that combines three distinct deep learning paradigms, each addressing different aspects of the brain tumor classification problem. Rather than selecting a single "best" architecture, this approach recognizes that different models capture complementary information: volumetric CNNs excel at extracting spatial features from 3D structures, transformers capture long-range dependencies through self-

attention, and multiple instance learning provides interpretable reasoning at the slice level while handling weakly labeled data effectively.

5.2.1 Complementary Model Selection

The selection of ResNet50-3D, SwinUNETR-3D, and DualStreamMIL-3D was motivated by their complementary strengths and their ability to address distinct limitations observed in prior studies:

ResNet50-3D serves as the volumetric CNN baseline, leveraging pretrained weights from MedicalNet to transfer knowledge from diverse medical imaging datasets. This model processes full 3D volumes through early fusion of all four modalities, capturing spatial relationships across the entire brain volume. Its strength lies in extracting hierarchical volumetric features through 3D convolutions, making it particularly effective at identifying tumor characteristics that manifest across multiple spatial scales.

SwinUNETR-3D introduces transformer-based reasoning to the ensemble, using the Swin Transformer encoder to capture long-range dependencies that may be missed by local convolutional operations. Unlike ResNet50-3D's fixed receptive fields, the self-attention mechanism allows the model to relate distant brain regions, potentially capturing subtle patterns in tumor morphology or surrounding tissue changes. This architectural diversity addresses the limitation of purely convolutional approaches in modeling global context.

DualStreamMIL-3D adopts a fundamentally different perspective by treating each patient as a bag of 2D slices (instances) rather than processing the full 3D volume directly. This approach provides two key advantages: first, it enables interpretable identification of critical slices that drive the classification decision, addressing the "black box" limitation of deep learning models; second, it naturally handles the weakly supervised nature of patient-level labels by learning to aggregate instance-level evidence. The dual-stream design—combining critical instance selection with contextual attention—ensures that the model focuses on informative slices while maintaining awareness of their spatial context.

5.2.2 Ensemble Integration Strategy

The integration of these three models follows a stacking approach, where a meta-learner learns to optimally combine their predictions. This design choice is motivated by the observation that different models may excel on different subsets of patients or capture complementary aspects of

the classification problem. Rather than simple averaging or voting, the logistic regression meta-learner learns model-specific weights that reflect each base model's contribution to the final decision, as evidenced by the learned coefficients (SwinUNETR-3D: 4.06, DualStreamMIL-3D: 0.89, ResNet50-3D: 0.54) that indicate their relative importance.

The ensemble training process ensures that each base model contributes unique information: out-of-fold predictions are collected from validation sets across all folds, guaranteeing that the meta-learner learns from predictions made on data that was not used to train the corresponding base model. This prevents overfitting and ensures that the ensemble generalizes to unseen patients. The final system thus combines the volumetric reasoning of ResNet50-3D, the global context modeling of SwinUNETR-3D, and the interpretable slice-level analysis of DualStreamMIL-3D into a unified decision framework.

5.3 Deep Learning Architectures

5.3.1 ResNet50-3D: Volumetric Feature Extraction

ResNet50-3D represents the volumetric CNN paradigm, adapted from the 2D ResNet50 architecture to process 3D medical volumes. The model's design philosophy centers on hierarchical feature extraction through residual connections, enabling effective training of deep networks while preserving gradient flow. For brain tumor classification, this architecture is particularly well-suited because tumors exhibit multi-scale characteristics—from fine-grained texture differences to gross morphological changes—that require features at multiple resolutions.

The architecture processes multi-modal input through early fusion: all four MRI modalities (T1, T1ce, T2, FLAIR) are stacked as channels (batch size, 4, 128, 128, 128), allowing the model to learn cross-modal relationships directly through shared convolutional filters. The initial convolution layer (Conv3d (4, 64, kernel size=7, stride=2, padding=3)) extracts low-level features across all modalities simultaneously, followed by max pooling that reduces spatial dimensions while preserving the most salient features.

The core of the model consists of four residual stages using Bottleneck3D blocks with configurations [3, 4, 6, 3] layers per stage. Each bottleneck block employs a 1×1×1 convolution for dimensionality reduction, a 3×3×3 convolution for spatial feature extraction, and another 1×1×1 convolution for expansion, with residual connections that enable training of very deep networks. This design allows the model to progressively extract features from local patterns (early layers) to global structures (later layers), making it effective at identifying tumor characteristics that span multiple spatial scales.

A critical advantage of this implementation is its compatibility with MedicalNet pretrained weights, which were trained on 23 diverse medical imaging datasets. When pretrained weights are used with multi-modal input, the first convolutional layer weights are adapted from single-channel to four-channel input using mean replication: pretrained filters are replicated across all modality channels and averaged, preserving learned feature extraction patterns while accommodating multi-modal data. This transfer learning approach addresses the data scarcity challenge common in medical imaging by leveraging knowledge from related domains.

The model supports differential learning rates for the backbone (default: $1e-4$) and classifier head (default: $2e-4$), enabling fine-tuning strategies where pretrained features are adapted conservatively while the randomly initialized classification head learns more rapidly. Global average pooling `AdaptiveAvgPool3d((1, 1, 1))` reduces spatial dimensions to a single feature vector, followed by dropout (default rate: 0.5) and a fully connected layer that produces binary classification logits.

Rationale for Volumetric Processing: Unlike 2D approaches that process individual slices independently, ResNet50-3D maintains spatial relationships across all three dimensions, enabling it to capture volumetric tumor characteristics such as shape irregularity, infiltration patterns, and relationships between tumor regions and surrounding brain structures. This volumetric reasoning is essential for accurate glioma grading, as tumor grade is determined not only by appearance in individual slices but by three-dimensional morphological characteristics.

5.3.2 SwinUNETR-3D: Transformer-Based Global Context Modeling

SwinUNETR-3D introduces transformer-based architecture to the ensemble, addressing the limitation of convolutional networks in modeling long-range dependencies. While ResNet50-3D excels at extracting local spatial features through convolutional operations, its receptive field is inherently limited by the depth of the network and kernel sizes. The Swin Transformer encoder, in contrast, uses self-attention mechanisms to relate any pair of voxels in the volume, enabling the model to capture global patterns that may be crucial for tumor classification.

The architecture extracts the encoder portion from MONAI's SwinUNETR implementation, originally designed for segmentation tasks, and adapts it for classification by replacing the segmentation decoder with global pooling and a classification head. This design choice leverages the transformer's ability to model relationships between distant brain regions—for example, correlating tumor appearance in one hemisphere with contralateral effects or identifying subtle changes in surrounding tissue that may indicate tumor grade.

The Swin Transformer processes input volumes through a patch-based approach: volumes are divided into non-overlapping patches of size $2 \times 2 \times 2$ voxels, which are then linearly embedded and processed through multiple transformer stages. The architecture uses a hierarchical design with four stages, each with depth configuration $[2, 2, 2, 2]$ layers and increasing numbers of attention heads $[3, 6, 12, 24]$. This progressive expansion of attention heads allows the model to capture increasingly complex relationships as features flow through the network.

A key innovation of the Swin Transformer is its window-based self-attention mechanism: rather than computing attention over all patches simultaneously (which would be computationally prohibitive for 3D volumes), attention is computed within local windows of size $7 \times 7 \times 7$, with window shifting between layers to enable cross-window communication. This design balances computational efficiency with the ability to model long-range dependencies, making it feasible to apply transformer architectures to volumetric medical data.

The model uses a memory-efficient configuration with base feature size 48 (compared to typical values of 96 or 192), resulting in an encoder output dimension of 768 ($\text{feature_size} * (2^{**} \text{num_stages}) = 48 * 16$). Global average pooling reduces spatial dimensions, followed by a classification head that can be configured as either a single-layer ($\text{Dropout}(0.4) \rightarrow \text{Linear}(768, 2)$) or two-layer network with a hidden dimension of 256. The default configuration uses the single-layer head, prioritizing simplicity and reducing overfitting risk.

rationale for Transformer Architecture: The self-attention mechanism enables the model to identify relationships between spatially distant regions that may be diagnostically relevant—for instance, correlating tumor characteristics in one brain region with effects in another, or identifying subtle patterns that span large portions of the volume. This global context modeling complements the local feature extraction of ResNet50-3D, providing the ensemble with a different perspective on the classification problem. Furthermore, the transformer's ability to weight different regions adaptively allows it to focus on the most informative parts of the volume, potentially identifying subtle indicators of tumor grade that convolutional networks might miss.

5.3.3 DualStreamMIL-3D: Interpretable Slice-Level Reasoning

DualStreamMIL-3D adopts a fundamentally different approach by treating brain tumor classification as a multiple instance learning (MIL) problem, where each patient (bag) contains multiple 2D slices (instances) and only the patient-level label is available. This paradigm shifts addresses two critical limitations of volumetric approaches: first, it provides interpretability by

identifying which slices drive the classification decision; second, it naturally handles the weakly supervised nature of patient-level labels by learning to aggregate instance-level evidence.

The model processes each patient volume by extracting 2D slices along the axial axis, with all four modalities stacked as channels in each slice $(N, 4, H, W)$ where N is the number of slices. This slice-based representation enables the model to leverage 2D CNN architectures (ResNet18 or EfficientNet-B0) that are highly optimized and pretrained on large-scale natural image datasets, while still incorporating multi-modal information through early fusion at the slice level.

The architecture employs a dual-stream design that combines two complementary aggregation strategies:

Stream 1: Critical Instance Selection identifies the most informative slice(s) in each patient through a learned scoring network. The scoring network assigns importance scores to each slice using a two-layer MLP ($\text{Linear}(\text{feature_dim}, 128) \rightarrow \text{ReLU}() \rightarrow \text{Dropout}(0.2) \rightarrow \text{Linear}(128, 1) \rightarrow \text{Sigmoid}()$), producing scores in the range $[0, 1]$. Rather than hard selection (choosing only the highest-scored slice), the model uses soft selection with temperature-scaled Softmax, creating a differentiable weighted combination of instance features. This design enables gradient flow through the selection mechanism while allowing the model to focus on the most critical slices. The temperature parameter (default: 1.0) can be annealed during training, implementing a curriculum learning strategy where the model initially explores all slices uniformly and gradually sharpens its focus.

Stream 2: Contextual Aggregator complements critical instance selection by aggregating information from all slices using attention mechanisms. The attention is conditioned on the critical instance feature from Stream 1, allowing the model to identify slices that support or complement the critical instance. The default attention mechanism uses gated attention, where query-key similarity is combined with learned gate weights, providing a flexible mechanism for weighting instances based on their relationship to the critical slice. Alternatively, cosine similarity-based attention can be used, which computes normalized dot products between the critical instance and all instances.

The two streams are fused through concatenation (default), weighted combination, or cross-attention, producing a combined feature representation that captures both the most informative slices and their contextual relationships. The classification head uses a two-layer design ($\text{Dropout}(0.4) \rightarrow \text{Linear}(\text{fused_dim}, 256) \rightarrow \text{ReLU}() \rightarrow \text{Dropout}(0.4) \rightarrow \text{Linear}(256, 2)$) with dropout rate 0.4, providing sufficient capacity to learn complex decision boundaries while maintaining regularization.

Rationale for Slice-Based MIL Approach: The 2D slice-based approach offers several advantages over direct 3D processing. First, it enables interpretability: the model can identify which slices are most informative for classification, providing clinicians with insights into the decision-making process. Second, it leverages pretrained 2D CNNs that have been trained on millions of natural images, transferring rich feature representations to the medical domain. Third, it naturally handles variable-length sequences (different numbers of slices per patient) and can focus computational resources on the most informative regions. Finally, the MIL framework is well-suited for weakly supervised learning scenarios where only patient-level labels are available, learning to aggregate instance-level evidence without requiring slice-level annotations.

The dual-stream design addresses a common limitation of MIL approaches: over-reliance on a single critical instance. By combining critical instance selection (which identifies the strongest signal) with contextual aggregation (which incorporates supporting evidence), the model avoids the risk of making decisions based solely on potentially noisy or misleading slices. This design ensures robust patient-level predictions that consider both the most informative slices and their spatial context.

5.4 Unified Training Strategy

While the three base models employ different architectures, they share a common training strategy designed to ensure fair comparison, robust generalization, and effective ensemble integration. This unified approach addresses the challenges of medical image classification—limited data, class imbalance, and the need for robust evaluation—through systematic application of best practices.

5.4.1 Patient-Level Cross-Validation

All three models are trained using 5-fold stratified cross-validation at the patient level, ensuring that all images from a single patient remain in the same fold. This patient-level splitting is critical for preventing data leakage: if slices or volumes from the same patient appeared in both training and validation sets, performance metrics would be artificially inflated due to the model's ability to memorize patient-specific characteristics rather than learning generalizable tumor features.

The cross-validation splits are generated using ``StratifiedKFold (n_splits=5, shuffle=True, random_state=42) ``, which preserves the class distribution (HGG: LGG ratio $\approx 2.8:1$) in each

fold. This stratification ensures that each fold is representative of the overall dataset, preventing scenarios where one fold contains predominantly one class, which would lead to unreliable performance estimates. The fixed random seed (42) ensures reproducibility across all experiments.

For each fold, the training process follows a consistent protocol: models are trained on the training set, validated on the validation set, and out-of-fold predictions are collected from the validation set. These OOF predictions serve dual purposes: they enable fair comparison of model performance across folds, and they provide the training data for the ensemble meta-learner. Model selection within each fold uses the checkpoint with the highest validation AUC, ensuring that the best-performing model for each fold is used for ensemble construction.

5.4.2 Addressing Class Imbalance

The dataset exhibits significant class imbalance (HGG: 210 cases, LGG: 75 cases, ratio $\approx 2.8:1$), which can lead to models that are biased toward the majority class. The proposed methodology addresses this imbalance through runtime oversampling using `WeightedRandomSampler`, which ensures that minority class samples (LGG) are sampled more frequently during training.

Class weights are computed using inverse frequency weighting:

(Equation 4)

where N is the total number of samples, K is the number of classes, n_c is the count of class c , and $\epsilon=10^{-6}$ prevents division by zero. This formula assigns higher weights to minority class samples, ensuring that each class contributes approximately equally to the training process. The sampler is configured with `replacement=True`, enabling oversampling of the minority class within each epoch.

This data-level balancing approach is preferred over loss-level reweighting because it ensures that the model sees a balanced distribution of classes during training, rather than simply penalizing misclassifications of minority class samples more heavily. The approach is compatible with gradient accumulation and mixed-precision training, maintaining computational efficiency while addressing class imbalance effectively.

5.4.3 Geometric Data Augmentation

Geometric data augmentation is applied on-the-fly during training using MONAI transforms, increasing data diversity without requiring storage of augmented volumes. The augmentation pipeline includes random rotation ($\pm 11.5^\circ$ around each axis), random flips along each spatial axis (probability 0.5 per axis), random zoom ($\pm 8\%$ scaling), and random translation ($\pm 8\%$ of volume size). These transformations preserve the spatial relationships between modalities while introducing geometric variations that improve model robustness to patient positioning, scanner differences, and anatomical variations.

The augmentation strategy is designed to be realistic: rotation angles are limited to $\pm 11.5^\circ$ to avoid unrealistic brain orientations, and zoom factors are constrained to $\pm 8\%$ to prevent excessive distortion. The overall probability of applying augmentation is 0.6, meaning that 60% of training samples are augmented while 40% remain in their original form. This balance ensures that the model sees both augmented and original data, preventing over-reliance on augmented samples.

For validation and test sets, no spatial augmentation is applied, ensuring that performance metrics reflect the model's ability to classify unmodified volumes. This approach maintains the integrity of the evaluation while maximizing training data diversity.

5.4.4 Loss Functions and Optimization

All three models use `CrossEntropyLoss` (PyTorch `nn.CrossEntropyLoss`) for training, a choice made after empirical evaluation showed that alternative loss functions (e.g., LDAM with deferred re-weighting) caused training instability in volumetric CNN models. The cross-entropy loss provides stable gradients and is well-suited for binary classification tasks, while class balancing is handled at the data level through `WeightedRandomSampler` rather than loss-level reweighting.

Optimization strategies are tailored to each model's architecture and training dynamics:

- **ResNet50-3D** uses the Adam optimizer with learning rates of $1e-4$ for the backbone and $2e-4$ for the classifier head, enabling conservative fine-tuning of pretrained features while allowing rapid adaptation of the randomly initialized classifier.

- **SwinUNETR-3D** employs AdamW (weight-decay decoupled) with learning rates of $5e-5$ for the encoder and $1e-4$ for the classifier head. The lower encoder learning rate reflects the

transformer's sensitivity to learning rate and the need for careful fine-tuning of attention mechanisms.

- **DualStreamMIL-3D** also uses AdamW with learning rates of $5e-5$ for the instance encoder and $1e-4$ for the classifier head, with increased weight decay ($5e-4$) to provide stronger L2 regularization for the MIL aggregation mechanisms.

All models use cosine annealing with warmup for learning rate scheduling: a linear warmup phase for the first $\max(5, \text{epochs} // 10)$ epochs (minimum 5 epochs) is followed by cosine annealing that decays the learning rate smoothly to zero. This schedule provides stable training in early epochs while enabling fine convergence in later epochs. Differential scheduling is applied to backbone/encoder and classifier head parameter groups, allowing independent learning rate decay for different components of the model.

5.4.5 Early Stopping and Model Selection

Early stopping prevents overfitting by monitoring validation performance and stopping training when no improvement is observed for a specified number of epochs. The strategy uses validation AUC-ROC as the primary metric (reflecting the model's discriminative ability) with F1-score as a tie-breaker (ensuring balanced precision and recall). Patience values are set to 7 epochs for ResNet50-3D and SwinUNETR-3D, and 10 epochs for DualStreamMIL-3D, reflecting the different convergence dynamics of these architectures.

Minimum epoch requirements (10 for ResNet50-3D and SwinUNETR-3D, 15 for DualStreamMIL-3D) ensure that models train for a sufficient duration before early stopping can trigger, preventing premature termination during initial training phases. Model checkpoints are saved when validation AUC improves (or F1 improves if AUC is tied), ensuring that the best-performing model for each fold is preserved.

Exponential Moving Average (EMA) is employed for ResNet50-3D and SwinUNETR-3D with decay rate 0.995, maintaining a smoothed version of model parameters that often generalizes better than the final training parameters. EMA checkpoints are saved separately, providing an additional model variant for ensemble consideration.

5.4.6 Computational Efficiency Techniques

Several techniques are employed to ensure efficient training while maintaining model performance:

- **Mixed Precision Training:** PyTorch Automatic Mixed Precision (AMP) reduces memory usage and accelerates training on compatible GPUs by using FP16 for forward passes while maintaining FP32 precision for gradient accumulation.

- **Gradient Accumulation:** ResNet50-3D and SwinUNETR-3D use gradient accumulation (default: 2 steps) to simulate larger batch sizes without increasing memory requirements, improving training stability and convergence.

- **Gradient Clipping:** Optional gradient clipping (default: disabled) can be enabled to prevent gradient explosion, though it is not required for stable training with the current configurations.

These techniques enable training of large 3D models on limited GPU memory while maintaining training dynamics comparable to larger batch sizes, making the methodology accessible to a wider range of computational resources.

5.5 Ensemble Framework: Learning to Combine Predictions

5.5.1 Out-of-Fold Prediction Collection

The ensemble framework relies on out-of-fold (OOF) predictions—predictions made by each base model on validation sets from folds where that model was not trained. This approach ensures that the meta-learner learns from predictions made on data that was not used to train the corresponding base model, preventing overfitting and ensuring robust ensemble generalization.

The OOF prediction collection process operates systematically: for each model and each fold, validation predictions are loaded from the model's checkpoint directory (`run_dir/predictions/val_probs.npy`), and HGG probabilities (class 1 probabilities) are extracted. These probabilities are matched with patient IDs from the corresponding validation

split CSV file, ensuring correct patient-prediction associations. Predictions from all five folds are concatenated to create a complete OOF prediction set for each model.

Comprehensive verification ensures data integrity: patient ID uniqueness is checked (each patient must appear exactly once), all five folds must be represented, probability values must be in the valid range [0, 1], and label consistency is verified. This rigorous validation prevents errors that could compromise ensemble training.

The final step merges OOF predictions from all three models on patient ID, creating a unified dataset where each patient has three features: ``hgg_prob_resnet``, ``hgg_prob_swin``, and ``hgg_prob_mil``. This merged dataset (``ensemble/oof_predictions/merged_oof_predictions.csv``) contains 285 patients with their corresponding ground truth labels, serving as the training data for the meta-learner.

5.5.2 Meta-Learner Design and Training

The meta-learner is a Logistic Regression classifier (scikit-learn ``LogisticRegression``) that learns to optimally combine base model predictions. This choice is motivated by several factors: first, logistic regression is interpretable, allowing examination of learned coefficients to understand each model's contribution; second, with only three features and 285 samples, a simple linear model is sufficient and reduces overfitting risk; third, logistic regression naturally handles probability inputs and produces calibrated probability outputs.

The model is configured with L2 regularization ($C=1.0$), balanced class weights to address remaining imbalance, and the LBFGS solver, which is efficient for small datasets with few features. Training proceeds by extracting the three HGG probability features and the binary label, fitting the logistic regression model, and evaluating performance on the same dataset (standard practice for stacking, where OOF predictions serve as both training and evaluation data).

The learned coefficients reveal the relative importance of each base model: SwinUNETR-3D has the highest coefficient (4.06), indicating it is the most important contributor to ensemble predictions, followed by DualStreamMIL-3D (0.89) and ResNet50-3D (0.54). This ranking reflects the complementary strengths of the models: SwinUNETR-3D's global context modeling appears most valuable, while ResNet50-3D's volumetric features, though important, contribute less to the final decision. The negative intercept (-2.40) indicates a bias toward the negative class (LGG) in base model predictions, which the meta-learner corrects.

5.5.3 Ensemble Inference

For inference on new patients, the ensemble operates as follows: each base model generates predictions on the input volume, producing HGG probabilities. These three probabilities are concatenated into a feature vector and fed to the meta-learner, which produces the final ensemble prediction and probability. This two-stage process combines the diverse perspectives of the three base models through learned weights, producing more robust predictions than any individual model.

The ensemble framework thus transforms the challenge of selecting a single best model into a systematic approach for combining complementary architectures, leveraging their diverse strengths while mitigating their individual limitations. This approach addresses the fundamental uncertainty in model selection by acknowledging that different models may excel on different patients or capture complementary aspects of the classification problem.

5.6 Evaluation Metrics

The evaluation framework employs comprehensive metrics to assess model performance from multiple perspectives, recognizing that different metrics capture different aspects of classification quality. All metrics are computed for both individual base models and the ensemble, enabling systematic comparison and analysis.

5.6.1 Classification Performance Metrics

Accuracy

(Equation 1)

measures the overall proportion of correctly classified patients, providing a simple summary of performance. However, accuracy can be misleading in imbalanced datasets, as it may be dominated by majority class performance.

Precision and Recall (macro-averaged)

(Equation 2)· (Equation 3)

provide class-balanced measures of performance. Precision measures the proportion of positive predictions that are correct, while recall measures the proportion of positive cases that are identified. Macro-averaging ensures that both classes contribute equally to the metric, preventing majority class dominance.

F1-Score (macro-averaged)

(Equation 4)

is the harmonic mean of precision and recall, providing a single metric that balances both aspects of performance. It is particularly useful as a tie-breaker in model selection when AUC values are equal.

AUC-ROC (Area Under the Receiver Operating Characteristic curve) measures the model's ability to distinguish between classes across all possible classification thresholds. It is the primary metric for model selection (early stopping) because it is threshold-independent and provides a comprehensive measure of discriminative ability. AUC values range from 0.5 (random performance) to 1.0 (perfect discrimination), with values above 0.9 indicating excellent performance.

5.6.2 Diagnostic Metrics

Confusion Matrix provides a detailed breakdown of classification performance, showing true positives, true negatives, false positives, and false negatives. This matrix enables identification of specific failure modes—for example, whether the model tends to misclassify LGG as HGG or vice versa.

Classification Report extends the confusion matrix by providing per-class precision, recall, F1-score, and support (sample counts), enabling detailed analysis of performance for each class individually.

ROC Curve plots true positive rate against false positive rate at various thresholds, visualizing the trade-off between sensitivity and specificity. The area under this curve (AUC-ROC) quantifies overall discriminative ability.

Precision-Recall Curve plots precision against recall at various thresholds, providing an alternative visualization that is particularly informative for imbalanced datasets where the positive class (HGG) is more common.

5.6.3 Metric Computation and Storage

Implementation: `compute_metrics()` function in each training script

Metrics are computed using scikit-learn functions (`accuracy_score`, `precision_score`, `recall_score`, `f1_score`, `roc_auc_score`, `confusion_matrix`, `classification_report`), with zero-division handling to prevent errors when a class has no predictions. Binary predictions use the default threshold of 0.5 (argmax of probabilities), though the probability outputs enable threshold tuning if needed.

During training, metrics are computed at each validation epoch and stored in training history, enabling analysis of learning dynamics and convergence behavior. Final evaluation uses the best model checkpoint (highest validation AUC) evaluated on the validation set, with metrics saved to `metrics/metrics.json` for persistence and analysis. Visualization plots (confusion matrix, ROC curve, PR curve, learning curves) are saved to the `plots/` directory, providing intuitive representations of model performance.

This comprehensive evaluation framework ensures that model performance is assessed from multiple angles, enabling identification of strengths and weaknesses and supporting informed decisions about model selection and deployment.

5.7 Chapter Summary

This chapter presented a comprehensive methodology for brain tumor grade classification that addresses the limitations identified in prior studies through an integrated ensemble system. The approach combines three complementary deep learning architectures—ResNet50-3D, SwinUNETR-3D, and DualStreamMIL-3D—each contributing unique strengths to the classification problem.

Architectural Diversity: ResNet50-3D provides robust volumetric feature extraction through 3D convolutions and transfer learning from MedicalNet, capturing multi-scale spatial patterns essential for tumor characterization. SwinUNETR-3D introduces transformer-based global context modeling, enabling the capture of long-range dependencies that convolutional networks may miss. DualStreamMIL-3D offers interpretable slice-level reasoning through multiple instance learning, identifying critical slices while maintaining contextual awareness.

Unified Training Strategy: All models are trained using patient-level 5-fold stratified cross-validation, ensuring robust evaluation and preventing data leakage. Class imbalance is addressed through runtime oversampling via `WeightedRandomSampler`, while geometric augmentation increases data diversity. CrossEntropyLoss with differential learning rates enables effective fine-tuning, and cosine annealing with warmup provides smooth convergence. Early stopping based on validation AUC prevents overfitting, while computational efficiency techniques (mixed precision, gradient accumulation) enable training on limited resources.

Ensemble Integration: The ensemble framework combines base model predictions through a learned logistic regression meta-learner, trained on out-of-fold predictions to ensure robust generalization. The meta-learner learns model-specific weights that reflect each architecture's contribution, with learned coefficients revealing SwinUNETR-3D as the most important contributor, followed by DualStreamMIL-3D and ResNet50-3D.

Comprehensive Evaluation: Performance is assessed through multiple metrics (accuracy, precision, recall, F1-score, AUC-ROC) with detailed diagnostic tools (confusion matrix, ROC/PR curves), enabling thorough analysis of model behavior and identification of failure modes.

The methodology transforms brain tumor classification from a single-model optimization problem into a systematic exploration of complementary architectural paradigms, ultimately producing a robust ensemble system that leverages the strengths of volumetric CNNs, transformers, and multiple instance learning. This integrated approach addresses the gaps identified in prior work—limited use of 3D architectures, reliance on single models, and lack of interpretability—while providing a principled framework for combining diverse deep learning paradigms in medical image classification.

CHAPTER 6: EXPERIMENTAL RESULTS AND DISCUSSION

6.1 Introduction

This chapter presents the experimental results obtained from training and evaluating the proposed ensemble system for brain tumor grade classification. The evaluation follows the methodology outlined in Chapter 5, employing patient-level 5-fold stratified cross-validation across all three base models (ResNet50-3D, SwinUNETR-3D, and DualStreamMIL-3D) and their ensemble combination via a logistic regression meta-learner. Results are reported on out-of-fold (OOF) predictions collected from validation sets across all folds, ensuring that performance metrics reflect generalization to unseen patients rather than training set memorization.

The chapter is organized as follows: Section 6.2 presents individual model performance analysis, focusing on the DualStreamMIL-3D model for which comprehensive per-fold results are available. Section 6.3 details the ensemble meta-learner performance, including learned combination weights and overall metrics. Section 6.4 provides comparative analysis of model predictions and their distributions. Section 6.5 discusses the results in the context of the research objectives, identifies limitations, and suggests directions for future work. Section 6.6 summarizes the key findings.

6.2 Individual Model Performance Analysis

6.2.1 DualStreamMIL-3D: Per-Fold Cross-Validation Results

Comprehensive per-fold results are available for the **DualStreamMIL-3D** model, providing detailed insights into training stability and performance variability across the five cross-validation folds. The model was trained using early stopping based on the validation AUC, and the final evaluation relied on exponential moving average (EMA) checkpoints selected at the best-performing epoch of each fold.

Table (1) summarizes the cross-validation performance of the DualStreamMIL-3D model, reported as mean $\pm$ standard deviation across the five folds, along with the minimum and maximum values observed.

The model demonstrates strong discriminative ability with a mean AUC of 0.9613, indicating excellent separation between HGG and LGG classes. However, substantial variability is observed in F1-score (standard deviation: 0.0487) and accuracy (standard deviation: 0.0502), suggesting inconsistent performance across folds that may be attributed to class imbalance effects and training instability.

Per-Fold Performance Details:

Fold 0 (run_20260109_032250):

- Best Epoch: 12 (selected based on Best Validation AUC: 0.9095)
- Best Validation AUC: 0.9095
- Final Metrics: AUC=0.9095, F1=0.7689, Accuracy=0.7895
- Confusion Matrix: LGG: 15/15 correct (100%), HGG: 3/42 correct (7.1%)
- **Issue:** Severe class imbalance—model heavily biased toward LGG predictions
- **Visualizations:** The confusion matrix at the best validation epoch (epoch 12) reveals a strong bias toward LGG predictions, with very limited sensitivity to HGG cases (**Figure 1**). The learning curves illustrate a clear divergence between training and validation performance, indicating overfitting behavior early during training (**Figure 2**). This trend is further supported by the loss curve, which shows a continuous decrease in training loss while validation loss remains unstable (**Figure 3**). The ROC curve corresponding to the best epoch demonstrates moderate discriminative ability (AUC = 0.9095) despite the severe class imbalance observed in the confusion matrix (**Figure 4**).

Fold 1 (run_20260109_145215):

- Best Epoch: 22 (selected based on Best Validation AUC: 0.9825)
- Best Validation AUC: 0.9825
- Final Metrics: AUC=0.9825, F1=0.8505, Accuracy=0.8947
- Confusion Matrix: LGG: 9/15 correct (60.0%), HGG: 38/42 correct (90.5%)
- **Observation:** High AUC with improved balance compared to other folds
- **Visualizations:** The confusion matrix at the best validation epoch (epoch 22) shows a more balanced classification behavior compared to other folds, with high sensitivity to HGG cases and moderate performance on LGG samples (**Figure 5**). The learning curves indicate improved generalization relative to Fold 0, although a noticeable gap between training and validation performance is still present (**Figure 6**). The loss curve suggests partial stabilization during later epochs, yet mild overfitting remains evident (**Figure 7**). The ROC curve demonstrates excellent discriminative performance with a high AUC value of 0.9825, confirming the model's strong separation capability for this fold (**Figure 8**).

Fold 2 (run_20260109_150832):

- Best Epoch: 14 (selected based on Best Validation AUC: 0.9714)

- Best Validation AUC: 0.9714
- Final Metrics: AUC=0.9714, F1=0.8697, Accuracy=0.8947
- Confusion Matrix: LGG: 15/15 correct (100%), HGG: 9/42 correct (21.4%)
- **Observation:** High AUC but severe class imbalance—model biased toward LGG predictions
- **Visualizations:** The confusion matrix for the best validation epoch (epoch 14) reveals perfect classification of LGG cases but substantially reduced sensitivity to HGG samples, indicating a strong class bias (**Figure 9**). The learning curves show early divergence between training and validation performance, reflecting limited generalization (**Figure 10**). This behavior is further supported by the loss curve, where validation loss fluctuates despite continuous training loss reduction (**Figure 11**). Nevertheless, the ROC curve presents a high AUC value of 0.9714, suggesting that the predicted probabilities still provide strong ranking capability despite poor threshold-based classification (**Figure 12**).

Fold 3(run_20260109_152503):

- Best Epoch: 7 (selected based on Best Validation AUC: 0.9778)
- Best Validation AUC: 0.9778
- Final Metrics: AUC=0.9778, F1=0.7738, Accuracy=0.7895
- Confusion Matrix: LGG: 15/15 correct (100%), HGG: 0/42 correct (0.0%)
- **Severe Degenerate Behavior:** Model exhibits complete collapse to single-class prediction (all samples predicted as LGG) despite high AUC
- **Visualizations:** At the best validation epoch (epoch 7), the confusion matrix indicates complete collapse to single-class prediction, where all samples are classified as LGG and no HGG cases are detected (**Figure 13**). The learning curves reveal rapid overfitting, with validation performance degrading shortly after early epochs (**Figure 14**). The loss curve further confirms training instability, as validation loss fails to converge (**Figure 15**). Despite this degenerate classification behavior, the ROC curve achieves a high AUC of 0.9778, highlighting a discrepancy between probability ranking performance and discrete classification outcomes (**Figure 16**).

Fold 4 (run_20260109_153707):

- Best Epoch: 10 (selected based on Best Validation AUC: 0.9651)
- Best Validation AUC: 0.9651
- Final Metrics: AUC=0.9651, F1=0.8580, Accuracy=0.8947

- Confusion Matrix: LGG: 15/15 correct (100%), HGG: 0/42 correct (0.0%)
- **Severe Degenerate Behavior:** Model exhibits complete collapse to single-class prediction (all samples predicted as LGG) despite high AUC
- **Visualizations:** Similar to Fold 3, the confusion matrix at the best validation epoch (epoch 10) shows a complete collapse to LGG-only predictions, with no correctly identified HGG cases (**Figure 17**). The learning curves demonstrate persistent overfitting, with a widening gap between training and validation performance (**Figure 18**). The loss curve confirms this instability through fluctuating validation loss values (**Figure 19**). Nevertheless, the ROC curve maintains a high AUC of 0.9651, reinforcing the observation that probability-level discrimination does not translate into effective threshold-based classification (**Figure 20**).

Training Dynamics Observations:

All folds exhibit consistent overfitting patterns, with training loss decreasing while validation loss increases or fluctuates. The train/val loss gap ranges from 0.29 to 0.40, indicating that the model memorizes training data characteristics that do not generalize to validation sets. Validation loss shows high variability across epochs, with fluctuations suggesting training instability. These dynamics are visualized in the loss curves and learning curves for each fold (see file paths above). Note that precision-recall curves are not available for DualStreamMIL-3D across any fold, as they were not generated during training.

The model consistently achieves perfect LGG recall (100% in all folds) but struggles with HGG classification, particularly in folds 0, 3, and 4. This pattern suggests that the MIL aggregation mechanism may be collapsing to focus on LGG-discriminative slices, ignoring HGG-relevant information. The complete collapse to single-class prediction observed in folds 3 and 4, where all predictions are LGG despite high AUC, indicates that the model learned to exploit class imbalance by defaulting to the minority class.

Root Cause Analysis:

The observed issues can be attributed to several factors:

1. **Class Imbalance:** The 2.8:1 HGG: LGG ratio, combined with bag-level supervision, makes it difficult for the model to learn balanced decision boundaries. WeightedRandomSampler may not be sufficient for MIL bag-level supervision.

2. **Attention/Selection Collapse:** The dual-stream mechanism may be collapsing to focus on a small subset of slices, losing contextual information. Without attention entropy logging, this hypothesis cannot be directly verified from available results.

3. **Training Instability:** High variance in validation metrics across epochs and folds suggests that the model's training dynamics are sensitive to initialization and data sampling.

4. **Insufficient Regularization:** Despite multiple regularization mechanisms (dropout, label smoothing, temperature annealing), the model still overfits, indicating that stronger regularization or architectural modifications may be needed.

6.2.2 ResNet50-3D and SwinUNETR-3D: Out-of-Fold Prediction Analysis

While comprehensive per-fold metrics are not available in aggregated form for ResNet50-3D and SwinUNETR-3D, their out-of-fold predictions have been collected and verified across all 5 folds. Analysis of these OOF predictions provides insights into model behavior and prediction characteristics.

Available Visualizations:

For ResNet50-3D, training visualizations are available for individual folds. In Fold 0 (best epoch: 18), the confusion matrix at the selected best validation epoch illustrates moderate discriminative performance, with noticeable misclassification of HGG samples (**Figure 21**). The corresponding learning curves reveal a widening gap between training and validation performance, indicating overfitting behavior emerging during later epochs (**Figure 22**). This trend is further supported by the loss curve, where validation loss plateaus while training loss continues to decrease (**Figure 23**). The ROC curve for this fold demonstrates limited separability between classes, achieving a best validation AUC of 0.8079 (**Figure 24**). The precision-recall curve highlights reduced precision at higher recall levels, reflecting challenges in handling class imbalance (**Figure 25**).

In Fold 1 (best epoch: 28), the confusion matrix shows improved classification balance compared to Fold 0, particularly in identifying HGG cases (**Figure 26**). The learning curves suggest improved but still imperfect generalization, with reduced yet persistent divergence between training and validation accuracy (**Figure 27**). The loss curve indicates more stable optimization

dynamics relative to Fold 0, although mild overfitting remains evident (**Figure 28**). The ROC curve demonstrates substantially improved discriminative performance, achieving a best validation AUC of 0.9016 (**Figure 29**). This improvement is further reflected in the precision-recall curve, which exhibits higher precision across a broader recall range (**Figure 30**).

For SwinUNETR-3D, similar training visualizations are available for individual folds. In Fold 0 (best epoch: 10), the confusion matrix reflects improved sensitivity to HGG cases compared to ResNet50-3D, although some LGG misclassification persists (**Figure 31**). The learning curves show relatively stable training behavior, with a smaller gap between training and validation performance than observed in CNN-based models (**Figure 32**). The loss curve further confirms this stability, with smoother convergence patterns (**Figure 33**). The ROC curve demonstrates strong discriminative ability, achieving a best validation AUC of 0.9175 (**Figure 34**). The precision-recall curve indicates robust performance under class imbalance, maintaining higher precision across a wide recall range (**Figure 35**).

In Fold 1 (best epoch: 17), the confusion matrix shows consistent classification behavior with further improvements in HGG detection (**Figure 36**). The learning curves indicate stable optimization and reduced overfitting compared to earlier folds (**Figure 37**). The loss curve reflects consistent convergence between training and validation loss values (**Figure 38**). The ROC curve achieves a best validation AUC of 0.9238, confirming strong class separability (**Figure 39**). This performance is corroborated by the precision-recall curve, which maintains favorable precision-recall trade-offs (**Figure 40**).

OOF Prediction Statistics (285 patients, all folds combined):

To quantitatively compare prediction behavior across models, **Table (2)** summarizes descriptive statistics of the predicted HGG probabilities obtained from OOF predictions across all five folds (285 patients).

Interpretation

ResNet50-3D produces consistently high HGG probabilities with relatively low variance, indicating confident but potentially overconfident predictions. In contrast, SwinUNETR-3D exhibits the largest variability and widest probability range, reflecting more nuanced and diverse prediction behavior. DualStreamMIL-3D produces more conservative probability estimates, consistent with the class imbalance and stability issues observed in its per-fold analysis. These distinct prediction characteristics support the motivation for ensemble learning, as the models capture complementary aspects of the classification task.

Observations:

1. **ResNet50-3D** produces the highest average HGG probability (0.8884) with relatively low variance (std=0.1195), suggesting that the model is generally confident in its predictions and may be calibrated to produce higher probabilities overall. The narrow range (0.1652 to 0.9999) indicates consistent prediction behavior.
2. **SwinUNETR-3D** shows the largest variance (std=0.4406) and the widest range (0.0023 to 1.0000), indicating more diverse predictions. The mean probability (0.5799) is closer to the balanced threshold of 0.5, suggesting different calibration characteristics compared to ResNet50-3D.
3. **DualStreamMIL-3D** produces the most conservative probabilities (mean=0.4847) with a narrow range (0.1113 to 0.9497), consistent with the class imbalance issues observed in per-fold analysis. The conservative probabilities reflect the model's tendency to favor LGG predictions.

These different probability distributions suggest that the three models have distinct calibration characteristics and may capture complementary information, making them suitable candidates for ensemble combination.

OOF Prediction Verification:

All OOF predictions have been verified for data integrity:

- ✓ Each patient appears exactly once per model (no duplicates)
- ✓ All 5 folds are represented (57 patients per fold)
- ✓ Probability values are in valid range [0, 1]
- ✓ Labels are consistent across all models
- ✓ No data leakage detected (folds are mutually exclusive)

The verified OOF predictions serve as the training data for the ensemble meta-learner, ensuring that the meta-learner learns from predictions made on data that was not used to train the corresponding base models.

6.3 Ensemble Model Performance

6.3.1 Meta-Learner Training and Configuration

The ensemble meta-learner is a Logistic Regression classifier trained on merged OOF predictions from all three base models. The training dataset consists of 285 patients with three features (HGG probabilities from each base model) and binary labels (0=LGG, 1=HGG). The class distribution is 75 LGG (26.3%) and 210 HGG (73.7%), maintaining the original dataset imbalance.

Model Configuration:

- Algorithm: Logistic Regression (scikit-learn)
- Solver: LBFGS (efficient for small datasets)
- Regularization: L2 penalty (C=1.0)
- Class weights: Balanced (LGG weight: 1.90, HGG weight: 0.68)
- Max iterations: 1000
- Random state: 42 (reproducibility)

6.3.2 Learned Combination Weights

The contribution of each base model to the ensemble decision is quantified by the coefficients learned by the logistic regression meta-learner, as summarized in **Table (3)**. The magnitude of each coefficient reflects the relative importance of the corresponding model's predicted HGG probability in the final classification, while the sign indicates the direction of its influence. The meta-learner was trained using out-of-fold predictions from all five cross-validation folds (285 patients).

Interpretation:

1. **SwinUNETR-3D** has the highest coefficient (4.06), indicating it is the most important base model in the ensemble. This suggests that SwinUNETR-3D's predictions contain the most informative signal for final classification decisions, possibly due to its ability to capture long-range dependencies through self-attention mechanisms.

2. **DualStreamMIL-3D** has moderate importance (coefficient 0.89), ranking second. Despite the class imbalance issues observed in individual fold analysis, the MIL model contributes meaningful information to the ensemble.

3. **ResNet50-3D** has the lowest coefficient (0.54), suggesting it contributes less to the final prediction. This may be because its predictions are highly correlated with SwinUNETR-3D (both process full 3D volumes) or because its higher average probabilities require less weight to achieve optimal combination.

4. **Negative Intercept** (-2.40) indicates a bias toward the negative class (LGG) in base model predictions, which the meta-learner corrects to achieve balanced classification.

The learned weights demonstrate that the ensemble successfully identifies and leverages the complementary strengths of each base model, with SwinUNETR-3D's global context modeling being most valuable for final decisions.

6.3.3 Ensemble Performance Metrics

The overall performance of the ensemble meta-learner was evaluated using out-of-fold (OOF) predictions from all five cross-validation folds, comprising a total of 285 patients. **Table (4)** summarizes the global performance metrics of the ensemble model.

Per-Class Performance:

To further analyze class-specific behavior, **Table (5)** reports precision, recall, and F1-score separately for LGG and HGG classes based on the same OOF predictions.

Confusion Matrix:

The classification outcomes of the ensemble model are further summarized by the confusion matrix presented in **Table (6)**, which illustrates the distribution of correct and incorrect predictions across both classes.

The confusion matrix is visualized in **Figure 43**

Analysis:

1. **High Precision (0.96)** : When the ensemble predicts HGG, it is correct 96% of the time, indicating very few false positives. This is clinically valuable as it minimizes unnecessary alarm for patients who do not have high-grade gliomas.

2. **Moderate Recall (0.77)**: The ensemble identifies 77% of all HGG cases, meaning 23% of HGG patients are missed. The 48 false negatives (HGG predicted as LGG) represent a clinical concern, as these patients may not receive appropriate treatment.

3. **Strong AUC (0.91)** : The AUC-ROC of 0.9126 indicates excellent discriminative ability, with the model achieving strong separation between HGG and LGG classes across all possible classification thresholds.

4. **LGG Performance**: High recall (0.92) but lower precision (0.59), indicating that the ensemble is conservative in predicting LGG—it tends to predict HGG when uncertain. This conservative approach may be appropriate for clinical applications where missing HGG cases is more critical than false LGG predictions.

5. **Class Imbalance Effect**: The balanced class weights improved LGG recall (0.92) but resulted in some HGG false negatives (48 cases). The overall F1-score of 0.86 indicates a reasonable balance between precision and recall, considering the class imbalance.

Visual Results:

The following visualizations are available for ensemble performance analysis:

To further analyze ensemble performance beyond numerical metrics, several visual representations are provided. These figures offer complementary perspectives on discriminative ability, calibration behavior, class-wise performance, and the contribution of individual base models.

The ROC curve illustrates the trade-off between sensitivity and specificity across all possible decision thresholds, providing a global view of the ensemble's discriminative capacity (**Figure 41**). Complementarily, the precision-recall curve highlights the balance between precision and

recall, which is particularly informative under class imbalance conditions where ROC curves alone may be misleading (**Figure 42**).

The confusion matrix presents a direct visualization of classification outcomes, summarizing true positives, true negatives, false positives, and false negatives at the default decision threshold (**Figure 43**). To better understand how individual base models contribute to the ensemble decision, the feature importance plot visualizes the learned coefficients of the logistic regression meta-learner, reflecting the relative influence of each model's predicted probability (**Figure 44**).

Class-wise performance is further examined through a per-class metrics comparison, showing precision, recall, and F1-score separately for LGG and HGG classes, thereby highlighting asymmetries in model behavior between the two classes (**Figure 45**). The distribution of predicted probabilities provides insight into calibration characteristics and confidence levels of ensemble outputs across all patients (**Figure 46**). Finally, a consolidated performance summary visualization aggregates key evaluation metrics into a single overview, facilitating rapid qualitative comparison of overall performance indicators (**Figure 47**).

These visualizations provide intuitive representations of ensemble performance and enable detailed analysis of model behavior across different classification thresholds and class distributions. All ensemble visualizations are available in the ``ensemble/visualizations/`` directory.

6.4 Comparative Results and Analysis

6.4.1 Base Model Prediction Characteristics

Analysis of OOF predictions reveals distinct prediction patterns across the three base models, reflecting their different architectural approaches and training dynamics.

Probability Distribution Analysis:

ResNet50-3D produces the most confident predictions on average (mean=0.8884), with a relatively narrow distribution (std=0.1195). This suggests that the volumetric CNN model has learned strong discriminative features and is generally confident in its classifications. The high average probability may reflect the model's calibration characteristics or its ability to extract clear volumetric patterns that distinguish HGG from LGG.

SwinUNETR-3D shows the most diverse prediction distribution (std=0.4406), with probabilities spanning nearly the full range [0.0023, 1.0000]. This high variance suggests that the transformer-based model produces more nuanced predictions, potentially capturing subtle patterns that lead to varying confidence levels. The mean probability (0.5799) closer to 0.5 indicates different calibration compared to ResNet50-3D.

DualStreamMIL-3D produces the most conservative probabilities (mean=0.4847) with a narrow range, consistent with the class imbalance issues observed in per-fold analysis. The conservative probabilities reflect the model's tendency to favor LGG predictions, which is problematic for individual model performance but may still contribute useful information to the ensemble.

Complementary Information:

The distinct probability distributions suggest that the three models capture complementary information:

- ResNet50-3D provides confident volumetric feature-based predictions
- SwinUNETR-3D offers nuanced global context-aware predictions with high variance
- DualStreamMIL-3D contributes conservative slice-level reasoning

The ensemble meta-learner successfully combines these diverse perspectives, with SwinUNETR-3D receiving the highest weight (4.06), indicating that its diverse predictions are most informative for final decisions.

6.4.2 Ensemble vs. Individual Models

While direct quantitative comparison of ensemble performance against individual base models on the same evaluation set is not available in aggregated form, the learned ensemble weights provide insights into relative model contributions. The meta-learner's assignment of highest weight to SwinUNETR-3D (4.06) suggests that this model's predictions are most valuable for ensemble decisions, followed by DualStreamMIL-3D (0.89) and ResNet50-3D (0.54).

The ensemble achieves an AUC-ROC of 0.9126, which is competitive with the mean AUC of DualStreamMIL-3D (0.9609) when considering that ensemble evaluation is on OOF predictions (validation data) rather than a held-out test set. The ensemble's strength lies in its ability to combine diverse model perspectives, potentially achieving more robust performance than any individual model alone.

Ensemble Benefits:

1. **Robustness:** By combining three different architectural paradigms, the ensemble is less susceptible to failure modes specific to individual models (e.g., DualStreamMIL-3D's class imbalance issues in some folds).
2. **Calibration:** The meta-learner learns to combine differently calibrated predictions, producing well-calibrated ensemble outputs.
3. **Complementary Information:** Each model contributes unique information (volumetric features, global context, slice-level reasoning), which the ensemble leverages for improved performance.

6.5 Discussion

6.5.1 Achievement of Research Objectives

The experimental results demonstrate that the proposed ensemble methodology successfully addresses several key research objectives:

1. **Multi-Modal 3D Processing:** All three base models process full 3D volumes with early fusion of four MRI modalities (T1, T1ce, T2, FLAIR), addressing the limitation of 2D approaches identified in prior work.
2. **Architectural Diversity:** The ensemble combines three distinct paradigms (volumetric CNN, transformer, multiple instance learning), leveraging complementary strengths and mitigating individual model limitations.

3. **Robust Evaluation:** Patient-level 5-fold cross-validation ensures that performance metrics reflect generalization to unseen patients, addressing concerns about data leakage in prior studies.

4. **Ensemble Integration:** The logistic regression meta-learner successfully learns optimal combination weights, with SwinUNETR-3D identified as the most important contributor, demonstrating effective ensemble construction.

6.5.2 Key Findings

Strengths:

1. **Strong Discriminative Ability*:** The ensemble achieves an AUC-ROC of 0.9126, indicating excellent separation between HGG and LGG classes. This performance is competitive with state-of-the-art methods while using a more comprehensive evaluation protocol (patient-level cross-validation).

2. **High Precision for HGG:** The ensemble's precision of 0.9643 for HGG predictions means that when the model predicts high-grade glioma, it is correct 96% of the time. This high precision is clinically valuable as it minimizes false alarms.

3. **Effective Ensemble Combination:** The learned combination weights demonstrate that the meta-learner successfully identifies and leverages the most informative base model (SwinUNETR-3D), while still incorporating useful information from other models.

4. **Complementary Model Contributions:** The distinct probability distributions across base models confirm that they capture complementary information, validating the ensemble design choice.

Limitations:

1. Moderate HGG Recall: The ensemble's recall of 0.7714 for HGG means that 23% of HGG cases are missed. The 48 false negatives represent a clinical concern, as these patients may not receive appropriate treatment. This limitation may be addressed through threshold tuning or additional training data.

2. Class Imbalance Effects: Despite balanced class weights in the meta-learner, the ensemble still shows conservative LGG predictions (precision=0.59) and some HGG false negatives, reflecting the underlying 2.8:1 class imbalance in the dataset.

3. Individual Model Instability: DualStreamMIL-3D shows high variance across folds and class imbalance issues in several folds, indicating that individual model training may benefit from improved regularization or architectural modifications.

4. Limited Test Set Evaluation: Results are reported on OOF predictions (validation data) rather than a held-out test set. While this is standard practice for cross-validation, evaluation on a truly independent test set would provide additional confidence in generalization.

6.5.3 Comparison with Prior Work

The ensemble's AUC-ROC of 0.9126 is competitive with results reported in the literature review (Chapter 3), particularly considering that:

- The evaluation uses patient-level cross-validation, which is more rigorous than many prior studies
- The methodology addresses limitations identified in prior work (3D processing, multi-modal fusion, ensemble methods)
- The ensemble combines three distinct architectural paradigms, providing robustness not present in single-model approaches

However, direct comparison is challenging due to differences in datasets, evaluation protocols, and task definitions (some studies focus on segmentation rather than classification). The current results demonstrate that the proposed methodology achieves strong performance while addressing methodological limitations of prior work.

6.5.4 Clinical Implications

The ensemble's high precision (0.96) for HGG predictions is clinically valuable, as it minimizes false positives that could lead to unnecessary patient anxiety or invasive procedures. However, the moderate recall (0.77) means that some HGG cases are missed, which is a concern for clinical deployment.

The conservative LGG predictions (precision=0.59) may be acceptable in clinical settings where erring on the side of caution (predicting HGG when uncertain) is preferred. However, this approach may lead to over-treatment of some LGG patients.

For clinical deployment, threshold tuning could be performed to optimize the precision-recall trade-off based on clinical priorities. Alternatively, the ensemble could be used as a screening tool, with uncertain cases flagged for additional review by radiologists.

6.5.5 Future Directions

Based on the experimental results, several directions for future improvement are identified:

1. **Address Class Imbalance:** Stronger class balancing techniques, such as focal loss or more aggressive oversampling, may improve HGG recall while maintaining precision.
2. **Improve MIL Training Stability:** The observed class imbalance issues in DualStreamMIL-3D suggest that architectural modifications or training strategies (e.g., stronger regularization, attention entropy monitoring) may be needed.
3. **Threshold Optimization:** Systematic threshold tuning based on clinical priorities could optimize the precision-recall trade-off for specific deployment scenarios.
4. **External Validation:** Evaluation on an independent test set from a different institution would provide additional confidence in generalization to new patient populations.

5. Interpretability Analysis: Detailed analysis of attention weights and critical instance selection in DualStreamMIL-3D could provide insights into model decision-making and identify failure modes.

5. **Calibration Analysis:** Systematic calibration analysis could identify whether probability outputs are well-calibrated and whether calibration techniques (e.g., Platt scaling, temperature scaling) could improve reliability.

6.

6.6 Chapter Summary

This chapter presented experimental results from training and evaluating the proposed ensemble system for brain tumor grade classification. Key findings include:

Individual Model Performance:

- DualStreamMIL-3D achieves strong discriminative ability (mean AUC=0.9609) but shows high variance across folds and class imbalance issues in several folds
- ResNet50-3D produces confident predictions (mean probability=0.8884) with low variance
- SwinUNETR-3D shows diverse predictions (high variance) with mean probability closer to balanced threshold

Ensemble Performance:

- The ensemble achieves AUC-ROC of 0.9126, precision of 0.9643, recall of 0.7714, and F1-score of 0.8571
- SwinUNETR-3D is identified as the most important base model (coefficient=4.06), followed by DualStreamMIL-3D (0.89) and ResNet50-3D (0.54)
- High precision for HGG (0.96) minimizes false positives, while moderate recall (0.77) indicates some HGG cases are missed

Key Observations:

- The three base models produce distinct probability distributions, confirming complementary information capture
- The ensemble successfully combines diverse model perspectives for robust performance

- Class imbalance effects persist despite balancing techniques, affecting LGG precision and HGG recall

Limitations and Future Work:

- Moderate HGG recall (0.77) may limit clinical utility
- Individual model training instability (particularly DualStreamMIL-3D) suggests need for improved regularization
- Evaluation on independent test set would provide additional confidence in generalization

The results demonstrate that the proposed ensemble methodology achieves competitive performance while addressing methodological limitations identified in prior work, providing a foundation for further development and potential clinical deployment.

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